

The coppe document class

Version 4.0

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Abstract

In this work, it is described the `coppe` document class as well as other files distributed by the COPPETEX project. This class is suitable for writing academic dissertations, thesis and qualifying exams according to the formatting rules of the Alberto Luiz Coimbra Institute for Graduate Studies and Research in Engineering. The minimalist set of macro commands allows its users to concentrate most of their efforts on text composition rather than on the document layout.

1 Introduction

Writing documents in L^AT_EX may be a laborious task when the authors have to prepare their manuscripts rigorously respecting formatting rules imposed by publishers. Regardless of difficulty, a lot of thesis presented to the Coordination of Graduate Studies and Research in Engineering of the Federal University of Rio de Janeiro (COPPE/UFRJ) is typesetted in L^AT_EX. This demand motivated the creation of the COPPETEX project, which tries to facilitate and encourage the use of L^AT_EX within the COPPE/UFRJ scope.

The `coppe` document class is the main product of COPPETEX. It was designed to be clear and succinct. It enables the creation of dissertations, qualifying exams and thesis in a simple and automatic way. The main goal of the `coppe` class is to maintain authors strictly focused on text composition without worrying about margins sizes, line spacing, paper size, vertical and horizontal alignment, etc. The COPPETEX project comprehends also BibT_EX and MakeIndex style files for creating lists of references, symbols and abbreviations. Although there aren't official guidelines to write qualifying exams, we provide this option just for convenience, as this exam is a requisite to obtain the DSc degree and, for some of the programs, the MSc degree.

In which follows, it is described the user interface of the `coppe` class. Some details about using the style files cited above are also given. We use the term *thesis* to generally refer to dissertation, qualifying exam, and thesis itself.

2 License

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copying, distributing or modifying the source code, among other acts covered by this license.

To see the full text of the GNU GPL license, go to the `COPYING` file attached to this package.

3 Support

We maintain a WhatsApp group where users can send questions, comments, and bugs to. More details can be found [here](#).

However, as the project maintenance was transferred to a new repository, bug reports, as well as new feature requests, should be directed to <https://github.com/COPPE-UFRJ/CoppeTeX>.

4 User interface

`\frontmatter` A thesis to be approved by the Academic Registry at COPPE/UFRJ must contain three-parts: *front*, *main* and *back* matters (COPPE/UFRJ Academic Registry guidelines). Each one of these parts is started by calling its corresponding macro `\frontmatter`, `\mainmatter` or `\backmatter`. The front matter of a thesis consists of front cover and face, cataloging page, dedication, acknowledgments, abstracts, table of contents, and lists of tables, algorithms, symbols and abbreviations. The main matter is just composed by chapters, while the back matter usually consists of bibliographic references, appendices and index.

You must invoke the `\frontmatter` macro immediately after the `\maketitle` one. The `\mainmatter` command comes right before the first chapter, and `\backmatter` must be typed before the list of references.

Cover (capa)

The ABNT cover suggested in the manual's Annex A. The `\maketitle` command builds it automatically, ahead of the title page: the university, the full COPPE name, and the graduate program (taken from `\department`), followed by the author, the title, and the city and year, all centred and in uppercase. The UFRJ and COPPE logos head the page. The capa is not counted.

Front face

The front face is unnumbered. There, it is not allowed to use hyphenation (COPPE/UFRJ Academic Registry guidelines). It is constructed by calling `\maketitle`. Next, it is described the commands used to enter the information required to create it.

`\author` The `\author` command was redefined. Here, it takes two arguments: the author's first names and surname, e.g., `\author{First Names}{Surname}`. The words should be typed with only first letters in uppercase.

`\title` The macros `\title` and `\foreigntitle` are used to enter the titles of your monograph in Brazilian Portuguese and English respectively (the two built-in languages of CoppeTeX 4.x). For documents whose main language is neither

brazilian nor english – e.g. spanish, french, italian, or a user-supplied language –, also call `\titlein{<lang>}{<text>}` to register the title in that language; the cover and folha de rosto print whichever title matches the main language declared in the class option. See the section “Multilingual support” below for the complete picture. The `babel` package is loaded automatically by `coppe.cls`, so you do not need to load it yourself.

`\advisor` Every COPPE student is coordinated by at least one advisor. M.Sc. and D.Sc. students can have at most 2 and 3 advisors, respectively. Their names must be provided by issuing the command `\advisor` as below:

```
\advisor{Title}{Advisor's Name}{Surname}{Degree}
\advisor{Title}{Second Advisor's Name}{Surname}{Degree}
\advisor{Title}{Third Advisor's Name}{Surname}{Degree}
```

The advisors are not necessarily members of the thesis examination board. Thus, it is required to enter the names of all examiners using the `\examiner` macro. The examiners' names are entered differently:

```
\examiner{Title}{First Examiner's Name Surname}{Degree}
\examiner{Title}{Second Examiner's Name Surname}{Degree}
...
\examiner{Title}{N-th Examiner's Name Surname}{Degree}
```

Remember that all names must be given before calling `\maketitle`.

`\department` The Alberto Luiz Coimbra institute is divided into 12 academic units: Biomedical Engineering (PEB), Civil Engineering (PEC), Electrical Engineering (PEE), Mechanical Engineering (PEM), Metallurgical and Materials Science Engineering (PEMM), Nuclear Engineering (PEN), Ocean Engineering (PENO), Energy Planning (PPE), Production Engineering (PEP), Chemical Engineering (PEQ), Systems Engineering and Computer Science (PESC), and Transportation Engineering (PET). You must specify your department using one of the above abbreviations, e.g., `\department{PEC}`.

`\date` This macro is used to set the month and year of defense. This information is required to create the front face, cataloging details page and abstracts. For example, October 2007 should be entered as `\date{10}{2007}`.

`\keyword` The keywords should describe the concentration areas of your work. You must provide them as follows:

```
\keyword{First Keyword}
\keyword{Second Keyword}
...
\keyword{N-th Keyword}
```

Usually, six words are enough.

Cataloging details

This page contains cataloging information useful for librarians. Fortunately, it is automatically generated from the data you entered at the time you call `\maketitle`. It is not needed in qualifying exams, though.

Dedication (optional)

`\dedication` This macro was added for convenience. The input text is placed at the right bottom of a blank page. It is emphasized and in normal size.

Abstracts

`abstract (env.)` As stated by the COPPE/UFRJ Academic Registry, abstracts must be in one page each, with at most 250 words. We recommended that they should be only one paragraph long. They must be defined inside the environments `abstract` and `foreignabstract`.

Lists of symbols and abbreviations (optional)

`\abbrev` The lists of symbols and abbreviations are optional, although highly recommended.
`\syml` It is a good practice to define a symbol/abbreviation in its first occurrence in the text. To define a symbol use `\syml[alphabetic symbol]{Symbol}{Symbol Definition}`, and for abbreviations `\abbrev[alphabetic symbol]{Abbreviation}{Abbreviation Definition}`. These commands are called *dummy*, since they don't output anything at the place they are executed, just an entry in the correspondent list.
`\makeloabbreviations` These lists are lexicographically sorted by using the MakeIndex program, which
`\makelosymbols` is part of any L^AT_EX implementation. For `\syml`, if the optional parameter is
`\printloabbreviations` provided, it will used as sort key. This was later, in 2024, implemented also for
`\printlosymbols` `\abbrev`, otherwise `Symbol`, or `Abbreviation` will be used as sort key, what can result in an undesirable order if it contains L^AT_EX commands, mathematical symbols, or mix of uppercase and lowercase. MakeIndex needs two commands to create a final sorted list: one which generates a list of entries and the other that indicates the position where the list will be printed out. To generate the lists of symbols and abbreviations, the `coppe` class provides the commands `\makeloabbreviations` and `\makelosymbols`, respectively. They must be called in the document preamble. The commands `\printlosymbols` and `\printloabbreviations` have to be invoked at the point where you want these lists appear, e.g., following the list of tables as showed in the example. Once you call `latex`, it will be created two files with extensions `abx` and `syx`, which contain MakeIndex input data. They must be processed with `makeindex` in order to get the lists correctly produced, redirecting the output to files with extension `lab` and `los` respectively:

```
makeindex -s coppe.ist -o example.lab example.abx
makeindex -s coppe.ist -o example.los example.syx
```

Note the `-s` option for specifying the style `coppe.ist`. Now, rerun `latex` twice to get the references solved and you are done.

References

COPPE_T_EX produces citations and the reference list with `biblatex` and the `biber` backend, through its own bibliography style implementing the post-2020 ABNT NBR 6023 rules. The class loads this style automatically: there is no `\bibliographystyle` to set and no Bib_T_EX `.bst` file to choose (the former `coppe-plain` and `coppe-unsrt` Bib_T_EX styles have been retired).

By default, citations follow the author–date style, e.g., “Chartier (2002)”. Passing the `numbers` class option switches to the numeric style, in which citations are bracketed numbers and the reference list is numbered in order of citation (unsorted).

Declare your bibliography database in the preamble with `\addbibresource` and print the list where it should appear with `\printbibliography`:

```
\addbibresource{example.bib}    % in the preamble
...
\printbibliography              % where the list goes
```

Entry types and fields may be written in English (`@book`, `title`, `author`) or with their unaccented Portuguese synonyms (`@livro`, `titulo`, `autor`).

Run in sequence `LATEX`, `biber`, and twice again `LATEX` to resolve the citations and references. The style is `natbib` compatible (via the `biblatex natbib=true` option), so you may freely issue `\citet` and `\citep`, as well as the other `natbib` commands.

Reference entry types

Every ABNT document category maps to a `biblatex` entry type, each with an unaccented Portuguese synonym (a worked example of every type is in the sample manuscript’s reference list):

Academic `@book` (`@livro`); `@article` (`@artigo`); `@artigodejornal` (newspaper); `@incollection` (`@partedelivro`); `@inbook` (`@emlivro`); `@inreference` (`@verbete`); `@proceedings` (`@anais`); `@inproceedings` (`@trabalhodeevento`); `@thesis` (`@tese/@dissertacao`, with the presets `@dissertacaomestrado`, `@tesedoutorado`, `@tesephds`, `@tcc`); `@report` (`@relatorio`); `@periodical` (`@periodico`); `@manual`; `@misc` (`@diverso`); `@online`.

Special (NBR 6023 §4.2.5–4.2.14) `@patent` (`@patente`); `@legislation` (`@legislacao/@atonormat`); `@jurisdiction` (`@jurisprudencia`, with a `relator` field); `@legal` (`@documentojuridico`); `@standard` (`@norma`); `@music` (`@partitura`); `@map` (`@mapa`); `@image` (`@iconografico`); `@artwork` (`@obradearte`); `@video` (`@audiovisual`); `@movie` (`@filme`); `@letter` (`@correspondencia`); `@dataset` (`@conjuntodedados`).

Games and TV `@game` (`@jogo`); `@boardgame` (`@jogodetabuleiro`); `@videogame` (`@jogoeletronico`); `@tvshow` (`@programadetv`).

When an entry has no `author` or `editor`, it is entered by title, with the first significant word (and a leading article) in capitals.

Appendix and Annex (Optional)

`\appendix` Appendices and annexes are optional chapters that are part of the back matter.

`\annex` The `\appendix` command is a standard `LATEX` command used before all appendices. `COPPETEX` introduces the `\annex` command, which should be used only in the back matter (i.e., after the `\backmatter` command) and only after all existing `\appendix` chapters. This restriction is due to implementation constraints.

Therefore, the order for backmatter is:

```
\backmatter
\printbibliography
\appendix
\chapter{An Appendix}
\chapter{Another Appendix}
\annex
\chapter{First Annex}
\chapter{Second Annex}
```

Annex was introduced in v3.5

Final colophon page (`\coppetexfinalpage`)

`\coppetexfinalpage` Optional closing page modelled on the printer's *colophon*: it records, in Portuguese, the tooling and edition used to typeset the thesis. Call it once, after every other piece of content, immediately before `\end{document}`. The macro

- forces a verso (even) page, so in a two-sided print the colophon sits opposite the rear cover;
- types the body at the foot of the page (`\vfill`);
- renders the text in Portuguese unconditionally (Brazilian `babel`), even when the thesis main language is English, Spanish, French or Italian — the colophon is institutional metadata, not body content.

Three slot macros are user-overridable. Redefine them with `\renewcommand` before `\coppetexfinalpage` if the document deviates from the defaults:

`\coppefinalengine` the T_EX engine identifier. Default: auto-detected via `iftex` (`\hologo{pdfLaTeX}`, `\hologo{LuaLaTeX}` or `\hologo{XeLaTeX}`).

`\coppefinalfont` the text body font. Default: *Computer Modern (padrão L^AT_EX)*. Redefine if you loaded `lmodern`, `mathptmx`, an OpenType math font, etc.

`\coppefinalmanual` the reference manual title (typeset in italics). Default: the current COPPE/UFRJ norm. Redefine if your work follows an earlier edition.

The macro also prints the COPPE_TE_X class version (captured into `\coppe@version` at class-load time, so it survives any later package that may redefine the global `\fileversion`) and the GitHub repository URL, <https://github.com/COPPE-UFRJ/CoppeTeX>.

5 Class options

There are some options users can specify in order to customize the appearance of the output produced by the `coppe` class. These options can be passed to `coppe` as follows: `\documentclass[option1, option2]{coppe}`. In which follows, we give a brief description of all supported options.

dsc, msc, dsccexam, msccexam The `coppe` class is able to produce thesis, dissertations, and qualifying exams, which are enabled by the `dsc`, `msc`, `msccexam`, and `dsccexam` options, respectively.

doublespacing The default line spacing is one-and-a-half. For enabling double spacing between lines, use the `doublespacing` option.

numbers The default citation style is the author–date scheme. For numbered citations, specify the `numbers` option to the `coppe` class; the reference list is then numbered in order of citation (unsorted). The matching `biblatex` style is selected automatically in either case.

brazilian, english, spanish, french, italian Pick the document’s *main* language. The class loads Babel with that language as the main one (the last in the Babel option list); inside the document `\selectlanguage{...}` and `\begin{otherlanguage}{...}` keep working as usual. The default, when no language option is given, is **brazilian**.

- **brazilian** and **english** are *built-in*: their strings ship inside `coppe.cls` and need no extra file.
- **spanish, french, italian** are shipped as *language packs*: two files per language (`coppe-lang-<lang>.def` and `<lang>-coppe.lbx`) that must sit next to `coppe.cls` (or in your project directory). They are auto-loaded by the class. Picking one of them as the main language without the matching pack files installed raises a clear class error.
- Any other Babel-known language can be plugged in by writing the same two files and either passing the language name as a class option or calling `\usecoppelanguage{<lang>}` in the preamble.

See the section “Multilingual support” below for the three-slot architecture, the `\titlein` / `\brazilianabstract` API, and how to create a new language pack.

5.1 Changing document identification

`\freeconfig` The user could *optionally* use the command `freeconfig` to modify the parameters that print the document identification. The command `freeconfig` needs all those parameters, which are degree initials, degree name, title, foreign title, local doctype, and foreign doctype as in the following example:

```
\freeconfig{Dr.}{Philosophiae Doctor}{PhD}{Doutor}{Dissertation}{Tese}
```

5.2 Multilingual support

CoppeTeX 4.x reserves three fixed language slots:

main the language of the body, captions, TOC labels and the `abstract` environment. Picked by the class option (**brazilian** by default; one of **english, spanish, french, italian** also accepted out of the box; any other Babel language possible with a user-supplied pack).

foreign the language of the `foreignabstract` environment. `english` by default for every main language; switches automatically to `brazilian` when `main = english`.

third optional. Loaded on demand with `\usecoppelanguage{<lang>}` in the preamble, useful when the body quotes or cites in a third language (e.g. a Spanish-main thesis citing a French source). Does not have a class option of its own.

`\titlein`
`brazilianabstract (env.)`
`\usecoppelanguage`

Two pieces of user-facing API come with the multilingual model:

- `\titlein{<lang>}{<text>}` records the title in any language. `\title{...}` still writes the Brazilian-Portuguese title and `\foreigntitle{...}` still writes the English one (back-compat); for a main language outside the `brazilian/english` pair, also call `\titlein{<lang>}{...}`. The cover and folha de rosto print whichever title matches `\coppe@mainlang`.
- The `brazilianabstract` environment is the optional third abstract used when the main language is neither `brazilian` nor `english` – typically a Spanish/French/Italian thesis that also wants a Portuguese resumo for UFRJ examiners. It wraps its body in `\begin{otherlanguage}{brazilian}` and uses the Portuguese template strings. A document with `main = brazilian` can simply keep using `abstract`; `brazilianabstract` is redundant then.
- `\usecoppelanguage{<lang>}` loads an extra language pack in the preamble without making it the main or foreign language.

Institutional names stay Portuguese

The cover, folha de rosto and ficha catalográfica are a Brazilian institutional template (NBR 14724 + COPPE manual). The strings `universityname`, `cityname`, `statename`, `countryname`, the departments declared with `\department` and the surrounding “apresentada/submetida ao...” wording are deliberately kept in Portuguese for every main language. Only the title (printed via `\coppe@selecttitle`) follows the main language.

Writing a new language pack

A language pack is just two files dropped next to `coppe.cls` (or anywhere on the `TEXINPUTS` path):

`coppe-lang-<lang>.def` class-level strings. Populate the dispatcher table with `\copperdefstring{<lang>}{<key>}{<value>}` for every key already populated for `brazilian/english` in `coppe.cls` (appendix, annex, frame, listframe, source, program, listprogram, references, in:, art, platform, directedby, producedby, listabbreviation, listsymbol, glossary, advisor, advisors, dept, approvedmale, approvedfemale, universityname, cityname, statename, countryname, monthname1...monthname12, algorithm2eopt, babelname). Finish with

```
\AtBeginDocument{\DeclareLanguageMapping{<lang>}{<lang>-coppe}}
```


so the `biblatex` mapping is registered after the bibliography engine is loaded by `coppe.cls` (which happens *after* the language pack is read).

`<lang>-coppe.lbx` `biblatex` localization strings. `\InheritBibliographyExtras{<lang>}` then `\DeclareBibliographyStrings` for the keys `availablefrom`, `urlseen`, `in`, `editor`, `editors`, `depositor`, `coppescale`, `mscdiss`, `dscthesi`, `phdthesis`, `tcc`. The shipped `spanish-coppe.lbx`, `french-coppe.lbx` and `italian-coppe.lbx` are the reference templates to copy from.

Tip: the `algorithm2eopt` key holds whatever `algorithm2e` calls the language. Mind the spelling: `algorithm2e` ships `portuguese`, `english`, `spanish`, `french` and `italiano` – not `italian`.

Example: a Spanish-main thesis

```
\documentclass[spanish,dsc]{coppe}
\addbibresource{example.bib}
\begin{document}
  \title{T'\i tulo em portugu\~es}      % Brazilian abstract slot
  \foreigntitle{English title}          % English abstract slot
  \titlein{spanish}{T'\i tulo en espa\~nol} % cover/folha
  \author{Juan}{P'\erez}
  \advisor{Prof.}{Mar'\i a}{Garc'\i a}{D.Sc.}
  \department{PESC}\date{01}{2026}
  \keyword{multilenguaje}
  \maketitle
  \frontmatter
  \begin{abstract}      Resumen principal en espa\~nol.      \end{abstract}
  \begin{foreignabstract} English foreign abstract.          \end{foreignabstract}
  \begin{brazilianabstract} Resumo em portugu\~es para a banca. \end{brazilianabstract}
  \tableofcontents
  \mainmatter
  \chapter{Introducci\on} Cuerpo en espa\~nol\dots
\end{document}
```

6 Files in your project folder

In a normal install where CoppeTeX is part of your `TEXINPUTS` path (e.g. inside the local `texmf` tree of MiKTeX or TeX Live), you do not need to think about this list at all — the files travel with the class. This section is for two situations where you do:

- shipping the class together with your manuscript (a self-contained submission package); or
- installing CoppeTeX by hand, dropping the files next to `coppe.cls` in the project directory.

Always required

These files must be findable at every compilation, regardless of the main language or any class option:

`coppe.cls` the document class itself.

`coppe.dbx` custom `biblatex` datamodel (declares the ABNT-specific entry types `@game`, `@videogame`, `@boardgame`, `@tvshow` and `@map`, plus a few custom fields).

`coppe.bbx` and `coppe.cbx` `biblatex` bibliography and citation styles for the author–date scheme (the default).

`brazilian-coppe.lbx` and `english-coppe.lbx` `biblatex` localisation tables for Portuguese and English. Both are required *even when neither language is the main one*, because `coppe.bbx` registers

```
\DeclareLanguageMapping{brazilian}{brazilian-coppe}
\DeclareLanguageMapping{english}{english-coppe}
```

unconditionally, so `biblatex` tries to read both LBX files at every compilation.

Conditional on class options

`coppe-numeric.bbx` and `coppe-numeric.cbx` only when the `numbers` class option is given (`\documentclass[numbers]{coppe}`). Without it, `biblatex` never tries to read them.

`coppe.ist` only when the glossary, list-of-symbols, or list-of-abbreviations features are used. It is the `makeindex` style file invoked by the `makeindex -s coppe.ist ...` step described in subsection *Lists of symbols and abbreviations* above.

Conditional on the main language

The main language is chosen by the matching class option (`brazilian`, `english`, `spanish`, `french`, `italian`). The two built-in languages ship their strings inside `coppe.cls` and so need no extra file. The three language packs add two files each — one class-level (`coppe-lang-<lang>.def`) and one `biblatex`-level (`<lang>-coppe.lbx`):

Main language	Extra files needed
<code>brazilian</code> (default)	— (built into <code>coppe.cls</code>)
<code>english</code>	— (built into <code>coppe.cls</code>)
<code>spanish</code>	<code>coppe-lang-spanish.def</code> , <code>spanish-coppe.lbx</code>
<code>french</code>	<code>coppe-lang-french.def</code> , <code>french-coppe.lbx</code>
<code>italian</code>	<code>coppe-lang-italian.def</code> , <code>italian-coppe.lbx</code>

Loading a third language at preamble time via `\usecoppelanguage{<lang>}` brings in the same pair of files for that language. A user-supplied language pack follows the same scheme: drop `coppe-lang-<lang>.def` and `<lang>-coppe.lbx` next to `coppe.cls`.

Cover assets

The institutional title page produced by `\maketitle` needs two PDF logo files at compile time:

`ufrj-logo.pdf` and `coppe-logo.pdf` the UFRJ shield and the COPPE/UFRJ wordmark. Both ship with the CoppeTeX distribution; they must be findable from the project directory (or on `TEXINPUTS`).

7 Required L^AT_EX packages

The class is built on the standard book class (`12pt`, `a4paper`, `twoside`) and brings in the stock CTAN packages listed below with `\RequirePackage`. On a full T_EX Live or MiK_TE_X install they are all present already, so this list matters only for a minimal install — or when you simply want to know which commands are already in scope. You do *not* need to `\usepackage` any of these yourself; every command and environment they provide is available in your manuscript. Options in brackets are the ones the class passes.

Engine and encoding

`iftex` detects the running engine, so the class can branch its setup between pdfT_EX and the Unicode engines (LuaT_EX/XeT_EX).

`inputenc` (`utf8`) UTF-8 input; loaded *only under pdfT_EX* (the Unicode engines read UTF-8 natively).

`fontenc` (`T1`) T1 font encoding, for correctly composed accented glyphs and hyphenation.

Mathematics and symbols

`amsmath` the AMS mathematics environments.

`amssymb` the AMS symbol fonts; loaded *only under pdfT_EX*. Under a Unicode engine load `unicode-math` (or `amssymb`) yourself — the class warns at end of document if neither is present.

Page layout and spacing

`geometry` A4 two-side margins, binding offset and header height.

`fancyhdr` running headers and footers.

`titlesec` chapter and section title formatting.

`setspace` one-and-a-half line spacing (double with the `doublespacing` option).

Lists, tables and rules

`enumitem` customisable list spacing and labels.

`tabularx` tables with automatic-width columns.

`booktabs` professional horizontal rules (`\toprule`, `\midrule`, `\bottomrule`).

`longtable` tables that break across pages.

Floats, graphics and captions

`graphicx` image inclusion.

`float` float placement control (e.g. the `[H]` specifier).

`caption` caption typography (the ABNT caption-on-top layout).

`subcaption` subfigures and subtables.

Code, verbatim and algorithms

`listings` source-code listings.

`fancyvrb` extended verbatim.

`algorithm2e` (`linesnumbered`, `lined`, `algochapter`, `ruled`) typeset algorithms;
the natural-language option tracks the main language.

Bibliography, quotations and language

`babel` multilingual typesetting; the language options are computed from the main
and foreign language slots.

`csquotes` context-sensitive quotation marks (the recommended companion of
`biblatex`).

`biblatex` (`backend=biber`) the bibliography engine and the ABNT `coppe` styles
(`coppe` or, with the `numbers` option, `coppe-numeric`). The backend is
`biber`, not `bibTeX`.

Hyperlinks

`hyperref` clickable cross-references, PDF bookmarks and document metadata.

Low-level utilities

`etoolbox` e-TeX programming toolkit.

`ltxcmds` low-level command helpers (`\ltx@ifpackageloaded`, ...).

`ifthen` the `\ifthenelse` conditional.

`hyphenat` hyphenation control (`\nohyphens`, used on the cover).

`lastpage` `\pageref{LastPage}` for the cataloging page count.

hologo T_EX-family logos (`\hologo{LaTeX}, \dots`).

xcolor colour support (used by `listings` and `hyperref`).

8 Quick, useful tips

Pictures. The default picture format of L^AT_EX is the Encapsulated PostScript (EPS). If you use pdfL^AT_EX, the default format becomes the PDF, but you can equally load PNG files. For such, you must enter the name of your image file without extension, e.g., `\includegraphics{filename}`, and pdf_lat_ex will firstly look for a file called `filename.pdf` and after for file `filename.png`. For producing high quality pictures with embedded fonts we recommend the Ipe drawing software available [here](#).

Fonts. The default font in L^AT_EX is the Computer Modern. If you would like to try its enhanced version, consider using the `lmodern` package. To use Times, it is recommended to load the package `mathptmx`, rather than the deprecated `times`. There is also an enhanced Times version available with the `tgtermes` package. You can still use the Arial font face with the package `uarial`.

Hyperref. When working with PDF's, there is the possibility to add extra information to the file as the author's name, document title, subject, keywords, etc. This is easily done with the `hyperref` package, which is loaded by the `coppe` class itself (since v4.0): the user must *not* load it again in the manuscript preamble. Author, title, subject and keyword metadata are filled in automatically by `\maketitle`.

Math (amssymb vs. unicode-math). The class always loads `amsmath`. The companion AMS symbol package `amssymb`, however, is loaded *only* under pdfL^AT_EX: it is a legacy 8-bit package built around the `msam/msbm` Type 1 fonts and clashes with `unicode-math`, the modern OpenType-math package used on Unicode engines (LuaL^AT_EX, XeL^AT_EX). On those engines the class skips `amssymb` and schedules an *end-of-document* check: if by then the author has loaded neither `unicode-math` nor `amssymb` (which they could have loaded by hand, though it is discouraged), a `ClassWarning` reminds them to add

```
\usepackage{unicode-math}
\setmathfont{Latin Modern Math} % or STIX Two Math, ...
```

to the manuscript preamble. The class deliberately does *not* auto-load `unicode-math`, because picking an OpenType math font is a typographic choice that belongs to the author. **Never load `amssymb` together with `unicode-math`:** it would re-establish the 8-bit AMS fonts on top of the OpenType math font and split symbols across two incompatible encodings.

Printing. To get your work correctly printed, you must ensure that any page scaling option (e.g., fit or shrink to printable area) isn't enabled. This kind of option often comes in print dialogs of document visualization softwares.

longquote Quotation To quote text larger than three lines, according to ABNT, you must increase the left margin to 4 cm, do not use quotation marks, and use a smaller font. The `coppe` class provides the `longquote` environment to easily make these adjustments.

9 A simple example

```

1 \example
2 \documentclass[dsc]{coppe}
3
4 \usepackage{rotating}          % rodando coisas, como tabelas
5 \usepackage[most]{tcolorbox} % caixas de texto
6 \usepackage{tikz}             % gráficos vetoriais (o tcolorbox já o traz)
7 % Sob LuaLaTeX/XeLaTeX, descomente as duas linhas a seguir para obter os
8 % símbolos AMS (\mathbb, \hbar, \varnothing, ...) via unicode-math. Sob
9 % pdfLaTeX, a classe já carrega amssymb automaticamente; deixe comentado.
10 \usepackage{unicode-math}
11 %\setmathfont{Latin Modern Math} % ou STIX Two Math, TeX Gyre Termes Math...
12 \DeclareMathOperator{\RMSE}{RMSE} % operadores próprios (cap. de matemática)
13 \DeclareMathOperator{\diag}{diag}
14 % A classe coppe (v4.0) já carrega: amsmath, booktabs, longtable, subcaption,
15 % hyperref, biblatex, listings, fancyvrb, algorithm2e -- não recarregue esses
16 % pacotes aqui (em particular, hyperref deve ficar a cargo da classe para que
17 % a ordem com setspace e biblatex permaneça correta).
18
19 \addbibresource{example.bib}% base de referências (biblatex/biber)
20
21 \makelosymbols
22 \makeloabbreviations
23
24 \begin{document}
25   \title{Título da Tese}
26   \foreigntitle{Thesis Title}
27   \author{Nome do Autor}{Sobrenome}
28   \advisor{Prof.}{Nome do Primeiro Orientador}{Sobrenome}{D.Sc.}
29   \advisor{Prof.}{Nome do Segundo Orientador}{Sobrenome}{Ph.D.}
30   % A classe aceita de um a tres \advisor; se houver apenas um, o rotulo na
31   % capa e na folha de rosto sai como "Orientador" (singular), com dois ou
32   % mais sai como "Orientadores" (plural). Para o caso 1/3 orientadores,
33   % veja tests/test_brazilian_one_advisor.tex e tests/test_brazilian_three_advisors.tex.
34
35   \examiner{Prof.}{Nome do Primeiro Examinador Sobrenome}{D.Sc.}
36   \examiner{Prof.}{Nome do Segundo Examinador Sobrenome}{Ph.D.}
37   \examiner{Prof.}{Nome do Terceiro Examinador Sobrenome}{D.Sc.}
38   \examiner{Prof.}{Nome do Quarto Examinador Sobrenome}{Ph.D.}
39   \examiner{Prof.}{Nome do Quinto Examinador Sobrenome}{Ph.D.}
40   \department{PESC}
41   \date{06}{2026}
42
43   \keyword{Primeira palavra-chave}
44   \keyword{Segunda palavra-chave}
45   \keyword{Terceira palavra-chave}
46

```

```

47 \maketitle
48
49 \frontmatter
50 \dedication{A alguém cujo valor é digno desta dedicatória.}
51
52 \chapter*{Agradecimentos}
53
54 Gostaria de agradecer a todos os que contribu\iram para este trabalho:
55 orientadores, colegas, familiares e \as ag\ências de fomento. Lorem ipsum
56 dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt
57 ut labore et dolore magna aliqua.
58
59 \begin{abstract}
60
61 Apresenta-se, nesta tese, um exemplo de resumo. Lorem ipsum dolor sit amet,
62 consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et
63 dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation
64 ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor
65 in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla
66 pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui
67 officia deserunt mollit anim id est laborum. Sed ut perspiciatis unde omnis
68 iste natus error sit voluptatem accusantium doloremque laudantium, totam rem
69 aperiam, eaque ipsa quae ab illo inventore veritatis et quasi architecto beatae
70 vitae dicta sunt explicabo, nemo enim ipsam voluptatem quia voluptas sit
71 aspernatur aut odit aut fugit.
72
73 \end{abstract}
74
75 \begin{foreignabstract}
76
77 In this work, we present an example abstract. Lorem ipsum dolor sit amet,
78 consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et
79 dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation
80 ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor
81 in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla
82 pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui
83 officia deserunt mollit anim id est laborum. Sed ut perspiciatis unde omnis
84 iste natus error sit voluptatem accusantium doloremque laudantium, totam rem
85 aperiam, eaque ipsa quae ab illo inventore veritatis et quasi architecto beatae
86 vitae dicta sunt explicabo, nemo enim ipsam voluptatem quia voluptas sit
87 aspernatur aut odit aut fugit.
88
89 \end{foreignabstract}
90
91 % Ordem pre-textual (NBR 14724/6027, R55): listas (ilustracoes, tabelas,
92 % abreviaturas e siglas, simbolos) ANTES do sumario; sumario por ultimo.
93 \listoffigures
94 \listoftables
95 \listofquadros
96 \listofprogramas
97 \listofalgorithms
98 \printloabbreviations
99 \printlosymbols
100 \tableofcontents

```

101
102 `\mainmatter`
103 `\chapter{Introdução}`
104
105 Este é um documento exemplo para o uso da classe CoppeTeX, destinado a ajudar os alunos da
106
107 A classe `\verb|coppe|` foi criada originalmente por Vicente Helano e George Ainsworth, poré
108 Em 2026, Geraldo Xexéo, com forte ajuda das ferramentas Claude 4.7 Opus e Max e 4.8 Opus M
109
110 Provavelmente Geraldo Xexéo, professor do Programa de Engenharia de Sistemas e Computação
111
112 A versão mais atual dessa classe é mantida no GitHub, no repositório `\url{https://github.c`
113
114 Esse documento segue a norma de formatação de teses e dissertações da COPPE. Ele também po
115
116 Há um grupo de WhatsApp para perguntas e discussões do estilo: `\href{https://github.com/CO`
117
118 Bugs devem ser reportados na `\href{https://github.com/COPPE-UFRJ/CoppeTeX/issues}{página d`
119
120 Esse documento é usado como exemplo de coisas que podem ser feitas. Ele está configurado p
121
122 É importante de notar que essa classe não foi construída sobre a classe `\LaTeX \` para a A
123
124 Essa classe implementa citações ABNT modernas e vários tipos de entrada descritos na norma
125
126 Este documento não substitui, mas complementa, o documento que descreve a classe.
127
128
129
130 `\chapter{Configurações Iniciais}`
131
132 A primeira coisa a fazer é escolher o tipo de documento. Isso é feito como uma opção no c
133
134 Como pode ser visto nesse documento, muita coisa pode ser configurada, o que gerará o tra
135
136 Recomendo ler o documento “‘The `\verb*|coppe|` document class’” para entender melhor todas
137
138 `\section{Linguagem principal do texto}`
139
140 Essa classe considera que o texto principal está em português e algumas partes específicas
141 `\begin{itemize}`
142 `\item` A opção `\verb|english|` deve ser usada no comando `\verb|\documentclass|`.
143 `\item` A bibliografia segue automaticamente o idioma do Babel (português ou inglês); não é p
144 `\end{itemize}`
145
146 A variação de linguagem, em inglês ou português apenas, já é suportada pela classe `\CoppeT`
147
148
149 `\section{Por que usar o \LaTeX}`
150
151 Há uma grande discussão entre usuário de Word e `\LaTeX`, principalmente, quanto ao uso dess
152
153 Nós escolhemos o `\LaTeX` por alguns motivos: grande facilidade de seguir um estilo sem se p
154

155 As principais desvantagens são: idiossincrasias que podem gastar tempo, pouco controle sob
156
157 \section{Como e onde usar o \LaTeX}
158
159 Existem muitos tutoriais de \LaTeX, mas basicamente, em 2024, ele é usado em dois ambientes
160 \begin{enumerate}
161 \item Na sua máquina, instalando uma versão completa como o \hologo{MiKTeX}, típico do Windows
162 \item Usar um ambiente na rede, como o Overleaf.
163 \end{enumerate}
164
165 Em todo caso, recomendo fortemente que, ao mesmo tempo, mantenha versões no Git e faça o backup
166
167
168 \chapter{Algumas Regras da COPPE}
169
170 Todas abreviaturas e símbolos devem ser definidas antes de utilizadas. Isso é facilmente feito
171
172 É imprescindível definir os símbolos, tal como o
173 conjunto dos números reais $\mathbb{R}$ e o conjunto vazio $\emptyset$.
174 \syml{\mathbb{R}}{Conjunto dos números reais}
175 \syml{\emptyset}{Conjunto vazio}. Usamos esse exemplo aqui justamente para mostrar como
176
177 Para as listas de abreviaturas e símbolos funcionarem no Overleaf é necessário rodar o \verb|makeindex|
178
179 Como as listas de símbolos e de abreviaturas usamos o mesmo comando usado para criar índices
180
181 \section{Citações}
182 \label{sec:citacoes}
183
184 Citações curtas podem ser feitas \code{\quote{...}} ou direto com “duas crases e dois pontos”
185
186 \begin{longquote}
187 Um exemplo de citação longa nas regras da ABNT (4cm de recuo e fonte menor)
188 feita com o ambiente \code{\verb=longquote=}: The primary objective of this
189 investigation was to determine the feasibility of detecting corrosion in
190 aluminum Naval aircraft components with neutron radiographic interrogation
191 and the use of standard corrosion penetrameters. Secondary objectives
192 included the determination of the effect of object thickness on image quality,
193 the defining of minimum levels of detectability and a preliminary investigation
194 of a means whereby the degree of corrosion could be quantified with neutron
195 radiographic data. \code{\citep{article-example}}
196 \end{longquote}
197
198 Citações devem apontar as referências. Para isso, está disponível o ótimo pacote \code{\verb*|natbib|}
199
200 Em todo caso, \code{\textbf{deve}} se tomar enorme atenção com as citações, para evitar ocorrer erros
201
202 Quando se cita uma obra por meio de outra (\code{\textit{apud}}), pela ABNT entra na lista de referências
203
204 \chapter{Floats}
205
206 Grande parte dos problemas de iniciantes, e veteranos, em \LaTeX é da localização dos \code{\textit{floats}}
207
208 A regra geral de posicionamento é que uma figura ou quadro só pode aparecer a partir da mesma

```

209
210 \textbf{Pela norma atual (manual2024), a legenda} \verb|\caption| \textbf{de toda ilustração}
211
212 Quadros são ilustrações cujos dados são delimitados por linhas em \emph{todas} as bordas (ao
213
214
215 \section{Tabelas e Figuras Padrão}
216
217 Vamos ver uma tabela padrão, como a \autoref{tab:exemplo_numeros}.
218
219 \begin{table}[ht]
220 \centering % Centraliza a tabela
221 \caption{Exemplo de Tabela de Números}
222 \label{tab:exemplo_numeros}
223 \begin{tabular}{ccc} % Define a quantidade de colunas
224 \hline % Linha superior
225 \textbf{Coluna 1} & \textbf{Coluna 2} & \textbf{Coluna 3} \\ % Cabeçalhos
226 \hline % Linha média
227 1 & 2 & 3 \\ % Primeira linha de dados
228 \hline
229 4 & 5 & 6 \\ % Segunda linha de dados
230 \hline
231 7 & 8 & 9 \\ % Terceira linha de dados
232 \hline
233 10 & 11 & 12 \\ % Quarta linha de dados
234 \hline % Linha inferior
235 \end{tabular}
236 \fonte{Elabora\c{c}\~ao pr\'opria.} % fonte obrigatoria, abaixo do float
237 \end{table}
238
239
240
241 Já a \autoref{fig:exemplo_figura} é uma figura padrão, com controle da largura.
242
243 \begin{figure}[ht]
244 \centering % Centraliza a figura
245 \caption{Exemplo de Figura com Legenda Acima} % Legenda em cima (norma atual)
246 \label{fig:exemplo_figura} % Etiqueta para referência cruzada
247 \includegraphics[width=0.5\textwidth]{coppe-logo.pdf} % Inclui a imagem com metade da largura
248 \source{COPPE/UFRJ.} % fonte obrigatoria, abaixo da figura
249 \end{figure}
250
251 Quadros têm a mesma regra (legenda acima, fonte abaixo), mas com linhas em todas as bordas.
252
253 \begin{quadro}[ht]
254 \centering
255 \caption{Exemplo de Quadro}
256 \label{qua:exemplo}
257 \begin{tabular}{|l|l|}
258 \hline
259 \textbf{Ilustração} & \textbf{Delimitação} \\
260 \hline
261 Tabela & Linhas apenas no topo e na base. \\
262 \hline

```

```

263 Quadro & Linhas em todas as bordas. \\
264 \hline
265 \end{tabular}
266 \fonte{Elabora\c{c}\~ao pr\'opria.}
267 \end{quadro}
268
269
270
271
272
273 \section{Tabelas mais elegantes}
274
275 Atualmente a tend\^encia \e usar tabelas mais leves, como \autoref{tab:exemplo_numerosbom}. Iss
276
277 \begin{table}[ht]
278 \centering % Centraliza a tabela
279 \caption{Exemplo de Tabela de N\^umeros mais elegantes}
280 \label{tab:exemplo_numerosbom}
281 \begin{tabular}{ccc} % Define a quantidade de colunas
282 \toprule % Linha superior
283 \textbf{Coluna 1} & \textbf{Coluna 2} & \textbf{Coluna 3} \\ \textbf{Cabe\c{a}lhos}
284 \midrule % Linha m\^edia
285 1 & 2 & 3 \\ % Primeira linha de dados
286 4 & 5 & 6 \\ % Segunda linha de dados
287 7 & 8 & 9 \\ % Terceira linha de dados
288 10 & 11 & 12 \\ % Quarta linha de dados
289 \bottomrule % Linha inferior
290 \end{tabular}
291 \fonte{Elabora\c{c}\~ao pr\'opria.}
292 \end{table}
293
294 \section{Tabelas Longas ou Largas}
295
296 Se sua tabela \e muito longa ou larga, existem v\arias op\c{c}o\~es.
297 \begin{itemize}
298 \item alterar o tamanho da letra
299 \item Usar o longtable
300 \item rodar a tabela, fazendo ela em \textit{landscape}
301 \item fazer a tabela dentro de um minibox
302 \end{itemize}
303
304
305 \subsection{Tabelas largas demais}
306
307 \E comum em teses que as tabelas sejam largas demais. H\aa v\arias formas de resolver isso.
308
309 A \autoref{tab:tabela_largafns} \e larga demais, e nela isso \e resolvido diminuindo a fonte p
310
311 \begin{table}[ht]
312 \centering % Centraliza a tabela
313 \caption{Exemplo de Tabela Larga com Fonte Menor}
314 \label{tab:tabela_largafns}
315 \footnotesize % Aplica uma fonte menor para a tabela
316 \begin{tabular}{ccccccc} % Aumente o n\^umero de colunas conforme necess\aaio

```

```

317 \toprule
318 \textbf{Coluna 1} & \textbf{Coluna 2} & \textbf{Coluna 3} & \textbf{Coluna 4} & \textbf{Coluna 5} \\
319 \midrule
320 Dado 1.1 & Dado 1.2 & Dado 1.3 & Dado 1.4 & Dado 1.5 & Dado 1.6 & Dado 1.7 & Dado 1.8 \\
321 Dado 2.1 & Dado 2.2 & Dado 2.3 & Dado 2.4 & Dado 2.5 & Dado 2.6 & Dado 2.7 & Dado 2.8 \\
322 Dado 3.1 & Dado 3.2 & Dado 3.3 & Dado 3.4 & Dado 3.5 & Dado 3.6 & Dado 3.7 & Dado 3.8 \\
323 \bottomrule
324 \end{tabular}
325 \fonte{Elabora\c{c}\~ao pr\'opria.}
326 \end{table}
327
328 O comando \verb|\resizebox{width}{height}{content}| permite ajustar o tamanho de qualquer co
329
330 \begin{table}[ht]
331 \centering
332 \caption{Exemplo de Tabela Redimensionada}
333 \label{tab:examplerb}
334 \resizebox{\textwidth}{!}{%
335 \begin{tabular}{lllll}
336 \toprule
337 Coluna 1 & Coluna 2 & Coluna 3 & Coluna 4 & \\
338 \midrule
339 Dados 1 & Dados 2 & Dados 3 & Dados 4 & \\
340 Dados 5 & Dados 6 & Dados 7 & Dados 8 & \\
341 \bottomrule
342 \end{tabular}%
343 }
344 \fonte{Elabora\c{c}\~ao pr\'opria.}
345 \end{table}
346
347
348 Para rodar uma tabela muito larga em 90 graus no LaTeX, voc\~e pode usar o pacote \verb*|rotat
349
350 Aqui est\~a um exemplo de como usar o ambiente \verb*|sidewaystable| para girar uma tabela. Pr
351
352 \begin{sidewaystable}
353 \centering
354 \caption{Sua Legenda Aqui}
355 \label{tab:sua_tabela}
356 \begin{tabular}{lll}
357 \toprule
358 Coluna 1 & Coluna 2 & Coluna 3 \\
359 \midrule
360 Item 1 & Item 2 & Item 3 \\
361 Item 4 & Item 5 & Item 6 \\
362 \bottomrule
363 \end{tabular}
364 \fonte{Elabora\c{c}\~ao pr\'opria.}
365 \end{sidewaystable}
366
367 Se a tabela for muito longa, o ambiente \verb|longtable| \~e o ideal. Ele fornece comandos par
368
369 % Exemplo de tabela longa que se estende por v\~arias p\~aginas
370 \begin{longtable}{|c|c|c|}

```

```

371 % primeiro cabeçalho (é o caption)
372 \caption{Exemplo de Tabela Longa}\label{tab:longa} \\
373 \hline \textbf{Coluna 1} & \textbf{Coluna 2} & \textbf{Coluna 3} \\ \hline
374 \endfirsthead
375 % cabeçalho normal
376 \multicolumn{3}{c}%
377 {\table\thetable{} -- continuação da página anterior} \\
378 \hline \textbf{Coluna 1} & \textbf{Coluna 2} & \textbf{Coluna 3} \\ \hline
379 \endhead
380 % pé normal
381 \hline \multicolumn{3}{r}{{Continua na próxima página}} \\ \hline
382 \endfoot
383 \hline
384 % último pé
385 \multicolumn{3}{r}{{Continua na próxima página}} \\
386 \hline \hline
387 \endlastfoot
388
389 % Conteúdo da tabela
390 1 & 2 & 3 \\
391 4 & 5 & 6 \\
392 1 & 2 & 3 \\
393 4 & 5 & 6 \\
394 1 & 2 & 3 \\
395 4 & 5 & 6 \\
396 1 & 2 & 3 \\
397 4 & 5 & 6 \\
398 1 & 2 & 3 \\
399 4 & 5 & 6 \\
400 1 & 2 & 3 \\
401 4 & 5 & 6 \\
402 1 & 2 & 3 \\
403 4 & 5 & 6 \\
404 1 & 2 & 3 \\
405 4 & 5 & 6 \\
406 1 & 2 & 3 \\
407 1 & 2 & 3 \\
408 4 & 5 & 6 \\
409 1 & 2 & 3 \\
410 4 & 5 & 6 \\
411 1 & 2 & 3 \\
412 4 & 5 & 6 \\
413 1 & 2 & 3 \\
414 4 & 5 & 6 \\
415 1 & 2 & 3 \\
416 4 & 5 & 6 \\
417 1 & 2 & 3 \\
418 4 & 5 & 6 \\
419 1 & 2 & 3 \\
420 4 & 5 & 6 \\
421 1 & 2 & 3 \\
422 4 & 5 & 6 \\
423 1 & 2 & 3 \\
424 4 & 5 & 6

```

425 1 & 2 & 3 \\
 426 4 & 5 & 6 \\
 427 1 & 2 & 3 \\
 428 4 & 5 & 6 \\
 429 1 & 2 & 3 \\
 430 4 & 5 & 6 \\1 & 2 & 3 \\
 431 4 & 5 & 6 \\
 432 1 & 2 & 3 \\
 433 4 & 5 & 6 \\
 434 1 & 2 & 3 \\
 435 1 & 2 & 3 \\
 436 4 & 5 & 6 \\
 437 1 & 2 & 3 \\
 438 4 & 5 & 6 \\
 439 1 & 2 & 3 \\
 440 4 & 5 & 6 \\
 441 1 & 2 & 3 \\
 442 4 & 5 & 6 \\
 443 1 & 2 & 3 \\
 444 4 & 5 & 6 \\
 445 1 & 2 & 3 \\
 446 4 & 5 & 6 \\
 447 1 & 2 & 3 \\
 448 4 & 5 & 6 \\
 449 1 & 2 & 3 \\
 450 4 & 5 & 6 \\
 451 1 & 2 & 3 \\
 452 4 & 5 & 6 \\
 453 1 & 2 & 3 \\
 454 4 & 5 & 6 \\
 455 1 & 2 & 3 \\
 456 4 & 5 & 6 \\
 457 1 & 2 & 3 \\
 458 4 & 5 & 6 \\
 459 1 & 2 & 3 \\
 460 4 & 5 & 6 \\
 461 1 & 2 & 3 \\
 462 4 & 5 & 6 \\
 463 1 & 2 & 3 \\
 464 4 & 5 & 6 \\
 465 1 & 2 & 3 \\
 466 4 & 5 & 6 \\
 467 1 & 2 & 3 \\
 468 4 & 5 & 6 \\
 469 1 & 2 & 3 \\
 470 4 & 5 & 6 \\
 471 1 & 2 & 3 \\
 472 4 & 5 & 6 \\
 473 1 & 2 & 3 \\
 474 4 & 5 & 6 \\
 475 1 & 2 & 3 \\
 476 4 & 5 & 6 \\
 477 1 & 2 & 3 \\
 478 4 & 5 & 6 \\

```

479 1 & 2 & 3 \\
480 4 & 5 & 6 \\
481 1 & 2 & 3 \\
482 4 & 5 & 6 \\
483 1 & 2 & 3 \\
484 4 & 5 & 6 \\
485 1 & 2 & 3 \\
486 4 & 5 & 6 \\
487 1 & 2 & 3 \\
488 4 & 5 & 6 \\
489 1 & 2 & 3 \\
490 4 & 5 & 6 \\
491 1 & 2 & 3 \\
492 4 & 5 & 6 \\
493 1 & 2 & 3 \\
494 4 & 5 & 6 \\
495 1 & 2 & 3 \\
496 4 & 5 & 6 \\
497
498 % Repetir linhas semelhantes conforme necessário para estender a tabela por 3 páginas
499 \end{longtable}
500 \fonte{Elabora\c{c}\~ao pr\'opria.}% a longtable não é float: a fonte vem após o ambiente
501
502 \section{Subfiguras (subcaption)}
503
504 Para colocar várias imagens (ou tabelas) lado a lado, cada uma com sua própria legenda, use
505
506 \begin{figure}[ht]
507   \centering
508   \begin{subfigure}[b]{0.42\textwidth}
509     \centering\includegraphics[width=\linewidth]{example-image-a}
510     \caption{Primeira vista}\label{fig:sub-a}
511   \end{subfigure}
512   \hfill
513   \begin{subfigure}[b]{0.42\textwidth}
514     \centering\includegraphics[width=\linewidth]{example-image-b}
515     \caption{Segunda vista}\label{fig:sub-b}
516   \end{subfigure}
517   \caption{Figura com duas subfiguras}
518   \label{fig:subs}
519   \fonte{Elabora\c{c}\~ao pr\'opria.}
520 \end{figure}
521
522 Referencie a figura inteira com \verb|\autoref{fig:subs}| e cada parte com \verb|\autoref{fig:sub-a}|
523
524 \section{Imagens (graphicx)}
525
526 A classe já carrega o \verb|graphicx|. Insira imagens com \verb|\includegraphics|; as opções
527 \begin{itemize}
528   \item \verb|[width=0.8\textwidth]| ou \verb|[scale=0.5]| --- tamanho;
529   \item \verb|[angle=90]| --- rotação;
530   \item \verb|[trim=l b r t, clip]| --- recorte (corta as bordas \emph{esquerda, baixo, direita, topo});
531 \end{itemize}
532 Formatos aceitos com \hologo{pdfLaTeX}/\hologo{LuaLaTeX}: \texttt{pdf}, \texttt{png}, \texttt{eps}

```

```

533
534 \section{Introdução ao tikz}
535
536 0 \verb|tikz| (disponível também via \verb|tcolorbox|) desenha gráficos vetoriais no próprio
537
538 \begin{figure}[ht]
539   \centering
540   \begin{tikzpicture}
541     \node[draw, rounded corners] (a) at (0,0) {Início};
542     \node[draw, rounded corners] (b) at (4,0) {Fim};
543     \draw[->, thick] (a) -- (b) node[midway, above] {processo};
544   \end{tikzpicture}
545   \caption{Diagrama simples em tikz}
546   \label{fig:tikz}
547   \fonte{Elabora\c{c}\~ao pr\'opria.}
548 \end{figure}
549
550 Para gráficos de dados (curvas, barras), veja o \verb|pgfplots|, construído sobre o \verb|tikz|
551
552 \section{Float ou direto no texto?}
553
554 Colocar a tabela ou imagem dentro de \verb|figure|/\verb|table| (um \emph{float}) deixa o \verb|L|
555
556 \chapter{Using BibLaTeX}
557 \label{cap:biblatex}
558
559 Esta classe gera as referências com BibLaTeX e o \emph{backend} \texttt{biber}, no padrão
560
561 \textbf{É} obrigatório compilar com \texttt{biber} (e não com \hologo{BibTeX}). A sequênci
562
563 Os comandos do \texttt{natbib} (\verb|\citet|, \verb|\citep|) continuam disponíveis porque
564
565 \begin{table}[h]
566 \caption{Exemplos de cita\c{c}\~oes utilizando o comando padr\~ao}
567   \texttt{\textbackslash cite} do \LaTeX\ e
568   o comando \texttt{\textbackslash citep},
569   disponível pela opção \texttt{natbib} do BibLaTeX.
570 \label{tab:citation}
571 \centering
572 {\footnotesize
573 \begin{tabular}{|c|c|c|}
574   \hline
575   Tipo da Publicação & \verb|\cite| & \verb|\citep|\\
576   \hline
577   Livro & \cite{book-example} & \citep{book-example}\\
578   Artigo & \cite{article-example} & \citep{article-example}\\
579   Relatório & \cite{techreport-example} & \citep{techreport-example}\\
580   Relatório & \cite{techreport-exampleIn} & \citep{techreport-exampleIn}\\
581   Anais de Congresso & \cite{inproceedings-example} & \\
582   & \citep{inproceedings-example}\\
583   Séries & \cite{incollection-example} & \citep{incollection-example}\\
584   Em Livro & \cite{inbook-example} & \citep{inbook-example}\\
585   Dissertação de mestrado & \cite{mastersthesis-example} & \\
586   & \citep{mastersthesis-example}

```



```

587     Tese de doutorado & \cite{phdthesis-example} & \citep{phdthesis-example}\\
588     \hline
589 \end{tabular}}
590 \fonte{Elabora\c{c}\~ao pr\'opria.}
591 \end{table}
592
593 \begin{table}[h]
594 \caption{Exemplos de cita\c{c}\~oes utilizando o comando padr\~ao
595 \texttt{\textbackslash cite} do \LaTeX\ e
596 o comando \texttt{\textbackslash citet},
597 disponivel pela op\c{c}\~ao \texttt{\textbackslash natbib} do BibLaTeX. Al\~em disso, usando o booktabs.}
598 \label{tab:citation1}
599 \centering
600 {\footnotesize
601 \begin{tabular}{ccc}
602 \toprule
603 Tipo da Publica\c{c}\~ao & \verb|\cite| & \verb|\citet|\\
604 \midrule
605 Livro & \cite{book-example} & \citet{book-example}\\
606 Artigo & \cite{article-example} & \citet{article-example}\\
607 Relat\~orio & \cite{techreport-example} & \citet{techreport-example}\\
608 Relat\~orio & \cite{techreport-exampleIn} & \citet{techreport-exampleIn}\\
609 Anais de Congresso & \cite{inproceedings-example} &
610 \citet{inproceedings-example}\\
611 S\~eries & \cite{incollection-example} & \citet{incollection-example}\\
612 Em Livro & \cite{inbook-example} & \citet{inbook-example}\\
613 Disserta\c{c}\~ao de mestrado & \cite{mastersthesis-example} &
614 \citet{mastersthesis-example}\\
615 Tese de doutorado & \cite{phdthesis-example} & \citet{phdthesis-example}\\
616 \bottomrule
617 \end{tabular}}
618 \fonte{Elabora\c{c}\~ao pr\'opria.}
619 \end{table}
620
621 \section{Comandos de cita\c{c}\~ao}
622
623 Os principais comandos s\~ao: \verb|\cite| (refer\~encia simples), \verb|\citet| (autor por ex
624
625 \subsection{Par\~ametro opcional: p\~agina e ‘‘tradu\c{c}\~ao nossa’’}
626
627 Todos os comandos aceitam um par\~ametro opcional para a localiza\c{c}\~ao (p\~agina, se\c{c}\~ao) e obser
628
629 \subsection{Cita\c{c}\~ao de cita\c{c}\~ao (\emph{apud})}
630
631 Para citar uma obra atrav\~es de outra, use \verb|\citetapud| e \verb|\citepapud| (demonstra
632
633 \subsection{M\~ultiplas bibliografias}
634
635 Para separar as refer\~encias (por tipo, cap\~itulo ou tema), use o ambiente \verb|\refsection|
636
637 \section{Tipos de entrada bibliogr\~afica}
638
639 Cada tipo de documento da ABNT corresponde a um tipo de entrada do BibLaTeX, com um sin\~oni
640

```

```

641 \subsection{Tipos acadêmicos principais}
642 \begin{itemize}
643   \item Livro / monografia --- \verb|@book| (\verb|@livro|)
644   \item Artigo de periódico --- \verb|@article| (\verb|@artigo|); matéria de jornal --- \verb|@journal|
645   \item Capítulo de livro --- \verb|@incollection| (\verb|@partedelivro|), \verb|@inbook|
646   \item Evento no todo (anais) --- \verb|@proceedings| (\verb|@anais|); trabalho em evento --- \verb|@inproceedings|
647   \item Tese/dissertação --- \verb|@thesis| (\verb|@tese|, \verb|@dissertacao|); com tipo --- \verb|@phdthesis|
648   \item Relatório --- \verb|@report| (\verb|@relatorio|, \verb|@relatoriotecnico|); periódico --- \verb|@article|
649   \item Manual --- \verb|@manual|; diversos --- \verb|@misc| (\verb|@diverso|); página/site --- \verb|@misc|
650 \end{itemize}
651
652 \subsection{Tipos especiais (ABNT §4.2.5--4.2.14)}
653 \begin{itemize}
654   \item Patente --- \verb|@patent| (\verb|@patente|)
655   \item Legislação / ato normativo --- \verb|@legislation| (\verb|@legislacao|, \verb|@ato|)
656   \item Norma técnica --- \verb|@standard| (\verb|@norma|); partitura --- \verb|@music| (\verb|@partitura|)
657   \item Documento cartográfico (mapa) --- \verb|@map| (\verb|@mapa|, \verb|@cartografico|)
658   \item Audiovisual --- \verb|@video| (\verb|@audiovisual|); filme --- \verb|@movie| (\verb|@filme|)
659 \end{itemize}
660
661 \subsection{Jogos e TV (tipos próprios da CoppeTeX)}
662 \begin{itemize}
663   \item Jogo --- \verb|@game| (\verb|@jogo|); de tabuleiro --- \verb|@boardgame| (\verb|@jogo|)
664   \item Programa de TV / série --- \verb|@tvshow| (\verb|@programadetelevisao|, \verb|@serietv|)
665 \end{itemize}
666
667 Os campos também aceitam sinônimos em português (sem acentos): \verb|autor|, \verb|titulo|, \verb|assunto|, \verb|assunto|
668
669 \chapter{Alguns outros exemplo úteis}
670
671 \begin{tcolorbox}[title=Meu Textbox]
672 Este é o conteúdo do meu textbox. Você pode adicionar qualquer texto aqui, bem como incluir
673 \end{tcolorbox}
674
675 \begin{tcolorbox}
676 Este é o conteúdo do meu textbox sem título. Você pode adicionar qualquer texto aqui, bem como
677 \end{tcolorbox}
678
679 \begin{figure}[ht]
680   \centering
681   \begin{tikzpicture}
682     \node[anchor=south west,inner sep=0] (image) at (0,0) {\includegraphics[width=0.5\textwidth]{image}}
683     \begin{scope}[x={(image.south east)},y={(image.north west)}]
684       % Definindo o textbox dentro da figura
685       \node[anchor=north west, text width=0.3\textwidth, fill=white, opacity=0.7, text=Este textbox fala sobre como inserir um textbox dentro de uma figura usada no documento]
686       \begin{tcolorbox}[colback=red!5!white,colframe=red!75!black,title=Textbox de exemplo]
687         Este textbox fala sobre como inserir um textbox dentro de uma figura usada no documento
688       \end{tcolorbox}
689     \end{scope}
690   \end{tikzpicture}
691   \caption{Figura com Textbox}
692   \label{fig:figura_com_textbox1}
693   \fonte{Elabora\c{c}\~ao pr\'opria.}

```

```

695 \end{figure}
696
697
698 \begin{figure}[ht]
699     \centering
700     \begin{tcolorbox}
701 Este é o conteúdo do meu textbox sem título. Você pode adicionar qualquer texto aqui, bem co
702 \end{tcolorbox}
703     \caption{Figura com Textbox simples}
704     \label{fig:figura_com_textbox}
705     \fonte{Elabora\c{c}\~ao pr\'opria.}
706 \end{figure}
707
708 \section{Listagens de código e algoritmos}
709
710 O CoppeTeX traz embutido o suporte a listagens de programas (pacote \verb|listings|) e a alg
711
712 \begin{python}[caption={Exemplo de programa em Python},label=prog:exemplo]
713 def saudacao(nome):
714     # imprime uma saudação acentuada
715     print(f"Olá, {nome}!")
716
717 saudacao("COPPE")
718 \end{python}
719 \source{Elaboração própria.}
720
721 Para inserir código no meio do texto use \verb|\pythoninline|, como em \pythoninline{saudaca
722
723 \begin{Verbatim}[frame=lines,numbers=left,numbersep=0.5em]
724 Olá, COPPE!
725 \end{Verbatim}
726
727 Há estilos prontos também para \verb|java|, \verb|xml|, \verb|html| e \verb|prolog|, comando
728
729 \begin{algorithm}[H]
730 \caption{Soma dos elementos de um vetor}
731 \label{alg:soma}
732 \Dados{vetor  $v$  com  $n$  elementos}
733 \Resultado{soma  $s$  dos elementos de  $v$ }
734  $s \leftarrow 0$ ;
735 \Para{ $i \leftarrow 1$  \Ate{ }  $n$ }{
736      $s \leftarrow s + v[i]$ ;
737 }
738 \Retorna  $s$ ;
739 \end{algorithm}
740
741 \chapter{Escrevendo matemática}
742
743 A classe carrega \texttt{amsmath} em qualquer motor. Sob \hologo{pdfLaTeX} ela também carr
744
745 \section{Alinhamento (align)}
746 O ambiente \verb|align| alinha equações pelo \verb|&| e numera cada linha (\verb|\\| quebr
747 \begin{align}
748     f(x) &= (x+1)^2 \setminus

```

```

749      &= x^2 + 2x + 1 .
750 \end{align}
751
752 \section{Matrizes (matrix)}
753 0 \texttt{amsmath} traz \verb|pmatrix|, \verb|bmatrix|, \verb|vmatrix|, entre outros:
754 \begin{equation}
755   A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}
756   \quad
757   \det A = \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix}
758 \end{equation}
759
760 \section{Subequações (subequations)}
761 Para numerar um grupo de equações sob o mesmo número, com letras (a), (b):
762 \begin{subequations}
763 \begin{align}
764   a &= b + c , \\
765   d &= e - f .
766 \end{align}
767 \end{subequations}
768
769 \section{Estruturas alinhadas (array)}
770 0 \verb|array| monta estruturas dentro do modo matemático, como definições por partes:
771 \begin{equation}
772   |x| = \left\{ \begin{array}{l}
773     x, \quad \text{se } x \geq 0 , \\
774     -x, \quad \text{caso contrário.}
775   \end{array} \right.
776 \end{equation}
777
778 \section{Operadores (\textbackslash DeclareMathOperator)}
779 Para operadores com a tipografia correta (romano, espaçamento adequado), declare-os no pre
780 \begin{verbatim}
781 \usepackage{amsmath}
782 \DeclareMathOperator{\RMSE}{RMSE}
783 \DeclareMathOperator{\diag}{diag}
784 \end{verbatim}
785 Assim, \verb|$\RMSE(y,\hat{y})$| produz  $\RMSE(y,\hat{y})$ , e \verb|$\diag(\lambda_1,\dots
786
787 \section{Unicode e LuaLaTeX (unicode-math)}
788 \label{sec:unicode-math}
789 Ao migrar para \hologo{LuaLaTeX} (ou \hologo{XeLaTeX}), \textbf{não carregue} \texttt{amss
790 \begin{verbatim}
791 \usepackage{amsmath} % a classe já carrega
792 \usepackage{unicode-math} % NÃO use \usepackage{amssymb} junto
793 \setmathfont{Latin Modern Math} % ou STIX Two Math, TeX Gyre Termes Math...
794 \end{verbatim}
795 0 código clássico continua valendo ( $\mathbb{R}$ ,  $\int_a^b f(x)dx$ ,  $\alpha$ ) e você ai
796
797 \chapter{Referências cruzadas, notas e índices}
798 \label{cap:refs}
799
800 Este capítulo reúne os recursos para \emph{apontar} para coisas: rótulos, notas de rodapé,
801
802 \section{Rótulos e referências cruzadas}

```

```

803
804 Marque um elemento com \verb|\label{tipo:nome}| e aponte para ele com \verb|\ref| (só o nú
805
806 \section{Notas de rodapé}
807
808 Use \verb|\footnote{texto}|\footnote{Como esta nota, que a classe imprime em espaçamento s
809
810 \section{hyperref e marcadores}
811
812 A classe carrega o \verb|hyperref|: o sumário, as citações e os \verb|\ref|/\verb|\autoref|
813
814 \section{URLs}
815
816 Para endereços, use \verb|\url{https://exemplo.org}|: o \verb|hyperref| cria o link e queb
817
818 \section{Índice remissivo}
819
820 Marque os termos com \verb|\index{termo}| e imprima o índice com \verb|\printindex| (carre
821
822 \section{Traduções (NBR 10520)}
823
824 Ao citar um trecho que você mesmo traduziu, coloque a tradução no corpo do texto e escreva
825
826 \chapter{Truques de \LaTeX}
827
828 \section{Comentários mágicos (\texttt{\% !TeX})}
829 Linhas \verb|\% !TeX| no topo do arquivo dizem ao editor (\hologo{TeX}studio, Overleaf, \ho
830 \begin{verbatim}
831 % !TeX program = lualatex
832 % !TeX encoding = UTF-8
833 % !TeX spellcheck = pt_BR
834 % !BIB program = biber
835 \end{verbatim}
836 Com isso o documento compila igual em qualquer ambiente. O \texttt{latexmkrc} cumpre papel
837
838 \section{Espaço engolido}
839 Um comando sem argumento engole o espaço seguinte: \verb|\LaTeX é bom| sai como “\LaTeX é
840
841 \section{Escopo (grupos)}
842 Chaves \verb|\{...}| (ou \verb|\begingroup|\dots\verb|\endgroup|) criam um escopo: mudanças
843
844 \section{Descarregar figuras: \texttt{\textbackslash clearpage} × \texttt{\textbackslash m
845 \verb|\newpage| apenas quebra a página e leva os floats pendentes adiante, acumulando-os.
846
847 \section{\texttt{latexmkrc}, perl e Overleaf}
848 O \verb|latexmk| automatiza a compilação (roda \LaTeX, \verb|biber| e \verb|makeindex| qua
849
850 \chapter{Método Proposto}
851 \chapter{Resultados e Discussões}
852
853 \section{Algumas Demonstrações}
854
855 A Lista de Símbolos precisa usar comandos específicos. Aqui vamos usar os símbolos $\alpha
856 \symbl[beta]{Beta}{A palavra Beta more e corrigida}

```

```

857 \syml[zzbeta]{$\beta$}{A letra $\beta$ corrigida}
858 \syml{beta}{A palavra beta}
859 \syml{alpha}{A palavra alpha}
860 \syml[alpha]{Alpha}{A palavra Alpha}
861 \syml[zzalpha]{$\alpha$}{A letra $\alpha$ corrigida}
862 \syml[marco]{Marco}{A palavra Marco corrigida}
863
864 A Lista de Abreviações segue, a partir de 2024, a mesma regra, e aqui seguem alguns exemplos
865 \abbrev{GoT}{Game of Thrones}
866 \abbrev[GOT]{GoT}{Game of Thrones ordenado como GOT}
867 \abbrev[iot]{IoT}{IoT ordenado como iot}
868 \abbrev[IoT]{IoT}{IoT ordenado como IoT}
869 \abbrev[IOT]{IoT}{IoT ordenado como IOT}
870 \abbrev{IoT}{IoT com ordenação default}
871 \abbrev[ITU]{ITU}{ITU mesmo}
872
873 \newsigla{ufrj}{UFRJ}{Universidade Federal do Rio de Janeiro}
874 \newsigla{cpgp}{CPGP}{Comissão de Pós-Graduação e Pesquisa de Engenharia}
875
876 As siglas (manual2024, seção 2.8) são expandidas na primeira ocorrência e registradas automaticamente
877
878
879
880
881 \chapter{Conclusões}
882
883 \backmatter
884
885 % Tipos exóticos da ABNT (NBR 6023): patente, legislação, norma, mapa,
886 % partitura --- declarados via sinônimos em português em example.bib.
887 \nocite{patente-example,legislacao-example,norma-example,mapa-example,partitura-example}
888 \nocite{jurisprudencia-example}
889 \nocite{videogame-example,filme-example,semautor-example,semautorart-example}
890 \nocite{jornalcaderno-example,jornal-example}
891 \printbibliography
892
893
894
895 \appendix
896
897 \chapter{Um apêndice}
898
899 Segundo a norma da ABNT (Associação Brasileira de Normas Técnicas), a definição e utilização de apêndices
900
901 Apêndice: O apêndice é um texto ou documento elaborado pelo autor do trabalho com o objetivo de complementar
902
903
904 \chapter{Outro apêndice}
905
906 \annex
907
908
909 \chapter{Um Anexo}
910 Segundo a norma da ABNT (Associação Brasileira de Normas Técnicas), a definição e utilização de anexos

```

```

911
912
913
914 Anexo: O anexo, por sua vez, consiste em um texto ou documento não elaborado pelo autor, que
915
916 No modelo \CoppeTeX os anexos devem obrigatoriedade vir depois dos apêndices e usam o comand
917
918
919 \chapter{Outro Anexo}
920
921
922 \coppetexfinalpage% colofão institucional (página par, face do contracapa)
923
924 \end{document}
925
926 \example

```

10 Implementation

10.1 The ‘coppe.cls’ file

```

927 \class
928 \def\filename{coppe.dtx}
929 \def\fileversion{v4.0}
930 \def\filedate{2026/05/28}
931 \NeedsTeXFormat{LaTeX2e}[1995/12/01]
932 \ProvidesClass{coppe}[\filedate\ \fileversion\ COPPE Dissertations and Thesis]
933 % Save the class version and date into private macros immediately, before any
934 % required package can redefine the very common \fileversion / \filedate
935 % global names (LaTeX has no convention reserving them for the host class).
936 % \coppe@version / \coppe@date are what user-facing macros (e.g.
937 % \coppetexfinalpage) should reference.
938 \let\coppe@version\fileversion
939 \let\coppe@date\filedate
940 % Document date stored in private slots so \date can set them without ever
941 % touching TeX's \month/\year primitives (which would break \today). They
942 % default to the current system month/year, so a document without \date
943 % still renders sensibly on the cover and folha de rosto.
944 \edef\coppe@docmonth{\the\month}
945 \edef\coppe@docyear{\the\year}
946 \LoadClass[12pt,a4paper,twoside]{book}
947 % setspace, hyperref and the bibliography engine (biblatex+biber) are loaded at
948 % the END of the class (after every class hook is defined), in that order:
949 % setspace -> hyperref -> biblatex. hyperref must come late so all the macros
950 % it patches (TOC, refs, lists) are already defined, and biblatex must come
951 % after hyperref so its citation links and back-references work.
952 \RequirePackage{hyphenat}
953 \RequirePackage{lastpage}
954 \RequirePackage{ifthen}
955 \RequirePackage{graphicx}
956 \RequirePackage{tabularx}
957 \RequirePackage{booktabs} % regras de tabela profissionais (toprule/midrule/bottomrule)
958 \RequirePackage{longtable} % tabelas que quebram entre páginas
959 \RequirePackage{etoolbox}

```

```

960 \RequirePackage{ltxcmds}
961 \RequirePackage{hologo} % TeX-family logos: \hologo{TeX}, \hologo{LaTeX}, ...
962 % Render \TeX, \LaTeX and friends via hologo (\CoppeTeX is a custom logo, kept).
963 \def\TeX{\hologo{TeX}}
964 \def\LaTeX{\hologo{LaTeX}}
965 \def\LaTeXe{\hologo{LaTeXe}}
966 \def\BibTeX{\hologo{BibTeX}}
967 \def\pdfLaTeX{\hologo{pdfLaTeX}}
968 \def\pdfTeX{\hologo{pdfTeX}}
969 \def\LuaLaTeX{\hologo{LuaLaTeX}}
970 \def\XeLaTeX{\hologo{XeLaTeX}}
971 \def\MiKTeX{\hologo{MiKTeX}}
972 % Engine-aware UTF-8 input: pdfLaTeX needs inputenc(utf8); LuaLaTeX (and XeTeX)
973 % are UTF-8 native, so inputenc must NOT be loaded there (it would clash). Source
974 % files may use accented Latin characters (áéíóú) directly. Under pdfLaTeX we also
975 % suggest LuaLaTeX, which handles Unicode/fonts better (at the cost of speed).
976 \RequirePackage{iftex}
977 \ifPDFTeX
978   \RequirePackage[utf8]{inputenc}
979   \AtEndOfClass{\ClassWarningNoLine{coppe}{%
980     Although slower, you should try compiling with LuaLaTeX}}
981 \fi
982 \RequirePackage[T1]{fontenc}
983 % Math support. amsmath is universal and works on every engine, so the class
984 % brings it in unconditionally (it must precede hyperref, which is loaded at
985 % the end of the class). amssymb is the legacy AMS symbol package, built
986 % around the 8-bit Type 1 fonts msam/msbm; on a Unicode engine (LuaLaTeX or
987 % XeLaTeX) it clashes with unicode-math, which already provides all of its
988 % symbols through Unicode math codes. So amssymb is loaded ONLY under
989 % pdfLaTeX; Unicode-engine users get a ClassWarning pointing them at
990 % unicode-math (the class does not auto-load unicode-math because choosing
991 % an OpenType math font is a typographic decision that belongs to the author).
992 \RequirePackage{amsmath}
993 \ifPDFTeX
994   \RequirePackage{amssymb}
995 \else
996   % Unicode engine: defer the check to end of document. Only warn if the
997   % user did NOT add either unicode-math (the modern OpenType-math route)
998   % or amssymb (the legacy route, which they would have loaded manually).
999   % This avoids nagging authors who already followed the documented
1000   % \usepackage{unicode-math} workflow.
1001   \AtEndDocument{%
1002     \@ifpackageloaded{unicode-math}{}%
1003     \@ifpackageloaded{amssymb}{}%
1004     \ClassWarningNoLine{coppe}{%
1005       You are compiling with a Unicode engine (LuaLaTeX or XeLaTeX) and\MessageBreak
1006       neither unicode-math nor amssymb is loaded. AMS-style symbols\MessageBreak
1007       (\string\mathbb, \string\hbar, \string\varnothing, ...) will be undefined.\MessageBreak
1008       Add to your preamble:\MessageBreak
1009       \space\space\string\usepackage{unicode-math}\MessageBreak
1010       \space\space\string\setmathfont{Latin Modern Math}\MessageBreak
1011       Do NOT load amssymb together with unicode-math -- it would clash\MessageBreak
1012       with the OpenType math font setup}}}%
1013 \fi

```



```

1014 % manual2024 margins: 3cm top, 2cm bottom, mirrored 3cm inner / 2cm outer for
1015 % two-sided printing -- realised as inner=outer=2cm plus a 1cm bindingoffset.
1016 % headheight=15pt is given here (not patched later) so geometry's own page
1017 % arithmetic stays self-consistent: fancyhdr needs >= 14.49998pt of header
1018 % height to host the running head, and the standard 12pt default triggers
1019 % its "headheight is too small" warning. Letting geometry know up front
1020 % keeps \topmargin a derived value -- we never touch it by hand.
1021 \RequirePackage[a4paper,twoside,bindingoffset=1cm,headheight=15pt,%
1022 top=3cm,bottom=2cm,inner=2cm,outer=2cm]{geometry}
1023 \RequirePackage{titlesec}
1024 % --- Section heading formats (manual2024 2.6 / proposta P22) -----
1025 % chapter (numbered): "N TITLE", ALL CAPS, bold, flush left (no "Capitolo N").
1026 % APENDICE/ANEXO chapters print "APENDICE A -- Title" (R75/R76); normal chapters
1027 % keep "N Title". \appendix and \annex raise the flag (see below).
1028 \newif\if@coppeappendix \@coppeappendixfalse
1029 \titleformat{\chapter}[hang]
1030   {\normalfont\Large\bfseries}
1031   {\if@coppeappendix\MakeUppercase{\appendixname}\ \thechapter\ \textendash\else\thechapter\
1032   {1ex}}{\MakeUppercase}
1033 \titlespacing*{\chapter}{0pt}{10pt}{1.5\baselineskip}
1034 \let\coppe@orig@appendix\appendix
1035 \renewcommand\appendix{\coppe@orig@appendix\@coppeappendixtrue}
1036 % Sumario (TOC) entries for APENDICE/ANEXO chapters read "APENDICE A -- Title"
1037 % / "ANEXO A -- Title" (R75/R76), not the bare "A". The label word is chosen by
1038 % which macro is written (so it is correct at .toc-render time even though the
1039 % appendixname has changed back); the chapter letter is baked in at write time.
1040 \newif\if@coppeannex \@coppeannexfalse
1041 \newcommand\coppe@tocapp[1]{%
1042   \MakeUppercase{\coppestring{appendix}}%
1043   \nobreakspace#1\nobreakspace\textendash\nobreakspace}
1044 \newcommand\coppe@tocanx[1]{%
1045   \MakeUppercase{\coppestring{annex}}%
1046   \nobreakspace#1\nobreakspace\textendash\nobreakspace}
1047 \def\@chapter[#1]#2{%
1048   \ifnum \c@secnumdepth >\m@ne
1049     \if@mainmatter
1050       \refstepcounter{chapter}%
1051       \typeout{\@chapapp\space\thechapter.}%
1052       \if@coppeappendix
1053         \if@coppeannex
1054           \addcontentsline{toc}{chapter}{\protect\coppe@tocanx{\thechapter}#1}%
1055         \else
1056           \addcontentsline{toc}{chapter}{\protect\coppe@tocapp{\thechapter}#1}%
1057         \fi
1058       \else
1059         \addcontentsline{toc}{chapter}{\protect\numberline{\thechapter}#1}%
1060       \fi
1061     \else
1062       \addcontentsline{toc}{chapter}{#1}%
1063     \fi
1064   \else
1065     \addcontentsline{toc}{chapter}{#1}%
1066   \fi
1067   \chaptermark{#1}%

```

```

1068 \addtocontents{lof}{\protect\addvspace{10\p@}}%
1069 \addtocontents{lot}{\protect\addvspace{10\p@}}%
1070 \if@twocolumn
1071 \topnewpage[\@makechapterhead{#2}]%
1072 \else
1073 \makechapterhead{#2}%
1074 \afterheading
1075 \fi}
1076 % chapter (unnumbered): centred ALL CAPS bold (errata, lists, resumo, sumario,
1077 % referencias, glossario, ... -- unnumbered headings are centred, P29-P30).
1078 \titleformat{name=chapter,numberless}[hang]
1079 {\normalfont\Large\bfseries\filcenter}{0pt}{\MakeUppercase}
1080 \titlespacing{name=chapter,numberless}{0pt}{10pt}{1.5\baselineskip}
1081 % section: ALL CAPS, normal weight.
1082 \titleformat{section}
1083 {\normalfont\large\bfseries}{\thesection}{1ex}{\MakeUppercase}
1084 % subsection: bold (author writes Title Case).
1085 \titleformat{subsection}
1086 {\normalfont\bfseries}{\thesubsection}{1ex}{}
1087 % subsubsection: bold italic.
1088 \titleformat{subsubsection}
1089 {\normalfont\bfseries\itshape}{\thesubsubsection}{1ex}{}
1090 % --- Captions (manual2024 2.10-2.11 / proposta P6): smaller font, ABNT
1091 % "Figura 1 -- titulo" travessao separator (sources handled by \source). ---
1092 \RequirePackage{caption}
1093 \captionsetup{font=footnotesize,labelsep=endash}
1094 % subcaption (NÃO subfig/subfigure) for subfigures / subtables; load right after
1095 % caption so it picks up the ABNT captionsetup above.
1096 \RequirePackage{subcaption}
1097 % --- Floats: caption on TOP and a mandatory "Fonte:" line at the bottom.
1098 % manual2024 2.10-2.11 (R46/R51: identification on top; R48/R52: the source
1099 % at the bottom is mandatory). The float package's plaintop style forces the
1100 % caption above the body for figure, table and the new quadro float. -----
1101 \RequirePackage{float}
1102 \floatstyle{plaintop}
1103 \restylefloat{figure}
1104 \restylefloat{table}
1105 % New "quadro" float -- "Quadro" (pt) / "Frame" (en) (R53) -- with its own list
1106 % file (.loq). A quadro's data is bordered on all sides (drawn by the author),
1107 % unlike a tabela (rules on top and bottom only).
1108 \providecommand{\quadroname}{\coppestring{frame}}
1109 \providecommand{\listquadroname}{\coppestring{listframe}}
1110 \newfloat{quadro}{htbp}{loq}[chapter]
1111 \floatname{quadro}{\quadroname}
1112 \providecommand{\quadroautorefname}{\quadroname}
1113 % Format the Lista de Quadros entries exactly like the Lista de Figuras ones
1114 % (float's own \l@quadro lays the number/title out incorrectly).
1115 \let\l@quadro\l@figure
1116 \newcommand\listofquadros{%
1117 \chapter*{\listquadroname}%
1118 \addcontentsline{toc}{chapter}{\listquadroname}%
1119 \mkboth{\MakeUppercase\listquadroname}{\MakeUppercase\listquadroname}%
1120 \starttoc{loq}}
1121 % \source / \fonte: the illustration source (ABNT-mandatory). Typeset below the

```

```

1122 % float as a smaller, single-spaced, centred line; it does NOT advance any
1123 % caption counter. \cpsource/\cpfonte are clash-safe aliases (the rename rule);
1124 % the label is "Fonte" (pt) / "Source" (en).
1125 \newcommand{\cpsourcenam}{\coppestring{source}}
1126 \newcommand{\cpsource}[1]{%
1127   \par\addvspace{\smallskipamount}%
1128   {\footnotesize\singlespacing\centering\cpsourcenam: #1\par}}
1129 \let\cpfonte\cpsource
1130 \@ifundefined{source}{\let\source\cpsource}{}%
1131 \@ifundefined{fonte}{\let\fonte\cpsource}{}%
1132 % \newcoppefloat{env}{caption name}{list title}: declare an extra caption-on-top
1133 % float (e.g. "picture") that supports \source and gains a \listof<env>s command.
1134 \newcommand{\newcoppefloat}[3]{%
1135   {\floatstyle{plaintop}\newfloat{#1}{htbp}{lo#1}}%
1136   \floatname{#1}{#2}%
1137   \expandafter\newcommand\csname listof#1s\endcsname{%
1138     \chapter*{#3}\addcontentsline{toc}{chapter}{#3}%
1139     \@mkboth{\MakeUppercase{#3}}{\MakeUppercase{#3}}%
1140     \@starttoc{lo#1}}}%
1141 % --- Footnotes (manual2024 / R16, R8, R21): single spacing and a 5 cm rule
1142 % (filete) from the left margin; footnotesize is the book default. -----
1143 \let\coppe@orig@footnotetext\@footnotetext
1144 \long\def\@footnotetext#1{\coppe@orig@footnotetext{\singlespacing#1}}
1145 \renewcommand\footnoterule{\kern-3\p@\hrule\@width 5cm\kern 2.6\p@}
1146 % --- alineas: list keyed by lowercase letters a) b) c); subitems with "--"
1147 % (manual2024 2.6 / proposta P24). -----
1148 \RequirePackage{enumitem}
1149 \newlist{alineas}{enumerate}{2}
1150 \setlist{alineas,1}{label=\alph*},leftmargin=1.8em,itemsep=0pt}
1151 \setlist{alineas,2}{label=--,leftmargin=1.8em,itemsep=0pt}
1152 % --- Page numbering: arabic in the top OUTER corner on textual pages
1153 % (manual2024 2.7 / proposta P25-P26). Pre-textual numbering is set by
1154 % \frontmatter (unnumbered by default; roman with the numeraisromanos
1155 % class option). -----
1156 \RequirePackage{fancyhdr}
1157 % \headheight is set to 15pt by geometry up above (see the geometry call),
1158 % which is what fancyhdr needs (>= 14.49998pt) for the running head.
1159 \fancypagestyle{coppe}{\fancyhf{}\fancyhead[R0,LE]{\thepage}%
1160   \renewcommand{\headrulewidth}{0pt}}
1161 % Blank pages inserted by \cleardoublepage (two-sided) must be empty: no page
1162 % number may leak onto them (e.g. the verso right after the cover).
1163 \renewcommand{\cleardoublepage}{%
1164   \clearpage
1165   \if@twoside
1166     \ifodd\c@page\else
1167       \hbox{}\thispagestyle{empty}\newpage
1168       \if@twocolumn\hbox{}\newpage\fi
1169     \fi
1170   \fi}
1171 \def\CoppeTeX{\rm C\kern-.05em{\sc o\kern-.025em p\kern-.025em
1172   p\kern-.025em e}}\kern-.08em
1173 T\kern-.1667em\lower.5ex\hbox{E}\kern-.125emX\spacefactor1000}
1174 \newboolean{maledoc}
1175 \setboolean{maledoc}{false}

```

```

1176 %
1177 % Class options.
1178 % Multilingual model (CoppeTeX 4.x, branch nlinguas). The class holds three
1179 % fixed language slots:
1180 % * main -- the language of the body, captions, and the main abstract.
1181 %          Set by a class option: brazilian (default), english,
1182 %          spanish, french, italian, or any user-supplied language
1183 %          that ships a coppe-lang-<name>.def file.
1184 % * foreign -- the language used by the foreignabstract environment and
1185 %          by \foreign@... registers. Defaults to english (or to
1186 %          brazilian when main=english) for back-compat.
1187 % * third -- optional, loaded on demand with \usecoppelanguage{<name>}
1188 %          in the preamble (for citations or quotations in a third
1189 %          language). Has no class option of its own.
1190 % Note: the legacy \if@english boolean (defined and toggled here in
1191 % v3.x for backward compatibility) was removed in v4.0 -- nothing
1192 % inside the class or in the biblatex style files (.bbx/.cbx/.dbx/.ltx)
1193 % tests it. Code that needs to branch on the main language should test
1194 % \ifx\coppe@mainlang\coppe@englishstr instead.
1195 \def\coppe@mainlang{brazilian}
1196 \def\coppe@foreignlang{english}
1197 \DeclareOption{brazilian}{%
1198   \def\coppe@mainlang{brazilian}\def\coppe@foreignlang{english}}
1199 \DeclareOption{english}{%
1200   \def\coppe@mainlang{english}\def\coppe@foreignlang{brazilian}}
1201 \DeclareOption{spanish}{%
1202   \def\coppe@mainlang{spanish}\def\coppe@foreignlang{english}}
1203 \DeclareOption{french}{%
1204   \def\coppe@mainlang{french}\def\coppe@foreignlang{english}}
1205 \DeclareOption{italian}{%
1206   \def\coppe@mainlang{italian}\def\coppe@foreignlang{english}}
1207 \DeclareOption{msc}{%
1208   \newcommand{\@degree}{M.Sc.}
1209   \newcommand{\@degreename}{Mestrado}
1210   \newcommand{\local@degname}{Mestre}
1211   \newcommand{\foreign@degname}{Master}
1212   \newcommand\local@doctype{Disserta{\c c}{\~ a}o}
1213   \newcommand\foreign@doctype{Dissertation}
1214 }
1215 \DeclareOption{dsccexam}{%
1216   \newcommand{\@degree}{D.Sc.}
1217   \newcommand{\@degreename}{Doutorado}
1218   \newcommand{\local@degname}{Doutor}
1219   \newcommand{\foreign@degname}{Doctor}
1220   \setboolean{maledoc}{true}
1221   \newcommand\local@doctype{Exame de Qualifica{\c c}{\~ a}o}
1222   \newcommand\foreign@doctype{Qualifying Exam}
1223 }
1224 \DeclareOption{mscexam}{%
1225   \newcommand{\@degree}{M.Sc.}
1226   \newcommand{\@degreename}{Mestrado}
1227   \newcommand{\local@degname}{Mestre}
1228   \newcommand{\foreign@degname}{Master}
1229   \setboolean{maledoc}{true}

```

```

1230 \newcommand\local@doctype{Exame de Qualifica{\c c}{\~ a}o}
1231 \newcommand\foreign@doctype{Qualifying Exam}
1232 }
1233 \DeclareOption{dsc}{%
1234 \newcommand{\@degree}{D.Sc.}
1235 \newcommand{\@degreename}{Doutorado}
1236 \newcommand{\local@degname}{Doutor}
1237 \newcommand{\foreign@degname}{Doctor}
1238 \newcommand\local@doctype{Tese}
1239 \newcommand\foreign@doctype{Thesis}
1240 }
1241 \newif\if@coppenumeric\@coppenumericfalse
1242 \DeclareOption{numbers}{\@coppenumerictrue}
1243 % numeraisromanos: show roman numerals on the pre-textual pages. Default: the
1244 % pre-textual pages are counted but not numbered (manual2024 2.7).
1245 \newif\if@copperoman\@copperomanfalse
1246 \DeclareOption{numeraisromanos}{\@copperomantrue}

```

Default line spacing is one-and-a-half; the `doublespacing` option switches to double spacing. Since `setspace` is loaded at the end of the class (after `hyperref`), the actual `\onehalfspacing`/`\doublespacing` call is deferred: the option only flips a flag, which the late-load block reads.

```

1247 \newif\if@coppedoublespace\@coppedoublespacefalse
1248 \DeclareOption{doublespacing}{\@coppedoublespacetrue}
1249 \ProcessOptions\relax
1250 % Built-in language names (used by the loader and the dispatcher below).
1251 \def\coppe@brazilianstr{brazilian}
1252 \def\coppe@englishstr{english}
1253 % Load Babel with the foreign language first and the main language last
1254 % (Babel treats the LAST option in the list as the main language). With
1255 % pt-main this produces [english,brazilian]{babel} and with en-main
1256 % [brazilian,english] -- identical to the historical \if@english calls.
1257 \edef\coppe@babelopts{\coppe@foreignlang,\coppe@mainlang}
1258 \expandafter\RequirePackage\expandafter[\coppe@babelopts]{babel}
1259 % --- Multilingual string dispatcher -----
1260 % \copperdefstring{<lang>}{<key>}{<value>} adds one entry to the table for
1261 % language <lang>. The three retrievers below look strings up either at the
1262 % current babel language (\coppestring -- follows \selectlanguage), or pinned
1263 % to the main slot (\coppemainstring) or the foreign slot (\coppeforeignstring).
1264 \providecommand{\copperdefstring}[3]{%
1265 \expandafter\gdef\csname coppe@str@#1@#2\endcsname{#3}}
1266 \providecommand{\coppestring}[1]{\@nameuse{coppe@str@\language@ #1}}
1267 \providecommand{\coppemainstring}[1]{\@nameuse{coppe@str@\coppe@mainlang @#1}}
1268 \providecommand{\coppeforeignstring}[1]{\@nameuse{coppe@str@\coppe@foreignlang @#1}}
1269 % Load an extra language pack at any point in the preamble (third-language
1270 % slot for citations or quotations beyond the main+foreign pair).
1271 \providecommand{\usecoppelanguage}[1]{%
1272 \def\coppe@@cur{#1}%
1273 \ifx\coppe@@cur\coppe@@brazilianstr\else
1274 \ifx\coppe@@cur\coppe@@englishstr\else
1275 \InputIfFileExists{coppe-lang-#1.def}{}
1276 {\ClassWarning{coppe}{Language pack coppe-lang-#1.def not found}}%
1277 \fi\fi}
1278 % \titlein{<lang>}{<text>}: record the title in any language. Will be

```

```

1279 % consulted by the cover/folha renderers once they are migrated in step 2.
1280 % In step 1 it is just a no-op store -- the historical \title / \foreigntitle
1281 % still drive every visible site, so behavior for pt/en is unchanged.
1282 \providecommand{\titlein}[2]{%
1283   \expandafter\gdef\csmname coppe@title@#1\endcsname{#2}}
1284 % --- Built-in string tables: brazilian + english. The values mirror the
1285 % literals currently scattered through the class; step-1 only populates the
1286 % tables -- no site reads them yet, so the rendered output is unchanged. ----
1287 \copperdefstring{brazilian}{appendix}{\Ap\~endice}
1288 \copperdefstring{english}{appendix}{Appendix}
1289 \copperdefstring{brazilian}{annex}{Anexo}
1290 \copperdefstring{english}{annex}{Annex}
1291 \copperdefstring{brazilian}{frame}{Quadro}
1292 \copperdefstring{english}{frame}{Frame}
1293 \copperdefstring{brazilian}{listframe}{Lista de Quadros}
1294 \copperdefstring{english}{listframe}{List of Frames}
1295 \copperdefstring{brazilian}{source}{Fonte}
1296 \copperdefstring{english}{source}{Source}
1297 \copperdefstring{brazilian}{program}{Programa}
1298 \copperdefstring{english}{program}{Program}
1299 \copperdefstring{brazilian}{listprogram}{Lista de Programas}
1300 \copperdefstring{english}{listprogram}{List of Programs}
1301 \copperdefstring{brazilian}{references}{Refer\~encias}
1302 \copperdefstring{english}{references}{References}
1303 \copperdefstring{brazilian}{in:}{em}
1304 \copperdefstring{english}{in:}{in}
1305 \copperdefstring{brazilian}{art}{Ilustra\c{c}\~ao}
1306 \copperdefstring{english}{art}{Art}
1307 \copperdefstring{brazilian}{platform}{Plataforma}
1308 \copperdefstring{english}{platform}{Platform}
1309 \copperdefstring{brazilian}{directedby}{Dire\c{c}\~ao}
1310 \copperdefstring{english}{directedby}{Directed by}
1311 \copperdefstring{brazilian}{producedby}{Produ\c{c}\~ao}
1312 \copperdefstring{english}{producedby}{Produced by}
1313 \copperdefstring{brazilian}{listabbreviation}{Lista de Abreviaturas e Siglas}
1314 \copperdefstring{english}{listabbreviation}{List of Abbreviations and Acronyms}
1315 \copperdefstring{brazilian}{listsymbol}{Lista de S{\~ i}mbolos}
1316 \copperdefstring{english}{listsymbol}{List of Symbols}
1317 \copperdefstring{brazilian}{glossary}{Gloss{\~ a}rio}
1318 \copperdefstring{english}{glossary}{Glossary}
1319 \copperdefstring{brazilian}{advisor}{Orientador}
1320 \copperdefstring{english}{advisor}{Advisor}
1321 \copperdefstring{brazilian}{advisors}{Orientadores}
1322 \copperdefstring{english}{advisors}{Advisors}
1323 \copperdefstring{brazilian}{dept}{Programa}
1324 \copperdefstring{english}{dept}{Department}
1325 \copperdefstring{brazilian}{approvedmale}{Examinado por}
1326 \copperdefstring{brazilian}{approvedfemale}{Examinada por}
1327 \copperdefstring{english}{approvedmale}{Examined by}
1328 \copperdefstring{english}{approvedfemale}{Examined by}
1329 % Institutional names: always Portuguese on the cover/folha (UFRJ template).
1330 \copperdefstring{brazilian}{universityname}{Universidade Federal do Rio de Janeiro}
1331 \copperdefstring{english}{universityname}{Universidade Federal do Rio de Janeiro}
1332 \copperdefstring{brazilian}{cityname}{Rio de Janeiro}

```

```

1333 \copperdefstring{english} {cityname} {Rio de Janeiro}
1334 \copperdefstring{brazilian}{statename} {RJ}
1335 \copperdefstring{english} {statename} {RJ}
1336 \copperdefstring{brazilian}{countryname} {Brasil}
1337 \copperdefstring{english} {countryname} {Brasil}
1338 % Month names.
1339 \copperdefstring{brazilian}{monthname1} {Janeiro}
1340 \copperdefstring{brazilian}{monthname2} {Fevereiro}
1341 \copperdefstring{brazilian}{monthname3} {Mar{\c c}o}
1342 \copperdefstring{brazilian}{monthname4} {Abril}
1343 \copperdefstring{brazilian}{monthname5} {Maio}
1344 \copperdefstring{brazilian}{monthname6} {Junho}
1345 \copperdefstring{brazilian}{monthname7} {Julho}
1346 \copperdefstring{brazilian}{monthname8} {Agosto}
1347 \copperdefstring{brazilian}{monthname9} {Setembro}
1348 \copperdefstring{brazilian}{monthname10}{Outubro}
1349 \copperdefstring{brazilian}{monthname11}{Novembro}
1350 \copperdefstring{brazilian}{monthname12}{Dezembro}
1351 \copperdefstring{english}{monthname1} {January}
1352 \copperdefstring{english}{monthname2} {February}
1353 \copperdefstring{english}{monthname3} {March}
1354 \copperdefstring{english}{monthname4} {April}
1355 \copperdefstring{english}{monthname5} {May}
1356 \copperdefstring{english}{monthname6} {June}
1357 \copperdefstring{english}{monthname7} {July}
1358 \copperdefstring{english}{monthname8} {August}
1359 \copperdefstring{english}{monthname9} {September}
1360 \copperdefstring{english}{monthname10}{October}
1361 \copperdefstring{english}{monthname11}{November}
1362 \copperdefstring{english}{monthname12}{December}
1363 % algorithm2e package option for this language (empty -> english default).
1364 \copperdefstring{brazilian}{algorithm2eopt}{portuguese}
1365 \copperdefstring{english} {algorithm2eopt}{}
1366 % Babel package name for the foreignabstract \begin{otherlanguage}{...}.
1367 \copperdefstring{brazilian}{babelname}{brazilian}
1368 \copperdefstring{english} {babelname}{english}
1369 % Auto-load language packs for any non-built-in language listed in babelopts.
1370 % Built-ins (brazilian, english) ship their string tables inside coppe.cls
1371 % (defined just above) and need no extra file; others (spanish/french/italian
1372 % /...) ship as coppe-lang-<name>.def alongside the class. The dispatcher
1373 % must already be defined when these run, so the loader sits AFTER the tables.
1374 \@for\coppe@@lang:=\coppe@babelopts\do{%
1375 \edef\coppe@@cur{\coppe@@lang}%
1376 \ifx\coppe@@cur\coppe@@brazilianstr\else
1377 \ifx\coppe@@cur\coppe@@englishstr\else
1378 \InputIfFileExists{coppe-lang-\coppe@@cur.def}
1379 {\ClassInfo{coppe}{Loaded language pack coppe-lang-\coppe@@cur.def}}
1380 {\ClassError{coppe}
1381 {Missing language pack 'coppe-lang-\coppe@@cur.def'.}
1382 {Install it next to coppe.cls, or pick a built-in language.}}%
1383 \fi\fi}
1384 % Quotation marks: csquotes gives the language-aware \enquote; \citacao is its
1385 % Portuguese alias for short, inline quotations (proposta P3).
1386 \RequirePackage{csquotes}

```

```

1387 \newcommand{\citacao}[1]{\enquote{#1}}
1388 % --- Code listings & algorithms (TODO3 "tratar listagens"): the listings
1389 % package gives the "Programa" listing (caption on top, ABNT R46) with
1390 % \listofprogramas; algorithm2e gives algorithms. Names are babel-aware and
1391 % the language styles below are based on the user's listagens.tex. -----
1392 \RequirePackage{xcolor}
1393 \RequirePackage{listings}
1394 \RequirePackage{fancyvrb}
1395 \DeclareFixedFont{\ttb}{T1}{txtt}{bx}{n}{10}
1396 \DeclareFixedFont{\ttm}{T1}{txtt}{m}{n}{10}
1397 \lstset{captionpos=t,extendedchars=true,basicstyle=\ttfamily,inputencoding=utf8,%
1398 literate=%
1399 {\~}{\~{a}}1 {\~}{\~{a}}1 {\~}{\~{a}}1 {\~}{\~{a}}1 {\~}{\~{e}}1%
1400 {\~}{\~{e}}1 {\~}{\~{e}}1 {\~}{\~{i}}1 {\~}{\~{i}}1 {\~}{\~{o}}1%
1401 {\~}{\~{o}}1 {\~}{\~{o}}1 {\~}{\~{u}}1 {\~}{\~{u}}1 {\~}{\~{c}}1%
1402 {\~}{\~{c}}1 {\~}{\~{A}}1 {\~}{\~{A}}1 {\~}{\~{A}}1 {\~}{\~{E}}1%
1403 {\~}{\~{E}}1 {\~}{\~{I}}1 {\~}{\~{I}}1 {\~}{\~{O}}1 {\~}{\~{O}}1 {\~}{\~{O}}1%
1404 {\~}{\~{U}}1%
1405 }
1406 % Babel-aware listing names, set after listings' own babel setup.
1407 \AtBeginDocument{%
1408 \renewcommand\lstlistingname{\coppestring{program}}%
1409 \renewcommand\lstlistlistingname{\coppestring{listprogram}}%
1410 % \listofprogramas in the coppe list style (heading + ToC + running head).
1411 \newcommand\listofprogramas{%
1412 \chapter*{\lstlistlistingname}%
1413 \addcontentsline{toc}{chapter}{\lstlistlistingname}%
1414 \mkboth{\MakeUppercase\lstlistlistingname}{\MakeUppercase\lstlistlistingname}%
1415 \starttoc{lol}}
1416 % Per-language styles + environments (python, java, xml, html, prolog).
1417 \newcommand\pythonstyle{\lstset{language=Python,basicstyle=\ttm,numbers=left,%
1418 numberstyle=\small\ttfamily,numbersep=0.5em,commentstyle=\color{purple},%
1419 morekeywords={self,as},keywordstyle=\ttb\color{rgb}{0,0,0.5},%
1420 emph={MyClass,__init__},emphstyle=\ttb\color{rgb}{0.6,0,0},%
1421 stringstyle=\color{rgb}{0,0.5,0},frame=tb,showstringspaces=false,%
1422 extendedchars=true,firstnumber=auto,inputencoding=utf8,breaklines=true,%
1423 escapechar=',postbreak=\mbox{\textcolor{red}{\hookrightarrow}}\space}}
1424 \lstnewenvironment{python}[1][\pythonstyle\lstset{#1}]%
1425 \csname\@lst @SetFirstNumber\endcsname{\csname\@lst @SaveFirstNumber\endcsname}
1426 \newcommand\pythoninline[1][\pythonstyle\lstinline!#1!]}
1427 \newcommand\pythonexternal[2][\pythonstyle\lstinputlisting[1]{#2}]}
1428 \newcommand\xmlstyle{\lstset{language=XML,basicstyle=\ttm,numbers=left,%
1429 numberstyle=\small\ttfamily,numbersep=0.5em,%
1430 morekeywords={xs:schema,xs:element,xs:sequence,xs:complexType},%
1431 keywordstyle=\ttb\color{rgb}{0,0,0.5},%
1432 emph={name,type,attributeFormDefault,elementFormDefault,xmlns:x},%
1433 emphstyle=\ttb\color{rgb}{0.6,0,0},stringstyle=\color{rgb}{0,0.5,0},frame=tb,%
1434 showstringspaces=false,extendedchars=true,inputencoding=utf8,breaklines=true,%
1435 postbreak=\mbox{\textcolor{red}{\hookrightarrow}}\space}}
1436 \lstnewenvironment{xml}[1][\xmlstyle\lstset{#1}]}
1437 \newcommand\htmlstyle{\lstset{language=HTML,basicstyle=\ttm,numbers=left,%
1438 numberstyle=\small\ttfamily,numbersep=0.5em,%
1439 keywordstyle=\ttb\color{rgb}{0,0,0.5},emphstyle=\ttb\color{rgb}{0.6,0,0},%
1440 stringstyle=\color{rgb}{0,0.5,0},frame=tb,showstringspaces=false,%

```



```

1441 extendedchars=true,inputencoding=utf8,breaklines=true,%
1442 postbreak=\mbox{\textcolor{red}{\hookrightarrow}}\space}}
1443 \lstnewenvironment{html}[1][\{\htmlstyle\lstset{#1}\}}{
1444 \newcommand\prologstyle{\lstset{language=Prolog,basicstyle=\ttm,numbers=left,%
1445 numberstyle=\small\ttfamily,numbersep=0.5em,commentstyle=\color{purple},%
1446 keywordstyle=\ttb\color{rgb}{0,0,0.5},emph={MyClass,__init__},%
1447 emphstyle=\ttb\color{rgb}{0.6,0,0},stringstyle=\color{rgb}{0,0.5,0},frame=tb,%
1448 showstringspaces=false,extendedchars=true,firstnumber=auto,inputencoding=utf8,%
1449 breaklines=true,escapechar=',%
1450 postbreak=\mbox{\textcolor{red}{\hookrightarrow}}\space}}
1451 \lstnewenvironment{prolog}[1][\{\prologstyle\lstset{#1}\}}{
1452 \csname\@lst @SetFirstNumber\endcsname\{\csname\@lst @SaveFirstNumber\endcsname\}
1453 \newcommand\javastyle{\lstset{language=Java,basicstyle=\ttm,numbers=left,%
1454 numberstyle=\small\ttfamily,numbersep=0.5em,commentstyle=\color{purple},%
1455 morekeywords={self,as},keywordstyle=\ttb\color{rgb}{0,0,0.5},%
1456 emphstyle=\ttb\color{rgb}{0.6,0,0},stringstyle=\color{rgb}{0,0.5,0},frame=tb,%
1457 showstringspaces=false,extendedchars=true,firstnumber=auto,inputencoding=utf8,%
1458 breaklines=true,escapechar=',%
1459 postbreak=\mbox{\textcolor{red}{\hookrightarrow}}\space}}
1460 \lstnewenvironment{java}[1][\{\javastyle\lstset{#1}\}}{
1461 \csname\@lst @SetFirstNumber\endcsname\{\csname\@lst @SaveFirstNumber\endcsname\}
1462 \newcommand\javainline[1]{\{\javastyle\lstinline!#1!\}}
1463 \newcommand\javaexternal[2][\{\javastyle\lstinputlisting[1]{#2}\}}
1464 % fancyvrb's Verbatim environment is available for program output; the
1465 % \DefineVerbatimEnvironment{gxoutput}/{gxoutputs} aliases that used to live
1466 % here were author-personal (gx prefix) and were removed in v4.0. Use, for
1467 % example,
1468 % \begin{Verbatim}[frame=lines,numbers=left,numbersep=0.5em]
1469 % ...
1470 % \end{Verbatim}
1471 % (or \DefineVerbatimEnvironment{...}{Verbatim}{...} in the manuscript preamble).
1472 \renewcommand{\theFancyVerbLine}{\small\ttfamily\arabic{FancyVerbLine}}
1473 % Algorithms: algorithm2e, language driven by the main-slot \coppemainstring
1474 % lookup (key "algorithm2eopt"). Empty means use algorithm2e's default
1475 % language (English) -- needed because algorithm2e errors on an empty option.
1476 \edef\coppe@@algotlang{\coppemainstring{algorithm2eopt}}
1477 \ifx\coppe@@algotlang\empty
1478 \RequirePackage[linesnumbered,lined,algochapter,ruled]{algorithm2e}
1479 \else
1480 \expandafter\RequirePackage\expandafter[\coppe@@algotlang,linesnumbered,lined,algochapter,r
1481 \fi
1482 \SetKwFor{Para}{para}{faça}{fim do para}
1483 \SetKwFor{Enquanto}{enquanto}{faça}{fim do enquanto}
1484 % setspace + hyperref + biblatex are loaded at the very end of the class
1485 % (see the block just before %</class>), so their hooks see every macro the
1486 % class defines, and the load order setspace -> hyperref -> biblatex is
1487 % respected.

```

`\department` This macro is used to set the author's affiliation. There are twelve options which correspond to all academic units at COPPE/UFRJ. It defines the current and the foreign names of these units.

```

1488 \newcommand\department[1]{%
1489 \ifthenelse{\equal{#1}{PEB}}{
1490 {\global\def\local@deptname{Engenharia Biom(\` e)dica}

```

```

1491 \global\def\foreign@deptname{Biomedical Engineering}}{}
1492 \ifthenelse{\equal{#1}{PEC}}
1493 {\global\def\local@deptname{Engenharia Civil}
1494 \global\def\foreign@deptname{Civil Engineering}}{}
1495 \ifthenelse{\equal{#1}{PEE}}
1496 {\global\def\local@deptname{Engenharia El{\'e}trica}
1497 \global\def\foreign@deptname{Electrical Engineering}}{}
1498 \ifthenelse{\equal{#1}{PEM}}
1499 {\global\def\local@deptname{Engenharia Mec{\'a}nica}
1500 \global\def\foreign@deptname{Mechanical Engineering}}{}
1501 \ifthenelse{\equal{#1}{PEMM}}
1502 {\global\def\local@deptname{Engenharia Metal{\'u}rgica e de Materiais}
1503 \global\def\foreign@deptname{Metallurgical and Materials Engineering}}{}
1504 \ifthenelse{\equal{#1}{PEN}}
1505 {\global\def\local@deptname{Engenharia Nuclear}
1506 \global\def\foreign@deptname{Nuclear Engineering}}{}
1507 \ifthenelse{\equal{#1}{PEN0}}
1508 {\global\def\local@deptname{Engenharia Oce{\'a}nica}
1509 \global\def\foreign@deptname{Ocean Engineering}}{}
1510 \ifthenelse{\equal{#1}{PPE}}
1511 {\global\def\local@deptname{Planejamento Energ{\'e}tico}
1512 \global\def\foreign@deptname{Energy Planning}}{}
1513 \ifthenelse{\equal{#1}{PEP}}
1514 {\global\def\local@deptname{Engenharia de Produ{\'c}{\~a}{o}}
1515 \global\def\foreign@deptname{Production Engineering}}{}
1516 \ifthenelse{\equal{#1}{PEQ}}
1517 {\global\def\local@deptname{Engenharia Qu{\'i}mica}
1518 \global\def\foreign@deptname{Chemical Engineering}}{}
1519 \ifthenelse{\equal{#1}{PESC}}
1520 {\global\def\local@deptname{Engenharia de Sistemas e Computa{\'c}{\~a}{o}}
1521 \global\def\foreign@deptname{Systems Engineering and Computer Science}}{}
1522 \ifthenelse{\equal{#1}{PET}}
1523 {\global\def\local@deptname{Engenharia de Transportes}
1524 \global\def\foreign@deptname{Transportation Engineering}}{}
1525 \ifthenelse{\equal{#1}{PENT}}
1526 {\global\def\local@deptname{Engenharia de Nanotecnologia}
1527 \global\def\foreign@deptname{Nanotechnology Engineering}}{}
1528 }

```

`\title` Used to enter the title in Brazilian Portuguese. Mirrors into `\coppe@title@brazilian` so that the per-language title slot used by the cover/folha renderers (`\coppe@selecttitle`) picks the right text for a pt-main thesis.

```

1529 \renewcommand\title[1]{%
1530 \global\def\local@title{#1}%
1531 \expandafter\gdef\csname coppe@title@brazilian\endcsname{#1}%
1532 }

```

`\foreigntitle` Used to enter the foreign title. Also mirrors into `\coppe@title@english` so that the cover renders the right text for an english-main thesis.

```

1533 \newcommand\foreigntitle[1]{%
1534 \global\def\foreign@title{#1}%
1535 \expandafter\gdef\csname coppe@title@english\endcsname{#1}%
1536 }

```

`\coppe@selecttitle` Expands to the title stored for the current main language. For pt-main this is `\local@title`; for en-main `\foreign@title`; for any other main language the user must call `\titlein{<lang>}{...}` to populate it.

```
1537 \newcommand\coppe@selecttitle{\@nameuse{coppe@title@\coppe@mainlang}}
```

`\advisor` Defines globally the title, name and academic degree of the advisors.

```
1538 \newcount\@advisor\@advisor0
1539 \newcommand\advisor[4]{%
1540   \global\@namedef{CoppeAdvisorTitle:\expandafter\the\@advisor}{#1}
1541   \global\@namedef{CoppeAdvisorName:\expandafter\the\@advisor}{#2}
1542   \global\@namedef{CoppeAdvisorSurname:\expandafter\the\@advisor}{#3}
1543   \global\@namedef{CoppeAdvisorDegree:\expandafter\the\@advisor}{#4}
1544   \global\advance\@advisor by 1
1545   \ifnum\@advisor>1
1546     \renewcommand\local@advisorstring{Orientadores}
1547     \renewcommand\foreign@advisorstring{Advisors}
1548   \fi
1549 }
```

`\examiner`

```
1550 \newcount\@examiner\@examiner0
1551 \newcommand\examiner[3]{%
1552   \global\@namedef{CoppeExaminer:\expandafter\the\@examiner}{#1\ #2}
1553   \global\advance\@examiner by 1
1554 }
```

`\author` It was redefined to allow the identification of the author's first names and surname.

```
1555 \renewcommand\author[2]{%
1556   \global\def\@authname{#1}
1557   \global\def\@authsurn{#2}
1558 }
```

`\date` Sets the *document's* nominal month and year (used by the cover, the folha de rosto, and the catalog). Stores them in private slots (`\coppe@docmonth` / `\coppe@docyear`) rather than mutating TeX's `\month` / `\year` primitives, which would clobber `\today`.

```
1559 \renewcommand\date[2]{%
1560   \def\coppe@docmonth{#1}%
1561   \def\coppe@docyear{#2}}
```

`\local@monthname`

```
1562 \newcommand\local@monthname{\ifcase\coppe@docmonth\or
1563   Janeiro\or Fevereiro\or Mar{c c}\or Abril\or Maio\or Junho\or
1564   Julho\or Agosto\or Setembro\or Outubro\or Novembro\or Dezembro\fi}
```

`\foreign@monthname`

```
1565 \newcommand\foreign@monthname{\ifcase\coppe@docmonth\or
1566   January\or February\or March\or April\or May\or June\or
1567   July\or August\or September\or October\or November\or December\fi}
```

`\keyword`

```
1568 \newcounter{keywords}
```

```

1569 \newcommand\keyword[1]{%
1570   \global\@namedef{CoppeKeyword:\expandafter\the\c@keywords}{#1}
1571   \global\addtocounter{keywords}{1}
1572 }

```

`\freeconfig` This command allows easy changing of core class parameters.

```

1573 \newcommand\freeconfig[6]{%
1574   \providecommand\@degree{}
1575   \renewcommand\@degree{#1}
1576   \providecommand\@degreename{}
1577   \renewcommand\@degreename{#2}
1578   \providecommand\local@degname{}
1579   \renewcommand\local@degname{#3}
1580   \providecommand\foreign@degname{}
1581   \renewcommand\foreign@degname{#4}
1582   \providecommand\local@doctype{}
1583   \renewcommand\local@doctype{#5}
1584   \providecommand\foreign@doctype{}
1585   \renewcommand\foreign@doctype{#6}%
1586 }
1587 %   \end{macrocode}
1588 % \end{macro}
1589 %
1590 % \begin{macro}{\frontmatter}
1591 % The number of pages for both frontmatter and mainmatter printed
1592 % in the cataloging details page is computed by means of simple
1593 % LaTeX\ labels.
1594 %   \begin{macrocode}
1595 \renewcommand\frontmatter{%
1596   \cleardoublepage
1597   \@mainmatterfalse
1598   \pagenumbering{roman}
1599   \thispagestyle{empty}
1600   % folha de rosto = counted page 1 (ABNT: the capa is not counted). An odd
1601   % start keeps the page-counter parity aligned with the physical (recto) side,
1602   % so \cleardoublepage lands every pre-textual element on an odd (front) page.
1603   \setcounter{page}{1}
1604   \makefrontpage
1605   \clearpage
1606   \if@copperoman
1607     \fancypagestyle{plain}{\fancyhf{}\fancyhead[R0,LE]{\thepage}}%
1608     \renewcommand{\headrulewidth}{0pt}}%
1609     \pagestyle{coppe}%
1610   \else
1611     \fancypagestyle{plain}{\fancyhf{}\renewcommand{\headrulewidth}{0pt}}%
1612     \pagestyle{empty}%
1613   \fi
1614 }

```

`\mainmatter`

```

1615 \renewcommand\mainmatter{%
1616   \coppe@mainBegin
1617   \cleardoublepage
1618   \@mainmattertrue

```

```

1619 \fancypagestyle{plain}{\fancyhf{}\fancyhead[R0,LE]{\thepage}%
1620 \renewcommand{\headrulewidth}{0pt}}%
1621 \pagestyle{coppe}%
1622 \pagenumbering{arabic}}

```

\backmatter

```

1623 \renewcommand\backmatter{%
1624 \if@openright
1625 \cleardoublepage
1626 \else
1627 \clearpage
1628 \fi}
1629 %

```

\coppetexfinalpage Optional colophon-style closing page intended to face the rear cover. The macro forces a verso (even) page so that, in a two-sided print, the reader sees the colophon on the left and the inside of the rear cover on the right; pushes the text to the bottom of the page with `\vfill`; and types it in Portuguese unconditionally (the colophon is institutional metadata, not body content, so it is rendered in pt-BR regardless of the thesis main language). Three slot macros are user-overridable: `\coppefinalengine` (engine name; auto-detected via `iftex`), `\coppefinalfont` (text body font; defaults to Computer Modern, the L^AT_EX default), and `\coppefinalmanual` (the reference manual title; defaults to the current COPPE/UFRJ norm). Override them before `\coppetexfinalpage` with `\renewcommand` if the document uses a non-default font or follows an earlier edition of the norm.

```

1630 \providecommand{\coppefinalengine}{%
1631 \ifPDFTeX\hologo{pdfLaTeX}\fi
1632 \ifLuaTeX\hologo{LuaLaTeX}\fi
1633 \ifXeTeX\hologo{XeLaTeX}\fi}
1634 \providecommand{\coppefinalfont}{Computer Modern (pdr~ao \LaTeX{})}
1635 \providecommand{\coppefinalmanual}{%
1636 \emph{Norma para a Elabora~ao Gr~afica de Teses e Disserta~oes
1637 da COPPE/UFRJ}}
1638 \newcommand{\coppetexfinalpage}{%
1639 \clearpage
1640 % Force the colophon onto a verso (even) page so it faces the rear cover.
1641 % If the current page is odd (recto), fill it with an empty verso first.
1642 \if@twoside
1643 \ifodd\c@page
1644 \hbox{}\thispagestyle{empty}\newpage
1645 \fi
1646 \fi
1647 \thispagestyle{empty}%
1648 \null\vfill
1649 \begin{otherlanguage}{brazilian}%
1650 \begin{center}
1651 {\footnotesize\setstretch{1.0}%
1652 \textsc{Colof~ao}\par
1653 \vspace{0.7em}
1654 \begin{minipage}{0.75\textwidth}\centering
1655 Este documento foi composto em \LaTeX{} com \coppefinalengine{}
1656 em \today, na fonte \coppefinalfont. A bibliografia foi processada

```

```

1657     por Bib\LaTeX{} e Biber. Foi utilizada a classe \CoppeTeX,
1658     vers\~ao\coppe@version, dispon\'i vel no reposit\'orio GitHub
1659     \mbox{\url{https://github.com/COPPE-UFRJ/CoppeTeX}}, em
1660     conformidade com a \coppefinalmanual.
1661     \end{minipage}\par}
1662   \end{center}
1663 \end{otherlanguage}%
1664 \vspace*{2cm}%
1665 \clearpage}

```

`\makecover` The ABNT cover (capa, NBR 14724) suggested in the manual's Anexo A. It is the institution block — the university, the full COPPE name, and the graduate program taken from `\department` — followed by the author, the title, and the city and year, all centred and uppercased. The optional UFRJ and COPPE logos head the page. The capa is not counted: it precedes the (roman) front matter and is produced before the `titlepage` (folha de rosto).

```

1666 \newcommand\makecover{%
1667   \begin{titlepage}%
1668   \centering
1669   \makebox[\linewidth]{%
1670     \includegraphics[height=2cm]{ufrj-logo}\hfill
1671     \includegraphics[height=2cm]{coppe-logo}}\par
1672   \vspace{1.5cm}
1673   {\singlespacing
1674   %% \MakeUppercase{\local@universityname}\par
1675   %% \nohyphens{\MakeUppercase{Instituto Alberto Luiz Coimbra de
1676   %%   P{\~o}s-Gradua{\c c}{\~a}o e Pesquisa de Engenharia (COPPE)}}\par
1677   \MakeUppercase{Programa de \local@deptname}\par}
1678   \vspace{\stretch{2}}
1679   \textbf{\MakeUppercase{\@authname\ \@authsur}}\par
1680   \vspace{\stretch{2}}
1681   \nohyphens{\MakeUppercase{\coppe@selecttitle}}\par
1682   \vspace{\stretch{3}}
1683   \MakeUppercase{\local@cityname}\par
1684   \coppe@docyear
1685   \end{titlepage}%
1686 }

```

`\maketitle`

```

1687 \renewcommand\maketitle{%
1688   \pagenumbering{alph}
1689   \ltx@ifpackageloaded{hyperref}{\coppe@hypersetup}{}%
1690   \makecover
1691   \begin{titlepage}
1692   \begin{flushleft}
1693     \vspace*{1.5mm}
1694     \setlength\baselineskip{0pt}
1695     \setlength\parskip{1mm}
1696     \makebox[\linewidth]{%
1697       \includegraphics[height=2cm]{ufrj-logo}\hfill
1698       \includegraphics[height=2cm]{coppe-logo}}
1699     \end{flushleft}
1700     \vspace{1.05cm}

```

```

1701 \begin{center}
1702 \nohyphens{\MakeUppercase{\coppe@selecttitle}}\par
1703 \vspace*{3cm}
1704 \nohyphens{\@authname\ \@authsurname}\par
1705 \end{center}
1706 \vspace*{2.1cm}
1707 \begin{flushright}
1708 \begin{minipage}{8.45cm}
1709 \frontcover@maintext
1710 \end{minipage}\par
1711 \vspace*{7.5mm}
1712 \nohyphens{%
1713 \begin{tabularx}{8.45cm}[b]{@{}l@{ }>{\raggedright\arraybackslash}X@{}}
1714 \local@advisorstring: &
1715 \count1=0
1716 \toks@={}
1717 \@whilenum \count1<\@advisor \do{%
1718 \ifcase\count1 % same as \ifnum0=\count1
1719 \toks@=\expandafter{\csname CoppeAdvisorName:\the\count1%
1720 \expandafter\endcsname\expandafter\space%
1721 \csname CoppeAdvisorSurname:\the\count1\endcsname\\}
1722 \else
1723 \toks@=\expandafter\expandafter\expandafter{%
1724 \expandafter\the\expandafter\toks@%
1725 \expandafter&\expandafter\space%
1726 \csname CoppeAdvisorName:\the\count1\expandafter\endcsname%
1727 \expandafter\space\csname CoppeAdvisorSurname:\the\count1\endcsname\\}
1728 }%
1729 \fi
1730 \advance\count1 by 1}
1731 \the\toks@
1732 \end{tabularx}}\par
1733 \end{flushright}
1734 \vspace*{\fill}
1735 \begin{center}
1736 \local@cityname\par
1737 \local@monthname\ de \coppe@docyear
1738 \end{center}
1739 \end{titlepage}
1740 % Ficha catalográfica on the verso of the folha de rosto (NBR 14724, R57):
1741 % the title page is a recto, so this lands on the immediately following verso,
1742 % ahead of the approval sheet produced by \frontmatter.
1743 \ifthenelse{\boolean{maledoc}}{\ifthispagestyle{empty}\makecatalog}%
1744 \global\let\maketitle\relax%
1745 \global\let\and\relax}

1746 \newcommand\makefrontpage{%
1747 \begin{center}
1748 \sloppy\nohyphens{\MakeUppercase{\coppe@selecttitle}}\par
1749 \vspace*{7mm}
1750 {\@authname\ \@authsurname}\par
1751 \end{center}\par
1752 \vspace*{4mm}
1753 \frontpage@maintext

```

```

1754 \vspace*{16mm}
1755 \nohyphens{%
1756 \noindent\begin{tabularx}{\textwidth}[b]{@{}l@{ }>{\raggedright\arraybackslash}X@{}}
1757 \local@advisorstring: &
1758 \count1=0
1759 \toks@={}
1760 \@whilenum \count1<\@advisor \do{%
1761 \ifcase\count1 % same as \ifnum0=\count1
1762 \toks@=\expandafter{\csname CoppeAdvisorName:\the\count1%
1763 \expandafter\endcsname\expandafter\space%
1764 \csname CoppeAdvisorSurname:\the\count1\endcsname\\}
1765 \else
1766 \toks@=\expandafter\expandafter\expandafter{%
1767 \expandafter\the\expandafter\toks@%
1768 \expandafter&\expandafter\space%
1769 \csname CoppeAdvisorName:\the\count1\expandafter\endcsname%
1770 \expandafter\space\csname CoppeAdvisorSurname:\the\count1\endcsname\\
1771 }%
1772 \fi
1773 \advance\count1 by 1}
1774 \the\toks@
1775 \end{tabularx}\par
1776 \vspace*{20mm}
1777 \noindent\begin{tabularx}{\textwidth}[b]{@{}l@{ }>{\raggedright\arraybackslash}X@{}}
1778 Aprovada por: &
1779 \count1=0
1780 \toks@={}
1781 \@whilenum \count1<\@examiner \do{%
1782 \ifcase\count1 % same as \ifnum0=\count1
1783 \toks@=\expandafter{\csname CoppeExaminer:\the\count1%
1784 \expandafter\endcsname\expandafter\\}
1785 \else
1786 \toks@=\expandafter\expandafter\expandafter{%
1787 \expandafter\the\expandafter\toks@%
1788 \expandafter&\expandafter\space%
1789 \csname CoppeExaminer:\the\count1\expandafter\endcsname%
1790 \expandafter\space\\
1791 }%
1792 \fi
1793 \advance\count1 by 1}
1794 \the\toks@
1795 \end{tabularx}}\par
1796 \vspace*{\fill}
1797 \frontpage@bottomtext}

1798 \newcommand\coppe@hypersetup{%
1799 \begin{group
1800 % changes to \toks@ and \count@ are kept local;
1801 % it's not necessary for them, but it is usually the case
1802 % for \count1, because the first ten counters are written
1803 % to the DVI file, thus you got lucky because of PDF output
1804 \toks@={} % in this special case not necessary
1805 \count@=0 %
1806 \@whilenum\count@<\value{keywords}\do{%

```



```

1807 % * a keyword separator is not necessary,
1808 % if there is just one keyword
1809 % * \csname CoppeKeyword:\the\count@\endcsname must be expanded
1810 % at least once, to get rid of the loop depended \count@
1811 \ifcase\count@ % same as \ifnum0=\count@
1812 \toks@=\expandafter{\csname CoppeKeyword:\the\count@\endcsname}%
1813 \else
1814 \toks@=\expandafter\expandafter\expandafter{%
1815 \expandafter\the\expandafter\toks@
1816 \expandafter;\expandafter\space
1817 \csname CoppeKeyword:\the\count@\endcsname
1818 }%
1819 \fi
1820 \advance\count@ by 1 %
1821 }%
1822 \edef\x{\endgroup
1823 \noexpand\hypersetup{%
1824 pdfkeywords={\the\toks@}%
1825 }%
1826 }%
1827 \x
1828 \hypersetup{%
1829 pdfauthor={\@authname\ \@authsur},
1830 pdftitle={\local@title},
1831 pdfsubject={\local@doctype\ de \@degree\ em \local@deptname\ da COPPE/UFRJ},
1832 pdfcreator={LaTeX with CoppeTeX toolkit},
1833 breaklinks={true},
1834 raiselinks={true},
1835 pageanchor={true},
1836 }}

```

`\makecatalog` When the document has illustrations, it is required to insert “: il;” between the number of pages of the textual part and the page dimension. We have created a label to flag the existence of lists of figures. It is checked to be undefined using the plain `TeX` command `\isundefined` (cf. the `TeX` FAQ).

```

1837 \newcommand\makecatalog{%
1838 \vspace*{\fill}
1839 \begin{center}
1840 \setlength{\fboxsep}{5mm}
1841 \framebox[120mm][c]{\makebox[5mm][c]{}%
1842 \begin{minipage}[c]{105mm}
1843 \setlength{\parindent}{5mm}
1844 \noindent\sloppy\nohyphens\@authsur,
1845 \nohyphens\@authname\par
1846 \nohyphens{%
1847 \coppe@selecttitle/\@authname\ \@authsur. -- \local@cityname:
1848 UFRJ/COPPE, \coppe@docyear.}\par
1849 \pageref{front:pageno},
1850 \pageref{LastPage}
1851 p.\@ifundefined{r@cat:lofflag}{\pageref{cat:lofflag}}{$29,7$cm.\par
1852 % There is an issue here. When the last entry must be split between lines,
1853 % the spacing between it and the next paragraph becomes smaller.
1854 % Should we manually introduce a fixed space? But how could we know that
1855 % a name was split? Is this happening yet?

```

```

1856 \nohyphens{%
1857 \begin{tabularx}{100mm}[b]{@{}l@{ }>{\raggedright\arraybackslash}X@{}}
1858 \local@advisorstring: &
1859 \count1=0
1860 \toks@={}
1861 \@whilenum \count1<\@advisor \do{%
1862 \ifcase\count1 % same as \ifnum0=\count1
1863 \toks@=\expandafter{\csname CoppeAdvisorName:\the\count1%
1864 \expandafter\endcsname\expandafter\space%
1865 \csname CoppeAdvisorSurname:\the\count1\endcsname\\}
1866 \else
1867 \toks@=\expandafter\expandafter\expandafter{%
1868 \expandafter\the\expandafter\toks@
1869 \expandafter&\expandafter\space
1870 \csname CoppeAdvisorName:\the\count1\expandafter\endcsname%
1871 \expandafter\space\csname CoppeAdvisorSurname:\the\count1\endcsname\\}
1872 }%
1873 \fi
1874 \advance\count1 by 1}
1875 \the\toks@
1876 \end{tabularx}}\par
1877 \nohyphens{\local@doctype\ ({\MakeLowercase\@degreename}) --
1878 UFRJ/COPPE/Programa de \local@deptname, \coppe@docyear.}\par
1879 Refer{\^ e}ncias Bibliogr{\' a}ficas: p. \pageref{bib:begin} -- \pageref{bib:end}.\par
1880 \count1=0
1881 \count2=1
1882 \nohyphens{\@whilenum \count1<\value{keywords} \do {%
1883 \number\count2. \csname CoppeKeyword:\the\count1 \endcsname.
1884 \advance\count1 by 1
1885 \advance\count2 by 1}
1886 I. \csname CoppeAdvisorSurname:0\endcsname,%
1887 \ \csname CoppeAdvisorName:0\endcsname%
1888 \ifthenelse{\@advisor>1}{\ \emph{et~al.}}{}}.
1889 II. \local@universityname, COPPE, Programa de \local@deptname.
1890 III. T{\' i}tulo.}
1891 \end{minipage}}
1892 \end{center}
1893 \vspace*{\fill}}

```

\dedication

```

1894 \newcommand\dedication[1]{
1895 \gdef\@dedic{#1}
1896 \cleardoublepage
1897 \vspace*{\fill}
1898 \begin{flushright}
1899 \begin{minipage}{60mm}
1900 \raggedleft \it \normalsize \@dedic
1901 \end{minipage}
1902 \end{flushright}}

```

abstract (*env.*) This is a specialization of the abstract in the article standard class.

```

1903 \newenvironment{abstract}{%
1904 \cleardoublepage
1905 \thispagestyle{plain}

```

```

1906 \abstract@toptext\par
1907 \vspace*{8.6mm}
1908 \begin{center}
1909 \sloppy\nohyphens{\MakeUppercase\local@title}\par
1910 \vspace*{13.2mm}
1911 \@authname\ \@authsur \par
1912 \vspace*{7mm}
1913 \local@monthname/\coppe@docyear
1914 \end{center}\par
1915 \vspace*{1.5cm}
1916 \noindent%
1917 \begin{tabularx}{\textwidth}[b]{@{}l@{ }>{\raggedright\arraybackslash}X@{}}
1918 \local@advisorstring: &
1919 \count1=0
1920 \toks@={}
1921 \@whilenum \count1<\@advisor \do{%
1922 \ifcase\count1 % same as \ifnum0=\count1
1923 \toks@=\expandafter{\csname CoppeAdvisorName:\the\count1%
1924 \expandafter\endcsname\expandafter\space%
1925 \csname CoppeAdvisorSurname:\the\count1\endcsname\\}
1926 \else
1927 \toks@=\expandafter\expandafter\expandafter{%
1928 \expandafter\the\expandafter\toks@
1929 \expandafter&\expandafter\space
1930 \csname CoppeAdvisorName:\the\count1\expandafter\endcsname%
1931 \expandafter\space\csname CoppeAdvisorSurname:\the\count1\endcsname\\}
1932 }%
1933 \fi
1934 \advance\count1 by 1}
1935 \the\toks@
1936 \end{tabularx}\par
1937 \vspace*{2mm}
1938 \noindent\local@deptstring: \local@deptname\par
1939 \vspace*{7mm}}{\vspace*{\fill}}

```

foreignabstract (*env.*)

```

1940 \newenvironment{foreignabstract}{%
1941 \cleardoublepage
1942 \thispagestyle{plain}
1943 \begin{otherlanguage}{\coppe@foreignlang}
1944 \foreignabstract@toptext\par
1945 \vspace*{8.6mm}
1946 \begin{center}
1947 \sloppy\nohyphens{\MakeUppercase\foreign@title}\par
1948 \vspace*{13.2mm}
1949 \@authname\ \@authsur \par
1950 \vspace*{7mm}
1951 \foreign@monthname/\coppe@docyear
1952 \end{center}\par
1953 \vspace*{1.5cm}
1954 \noindent%
1955 \begin{tabularx}{\textwidth}[b]{@{}l@{ }>{\raggedright\arraybackslash}X@{}}
1956 \foreign@advisorstring: &
1957 \count1=0

```

```

1958 \toks@={}
1959 \@whilenum \count1<\@advisor \do{%
1960 \ifcase\count1 % same as \ifnum0=\count1
1961 \toks@=\expandafter{\csname CoppeAdvisorName:\the\count1%
1962 \expandafter\endcsname\expandafter\space%
1963 \csname CoppeAdvisorSurname:\the\count1\endcsname\\}
1964 \else
1965 \toks@=\expandafter\expandafter\expandafter{%
1966 \expandafter\the\expandafter\toks@
1967 \expandafter&\expandafter\space
1968 \csname CoppeAdvisorName:\the\count1\expandafter\endcsname%
1969 \expandafter\space\csname CoppeAdvisorSurname:\the\count1\endcsname\\
1970 }%
1971 \fi
1972 \advance\count1 by 1}
1973 \the\toks@
1974 \end{tabularx}\par
1975 \vspace*{2mm}
1976 \noindent\foreign@deptstring: \foreign@deptname\par
1977 \vspace*{7mm}}{%
1978 \end{otherlanguage}
1979 \vspace*{\fill}
1980 \global\let\@author\@empty
1981 \global\let\@date\@empty
1982 \global\let\foreign@title\@empty
1983 \global\let\foreign@title\relax
1984 \global\let\local@title\@empty
1985 \global\let\local@title\relax
1986 \global\let\author\relax
1987 \global\let\author\relax
1988 \global\let\date\relax}

```

brazilianabstract (*env.*) Optional third abstract for non-pt main languages: a Brazilian Portuguese abstract printed in addition to **abstract** (main language) and **foreignabstract** (English by default). Wraps its body in `\begin{otherlanguage}{brazilian}` so the babel typography (hyphenation, quotation marks) is correct regardless of the current main language. Uses the same `\local@...` Portuguese strings the **abstract** env uses today. Meaningful only when main is neither **brazilian** nor **english**; harmless otherwise (the document just won't include it).

```

1989 \newenvironment{brazilianabstract}{%
1990 \cleardoublepage
1991 \thispagestyle{plain}
1992 \begin{otherlanguage}{brazilian}
1993 \abstract@toptext\par
1994 \vspace*{8.6mm}
1995 \begin{center}
1996 \sloppy\nohyphens{\MakeUppercase\local@title}\par
1997 \vspace*{13.2mm}
1998 \@authname\ \@authsurn \par
1999 \vspace*{7mm}
2000 \local@monthname/\coppe@docyear
2001 \end{center}\par
2002 \vspace*{1.5cm}

```

```

2003 \noindent%
2004 \begin{tabularx}{\textwidth}[b]{@{}l@{ }>{\raggedright\arraybackslash}X@{}}
2005 \local@advisorstring: &
2006 \count1=0
2007 \toks@={}
2008 \@whilenum \count1<\@advisor \do{%
2009 \ifcase\count1
2010 \toks@=\expandafter{\csname CoppeAdvisorName:\the\count1%
2011 \expandafter\endcsname\expandafter\space%
2012 \csname CoppeAdvisorSurname:\the\count1\endcsname\\}
2013 \else
2014 \toks@=\expandafter\expandafter\expandafter{%
2015 \expandafter\the\expandafter\toks@
2016 \expandafter&\expandafter\space
2017 \csname CoppeAdvisorName:\the\count1\expandafter\endcsname%
2018 \expandafter\space\csname CoppeAdvisorSurname:\the\count1\endcsname\\
2019 }%
2020 \fi
2021 \advance\count1 by 1}
2022 \the\toks@
2023 \end{tabularx}\par
2024 \vspace*{2mm}
2025 \noindent\local@deptstring: \local@deptname\par
2026 \vspace*{7mm}}{%
2027 \end{otherlanguage}
2028 \vspace*{\fill}}

```

\listoffigures

```

2029 \renewcommand\listoffigures{%
2030 \coppe@hasLof
2031 \if@twocolumn
2032 \@restonecoltrue\onecolumn
2033 \else
2034 \@restonecolfalse
2035 \fi
2036 \chapter*{\listfigurename}%
2037 \addcontentsline{toc}{chapter}{\listfigurename}%
2038 \@mkboth{\MakeUppercase\listfigurename}%
2039 {\MakeUppercase\listfigurename}%
2040 \@starttoc{lof}%
2041 \if@restonecol\twocolumn\fi
2042 }

```

\listoftables

```

2043 \renewcommand\listoftables{%
2044 \if@twocolumn
2045 \@restonecoltrue\onecolumn
2046 \else
2047 \@restonecolfalse
2048 \fi
2049 \chapter*{\listtablename}%
2050 \addcontentsline{toc}{chapter}{\listtablename}%
2051 \@mkboth{%
2052 \MakeUppercase\listtablename}%

```

```

2053         {\MakeUppercase\listtablename}%
2054     \@starttoc{lot}%
2055     \if@restonecol\twocolumn\fi
2056 }

```

\printlosymbols

```

2057 \newcommand\printlosymbols{%
2058 \renewcommand\glossaryname{\listsymbolname}%
2059 \@input@{\jobname.los}}

```

\makelosymbols

```

2060 \def\makelosymbols{%
2061   \newwrite\@losfile
2062   \immediate\openout\@losfile=\jobname.syx
2063   \newcommand\syml[3][\@bsphack\begin group
2064     \ifstrempy{##1}{\def\@tempsymb1{##2=}}{\def\@tempsymb1{##1=}}%
2065     \@sanitize%
2066     \@wrlos{\@tempsymb1}{##2}{##3}\typeout%
2067     {Writing index of symbols file \jobname.syx}%
2068     \let\makelosymbols\@empty%
2069 }%
2070 \@onlypreamble\makelosymbols

2071 \AtBeginDocument{%
2072 \ifpackage loaded{hyperref}{%
2073   \newcommand\@wrlos[3]{%
2074     \protected@write\@losfile{%
2075       {\string\indexentry{#1[#2] #3|hyperpage}{\thepage}}%
2076     \end group%
2077     \@esphack}}{%
2078   \newcommand\@wrlos[3]{%
2079     \protected@write\@losfile{%
2080       {\string\indexentry{#1[#2] #3}{\thepage}}%
2081     \end group%
2082     \@esphack}}}%

```

\printloabbreviations

```

2083 \newcommand\printloabbreviations{%
2084 \renewcommand\glossaryname{\listabbreviationname}%
2085 \@input@{\jobname.lab}}

```

\makeloabbreviations

```

2086 \def\makeloabbreviations{%
2087   \newwrite\@labfile
2088   \immediate\openout\@labfile=\jobname.abx
2089   \newcommand\abbrev[3][\@bsphack\begin group
2090     \ifstrempy{##1}{\def\@tempsymb1{##2=}}{\def\@tempsymb1{##1=}}
2091     \@sanitize
2092     \@wrlab{\@tempsymb1}{##2}{##3}\typeout
2093     {Writing index of abbreviations file \jobname.abx}%
2094     \let\makeloabbreviations\@empty
2095 }
2096 \@onlypreamble\makeloabbreviations

```

```

2097 \AtBeginDocument{%
2098 \@ifpackageloaded{hyperref}{%
2099   \newcommand\wrlab[3]{%
2100     \protected@write\@labfile{%
2101       {\string\indexentry{#1[#2] #3|hyperpage}{\thepage}}%
2102     \endgroup%
2103     \@esphack}}{%
2104   \newcommand\wrlab[3]{%
2105     \protected@write\@labfile{%
2106       {\string\indexentry{#1[#2] #3}{\thepage}}%
2107     \endgroup%
2108     \@esphack}}}%

\newsigla Acronyms (siglas), manual2024 2.8 / R43. \newsigla{key}{SIGLA}{full form}
\sigla declares one; \sigla{key} prints “full form (SIGLA)” on first use and registers it in
the list of abbreviations and acronyms (through \abbrev, so \makeloabbreviations
must be active), then just “SIGLA” on later uses. \sigla*{key} forces the full
form. The first-use flag is set globally so it survives groups (floats, footnotes).

2109 \newcommand\newsigla[3]{%
2110   \@namedef{coppe@sigla@s#1}{#2}%
2111   \@namedef{coppe@sigla@l#1}{#3}%
2112 \newcommand\coppe@sigla@reg[1]{%
2113   \@ifundefined{abbrev}{}%
2114   {\abbrev{\@nameuse{coppe@sigla@s#1}}{\@nameuse{coppe@sigla@l#1}}}%
2115 \newcommand\coppe@sigla@full[1]{%
2116   \@nameuse{coppe@sigla@l#1}\space(\@nameuse{coppe@sigla@s#1})}
2117 \newcommand\coppe@sigla@undef[1]{%
2118   \PackageWarning{coppe}{Sigla '#1' nao declarada com \string\newsigla}%
2119   \textbf{???#1???}}
2120 \newcommand\coppe@sigla@first[1]{%
2121   \global\@namedef{coppe@sigla@done#1}{\coppe@sigla@reg{#1}\coppe@sigla@full{#1}}
2122 \newcommand\coppe@sigla@norm[1]{%
2123   \@ifundefined{coppe@sigla@s#1}{\coppe@sigla@undef{#1}}%
2124   {\@ifundefined{coppe@sigla@done#1}{\coppe@sigla@first{#1}}%
2125     {\@nameuse{coppe@sigla@s#1}}}%
2126 \newcommand\coppe@sigla@star[1]{%
2127   \@ifundefined{coppe@sigla@s#1}{\coppe@sigla@undef{#1}}%
2128   {\global\@namedef{coppe@sigla@done#1}{\coppe@sigla@full{#1}}}%
2129 \newcommand\sigla{\@ifstar\coppe@sigla@star\coppe@sigla@norm}

2130 %%% \AtBeginDocument{%
2131 %%% \@ifpackageloaded{hyperref}{%
2132 %%%   \def\wrlab#1#2{%
2133 %%%     \protected@write\@labfile{%
2134 %%%       {\string\indexentry{[#1] #2|hyperpage}{\thepage}}%
2135 %%%     \endgroup
2136 %%%     \@esphack}}{%
2137 %%%   \def\wrlab#1#2{%
2138 %%%     \protected@write\@labfile{%
2139 %%%       {\string\indexentry{[#1] #2}{\arabic{page}}}%
2140 %%%     \endgroup
2141 %%%     \@esphack}}

2142 % Catalog-label writers. Two language-related traps must be avoided here:
2143 % 1. The label NAMES contain ':' and '.', which babel makes active in

```

```

2144 %      some languages (e.g. Brazilian's "shorthand" set). If the write
2145 %      argument is tokenised while one of those is active, pdfTeX aborts
2146 %      with "<inserted text> }\endwrite" because the shorthand expansion
2147 %      injects unbalanced braces. \detokenize emits its argument as
2148 %      catcode-12 characters regardless of the current activation, so the
2149 %      label NAMES survive whatever language is current at writer time.
2150 % 2. Spanish babel redefines \@roman (and therefore \roman{...}) to
2151 %      produce small-caps roman numerals via \es@scroman, whose expansion
2152 %      contains an \uppercase group that is unbalanced inside \write. The
2153 %      same crash signature appears. The fix is to bypass LaTeX's
2154 %      \roman/\arabic counter wrappers entirely and use the raw e-TeX
2155 %      primitives \romannumeral and \number directly on the \c@page count
2156 %      register --- both are fully expandable, language-agnostic, and
2157 %      always produce plain ASCII digits or lowercase roman letters.
2158 %      \Roman{page} is left in place because babel-spanish only redefines
2159 %      \@Roman under the (non-default) ucroman option.
2160 \AtBeginDocument{%
2161   \ltx@ifpackageloaded{hyperref}{%
2162     \def\coppe@bibEnd{%
2163       \immediate\write\@auxout{%
2164         \string\newlabel{\detokenize{bib:end}}{%
2165           {}{\number\c@page}}{\detokenize{page.}\number\c@page}}}%
2166     \def\coppe@bibBegin{%
2167       \immediate\write\@auxout{%
2168         \string\newlabel{\detokenize{bib:begin}}{%
2169           {}{\number\c@page}}{\detokenize{page.}\number\c@page}}}%
2170     \def\coppe@mainBegin{%
2171       \immediate\write\@auxout{%
2172         \string\newlabel{\detokenize{front:pageno}}{%
2173           {}{\Roman{page}}{\detokenize{page.}\romannumeral\c@page}}}%
2174     \def\coppe@hasLof{%
2175       \immediate\write\@auxout{%
2176         \string\newlabel{\detokenize{cat:lofflag}}{%
2177           {}{\detokenize{:~il.}}{\detokenize{page.}\romannumeral\c@page}}}%
2178   }%
2179   \def\coppe@bibEnd{%
2180     \immediate\write\@auxout{%
2181       \string\newlabel{\detokenize{bib:end}}{\detokenize{page.}}}%
2182   \def\coppe@bibBegin{%
2183     \immediate\write\@auxout{%
2184       \string\newlabel{\detokenize{bib:begin}}{\detokenize{page.}}}%
2185   \def\coppe@mainBegin{%
2186     \immediate\write\@auxout{%
2187       \string\newlabel{\detokenize{front:pageno}}{\detokenize{page.}}}%
2188   \def\coppe@hasLof{%
2189     \immediate\write\@auxout{%
2190       \string\newlabel{\detokenize{cat:lofflag}}{\detokenize{:~il.}}}%
2191   }%
2192 }
2193 % The reference list is now produced by biblatex's \printbibliography (see the
2194 % bibliography setup after babel, above). The former thebibliography
2195 % redefinition and its \bibindent are no longer used and have been removed; the
2196 % bib:begin/bib:end catalog labels are now written by \AtBeginBibliography /
2197 % \AtEndBibliography via \coppe@bibBegin and \coppe@bibEnd, defined just above.

```


longquote (*env.*)

```
2198 \newlength{\recuolongquote}%
2199 \setlength{\recuolongquote}{4cm}%
2200 \newenvironment*{longquote}[1][default]{%
2201   \list{}%
2202   \footnotesize%
2203   \addtolength{\leftskip}{\recuolongquote}%
2204   \item[]%
2205   \singlespacing%
2206   \ifthenelse{\not\equal{#1}{default}}{\itshape\selectlanguage{#1}}{}%
2207 }{\endlist}%

2208 \newenvironment{theglossary}{%
2209   \if@twocolumn%
2210     \@restonecoltrue\onecolumn%
2211   \else%
2212     \@restonecolfalse%
2213   \fi%
2214   \@mkboth{\MakeUppercase\glossaryname}%
2215   {\MakeUppercase\glossaryname}%
2216   \chapter*{\glossaryname}%
2217   \addcontentsline{toc}{chapter}{\glossaryname}
2218   \list{}
2219   {\setlength{\listparindent}{0in}%
2220    \setlength{\labelwidth}{1.0in}%
2221    \setlength{\leftmargin}{1.5in}%
2222    \setlength{\labelsep}{0.5in}%
2223    \setlength{\itemindent}{0in}}%
2224   \sloppy}%
2225   {\if@restonecol\twocolumn\fi%
2226 \endlist}
2227 %
2228 \renewenvironment{theindex}{%
2229   \if@twocolumn
2230     \@restonecolfalse
2231   \else
2232     \@restonecoltrue
2233   \fi
2234   \twocolumn[\@makeschapterhead{\indexname}]%
2235   \@mkboth{\MakeUppercase\indexname}%
2236   {\MakeUppercase\indexname}%
2237   \thispagestyle{plain}\parindent\z@
2238   \addcontentsline{toc}{chapter}{\indexname}
2239   \parskip\z@ \@plus .3\p@\relax
2240   \columnseprule \z@
2241   \columnsep 35\p@
2242   \let\item\@idxitem}
2243   {\if@restonecol\onecolumn\else\clearpage\fi}
2244 \newcommand\listabbreviationname{\coppemainstring{listabbreviation}}
2245 \newcommand\listsymbolname{\coppemainstring{listsymbol}}
2246 \newcommand\glossaryname{\coppemainstring{glossary}}
2247 %
2248 \newcommand\local@advisorstring{Orientador}
2249 \newcommand\foreign@advisorstring{Advisor}
```

```

2250 \ifthenelse{\boolean{maledoc}}{%
2251 \newcommand\local@approvedname{Examinado por}%
2252 }{%
2253 \newcommand\local@approvedname{Examinada por}%
2254 }
2255 \newcommand\local@universityname{Universidade Federal do Rio de Janeiro}
2256 \newcommand\local@deptstring{Programa}
2257 \newcommand\foreign@deptstring{Department}
2258 \newcommand\local@cityname{Rio de Janeiro}
2259 \newcommand\local@statename{RJ}
2260 \newcommand\local@countryname{Brasil}
2261 %
2262 \newcommand\frontcover@maintext{
2263 \sloppy\nohyphens{\local@doctype\ de \@degree\
2264 \ifthenelse{\boolean{maledoc}}{apresentado}{apresentada}
2265 ao Programa de P{\ ' o}s-gradua{\c c}{\~ a}o em \local@deptname,
2266 COPPE, da \local@universityname, como parte dos requisitos
2267 necess{\ ' a}rios {\ ' a} obten{\c c}{\~ a}o do t{\ ' i}tulo de
2268 \local@degname\ em \local@deptname.}
2269 }
2270 %
2271 \newcommand\frontpage@maintext{
2272 \noindent {\MakeUppercase\local@doctype}
2273 \ifthenelse{\boolean{maledoc}}{SUBMETIDO}{SUBMETIDA}
2274 \sloppy\nohyphens{AO CORPO DOCENTE DO INSTITUTO ALBERTO LUIZ COIMBRA
2275 DE P{\ ' O}S-GRADUA{\c C}{\~ A}O E PESQUISA DE ENGENHARIA DA
2276 UNIVERSIDADE FEDERAL DO RIO DE JANEIRO COMO PARTE DOS REQUISITOS
2277 NECESS{\ ' A}RIOS PARA A OBTEN{\c C}{\~ A}O DO GRAU DE
2278 {\MakeUppercase\local@degname} EM CI{\^E}NCIAS EM
2279 {\MakeUppercase\local@deptname.\par}}%
2280 }
2281 %
2282 \newcommand\frontpage@bottomtext{%
2283 \begin{center}
2284 {\MakeUppercase{\local@cityname, \local@statename\ -- \local@countryname}}\par
2285 {\MakeUppercase\local@monthname\ DE \coppe@docyear}
2286 \end{center}}%
2287 }
2288 %
2289 \newcommand\abstract@toptext{%
2290 \noindent Resumo \ifthenelse{\boolean{maledoc}}{do}{da}
2291 \local@doctype\ \ifthenelse{\boolean{maledoc}}{apresentado}{apresentada}
2292 \sloppy\nohyphens{\ ' a} COPPE/UFRJ como parte dos requisitos
2293 necess{\ ' a}rios para a obten{\c c}{\~ a}o do grau de
2294 \local@degname\ em Ci{\^e}ncias (\@degree)}
2295 }
2296 \newcommand\foreignabstract@toptext{%
2297 \noindent \sloppy\nohyphens{Abstract of \foreign@doctype\ presented to
2298 COPPE/UFRJ as a partial fulfillment of the requirements for the
2299 degree of \foreign@degname\ of Science (\@degree)}
2300 }
2301 %

```

Late-load packages: setspace, hyperref, biblatex

These three packages *must* be the last things the class loads, and in this exact order:

- **setspace** – supplies `\singlespacing`, `\onehalfspacing` and friends. It is loaded here so the line-spacing state it carries around is settled before **hyperref** patches all the cross-reference and TOC plumbing.
- **hyperref** – patches an enormous number of LaTeX internals (`\section`, `\label`, `\ref`, `\bibcite`, list-of-* commands, ...). Loading it as late as possible is the standard rule of thumb so it sees the final version of those macros, including the ones the **coppe** class redefines (e.g. `\@chapter`, `\listofquadros`).
- **biblatex** – must come after **hyperref** so its citation links and back-references hook into **hyperref** correctly; otherwise the `\cite` commands print as plain (non-clickable) text.

```
2302 \RequirePackage{setspace}
2303 % Apply the default line spacing now that setspace is available. The flag was
2304 % set above by \DeclareOption{doublespacing}; absent the option, one-and-a-half.
2305 \if@coppedoublespace\doublespacing\else\onehalfspacing\fi
2306 \RequirePackage{hyperref}
2307 % Bibliography engine: the full coppe BibLaTeX/biber ABNT style, replacing the
2308 % former natbib+bibtex setup. The numbers class option selects the numeric
2309 % citation style; otherwise the author-date default is used. natbib=true keeps
2310 % \citel/\citep available for users migrating from natbib.
2311 \if@coppenumeric
2312   \RequirePackage[backend=biber,datamodel=coppe,%
2313     bibstyle=coppe-numeric,citestyle=coppe-numeric,natbib=true]{biblatex}
2314 \else
2315   \RequirePackage[backend=biber,datamodel=coppe,%
2316     bibstyle=coppe,citestyle=coppe,natbib=true]{biblatex}
2317 \fi
2318 % Reference-list heading "Referencias" (pt) / "References" (en) -- manual2024
2319 % 3.1.4.1 requires the post-textual list to be titled "Referencias", not
2320 % "Bibliografia" (babel's default \bibname). The title is rendered through
2321 % \coppestring so it follows the current main language in BOTH the chapter
2322 % heading and the .toc entry (biblatex's \bibstring proved unreliable across
2323 % the .toc write, and babel's \bibname caption override was equally fragile).
2324 \defbibheading{bibliography}{%
2325   \chapter*{\coppestring{references}}%
2326   \addcontentsline{toc}{chapter}{\coppestring{references}}}
2327 % The catalog page range (bib:begin/bib:end) was written by the old
2328 % thebibliography redefinition, which biblatex's \printbibliography does not
2329 % use. biblatex offers \AtBeginBibliography but no end counterpart, so the
2330 % begin label uses that hook and the end label is emitted by wrapping
2331 % \printbibliography to write it just after the reference list.
2332 \AtBeginBibliography{\coppe@bibBegin}
2333 \let\coppe@orig@printbibliography\printbibliography
2334 \renewcommand\printbibliography[1][]{%
2335   \coppe@orig@printbibliography[#1]%
2336   \coppe@bibEnd}
2337 \</class>
```

```

2338 <*glossary>
2339 actual '='
2340 quote '!'
2341 level '>'
2342 %%%% delim_0 " , p. "
2343 delim_0 "\\dotfill "
2344 lethead_flag 0
2345 headings_flag 0
2346 preamble
2347 "\n\\begin{theglossary}\n \\makeatletter"
2348 postamble
2349 "\n \\end{theglossary}\n"
2350 </glossary>

2351 <*class>
2352 \newcommand{\annex}{
2353     \renewcommand{\appendixname}{\coppestring{annex}}%
2354     \appendix
2355     \@coppeannextrue
2356 }
2357 </class>

```

BibLaTeX style files

Generated from this .dtx by coppe.ins (one docstrip module each).

```

2358 <*dbx>
2359 \ProvidesFile{coppe.dbx}[2026/05/28 v4.0 CoppeTeX ABNT datamodel]
2360 % Custom entry types beyond biblatex's standard datamodel (games, TV, maps).
2361 % map = cartographic document (NBR 6023 4.2.12); biblatex has no native @map.
2362 % The remaining ABNT "exotic" types (legislation, jurisdiction, legal, patent,
2363 % standard, music, audio, image, artwork, video, dataset, letter, periodical,
2364 % performance) are already in biblatex's standard datamodel.
2365 \DeclareDatamodelEntrytypes{game,videogame,boardgame,tvshow,map}
2366 % Custom fields.
2367 \DeclareDatamodelFields[type=field,datatype=literal]{course} % "(... em <course>)"
2368 \DeclareDatamodelFields[type=field,datatype=literal]{coppedegree} % degree word
2369 \DeclareDatamodelFields[type=field,datatype=literal]{platform} % videogame platform
2370 \DeclareDatamodelFields[type=field,datatype=literal]{scale} % map scale (escala)
2371 \DeclareDatamodelFields[type=list,datatype=literal]{artist} % game/illustration art
2372 \DeclareDatamodelFields[type=list,datatype=literal]{director} % movie/tvshow/video
2373 \DeclareDatamodelFields[type=list,datatype=literal]{producer}
2374 \DeclareDatamodelFields[type=field,datatype=literal]{caderno} % newspaper section
2375 \DeclareDatamodelFields[type=field,datatype=literal]{relator} % jurisprudence rapporteur
2376 \DeclareDatamodelEntryfields{course,coppedegree}
2377 \DeclareDatamodelEntryfields[game,videogame,boardgame,software,movie,tvshow,video]%
2378 {artist,platform,director,producer,version}
2379 \DeclareDatamodelEntryfields[map]{scale}
2380 \DeclareDatamodelEntryfields[article]{caderno}
2381 \DeclareDatamodelEntryfields[jurisdiction]{relator}
2382 </dbx>
2383 <*bbx>
2384 \ProvidesFile{coppe.bbx}[2026/05/28 v4.0 CoppeTeX ABNT bibliography style]
2385 % The reference list is single-spaced (ABNT 2.4) via \singlespacing in
2386 % \AtBeginBibliography. coppe.cls already loads setspace, but the style must work

```

```

2387 % standalone too (e.g. \usepackage[bibstyle=coppe]{biblatex} in a plain article),
2388 % so load setspace here as well (idempotent).
2389 \RequirePackage{setspace}
2390 % ABNT reference list keeps the date at the end (imprenta), same for author-date
2391 % and numeric modes, so we derive from the standard style, not authoryear.
2392 \RequireBibliographyStyle{standard}
2393
2394 % Note: standard style never dashes repeated authors, which already matches the
2395 % post-2020 ABNT rule (repeat the entry in full), so no 'dashed' option is needed.
2396 \ExecuteBibliographyOptions{%
2397   maxbibnames=99,      % ABNT 4.3.1.3: list all authors by default
2398   giveninits=false,
2399   uniquename=false,
2400   uniquelist=false,
2401   labeldateparts=true, % needed for author-date extradata (1999a/1999b)
2402   sorting=nyt,
2403 }
2404
2405 % Language strings (Disponível em / Acesso em / In, etc.)
2406 \DeclareLanguageMapping{brazilian}{brazilian-coppe}
2407 \DeclareLanguageMapping{english}{english-coppe}
2408
2409 % --- Names: all reversed "FAMILY, Given", separated by ";" in the list ---
2410 \DeclareNameAlias{author}{family-given}
2411 \DeclareNameAlias{editor}{family-given}
2412 \DeclareNameAlias{translator}{family-given}
2413 \DeclareNameAlias{default}{family-given}
2414 \DeclareDelimFormat[bib,biblist]{multinamedelim}{\addsemicolon\space}
2415 \DeclareDelimFormat[bib,biblist]{finalnamedelim}{\addsemicolon\space}
2416
2417 % ABNT (NBR 6023 8.1.1.4): a collection entered under the name of its
2418 % organizer prints the kind of responsibility in parentheses after the
2419 % name, abbreviated, lowercase and in the singular even for several
2420 % organizers -- e.g. "SOUSA, A. M. de; OLIVEIRA, L. da C. (org.)". The
2421 % standard style prints ", Org." (comma, no parentheses) when the editor
2422 % takes the author slot, so override editor+others to wrap the role string
2423 % in "(...)". The host position ("In: NAME (org.)") already parentheseses
2424 % the role via coppe:bookeditor below. The role word itself is the
2425 % lowercase, singular "org." (and "ed.", "cur.", ...) set in the language
2426 % packs (brazilian-coppe.lbx et al.).
2427 \renewbibmacro*{editor+others}{%
2428   \ifboolexpr{ test \ifuseeditor and not test {\ifnameundef{editor}} }
2429     {\printnames{editor}%
2430      \setunit*{\addspace}%
2431      \printtext{\mkbibparens{\usebibmacro{editor+othersstrg}}}%
2432      \clearname{editor}}
2433   {}
2434 % Same parenthesised role for drivers that take the editor through the plain
2435 % editor macro instead of editor+others (e.g. @manual).
2436 \renewbibmacro*{editor}{%
2437   \ifboolexpr{ test \ifuseeditor and not test {\ifnameundef{editor}} }
2438     {\printnames{editor}%
2439      \setunit*{\addspace}%
2440      \printtext{\mkbibparens{\usebibmacro{editorstrg}}}%

```

```

2441 \clearname{editor}}
2442 {}
2443
2444 % Uppercase the surname (family + prefix) only inside the bibliography list;
2445 % citations keep normal caps (post-2020 rule).
2446 % ABNT date (NBR 6023 §4.3.5.5.2): "dia mês. ano" -- without the brazilian "de"
2447 % between the parts (e.g. "4 fev. 1992", "abr. 1996"). #1=year, #2=month, #3=day
2448 % field names; the month stays abbreviated via \mkbibmonth.
2449 \protected\def\coppe@abntdate#1#2#3{%
2450 \iffieldundef{#3}{\stripzeros{\thefield{#3}}\nobreakspace}%
2451 \iffieldundef{#2}{\mkbibmonth{\thefield{#2}}\iffieldundef{#1}{\space}}%
2452 \iffieldundef{#1}{\stripzeros{\thefield{#1}}}%
2453 \AtBeginBibliography{%
2454 \singlespacing % entries single-spaced internally (2.4)
2455 \setlength{\bibitemsep}{\baselineskip}% one blank line between entries (2.4)
2456 \renewcommand*{\mkbibnamefamily}[1]{\MakeUppercase{#1}}%
2457 \renewcommand*{\mkbibnameprefix}[1]{\MakeUppercase{#1}}%
2458 \let\mkbibdatelong\coppe@abntdate % ABNT date format (drop the "de")
2459 \let\mkbibdateshort\coppe@abntdate
2460 }
2461
2462 % --- Titles: bold for whole-document types; plain for parts; entry-by-title
2463 % (no author/editor) -> not bold, first word in CAPS (ABNT 4.2). ---
2464 % \coppeUCtitle uppercases the first SIGNIFICANT word of the title (ABNT 4.2).
2465 % If the title starts with an article (o/a/os/as/um/uma/uns/umas, the/an), the
2466 % article AND the next word are uppercased; otherwise just the first word. The
2467 % delimited macros split off words; \coppeUCkeep strips the trailing-space
2468 % sentinel so no spurious space is left before the following punctuation.
2469 \protected\def\coppeUCtitle#1{\coppeUCtitleA#1 \coppeUCstop}
2470 \protected\def\coppeUCtitleA#1 #2\coppeUCstop{%
2471 \coppeUCifart{#1}
2472 {\MakeUppercase{#1}\ifblank{#2}{\space\coppeUCtitleB#2\coppeUCstop}}%
2473 {\MakeUppercase{#1}\ifblank{#2}{\space\coppeUCkeep#2\coppeUCstop}}%
2474 \protected\def\coppeUCtitleB#1 #2\coppeUCstop{%
2475 \MakeUppercase{#1}\ifblank{#2}{\space\coppeUCkeep#2\coppeUCstop}}
2476 \protected\def\coppeUCkeep#1 \coppeUCstop{#1}
2477 % \coppeUCifart{word}{if-article}{if-not}: case-insensitive membership test.
2478 \protected\def\coppeUCifart#1{%
2479 \begingroup
2480 \lowercase{\def\coppeUC@w{#1}}%
2481 \edef\coppeUC@w{\coppeUC@w}%
2482 \expandafter\in@\expandafter{\coppeUC@w}{a,o,as,os,um,uma,uns,umas,the,an,}%
2483 \ifin@\aftergroup\@firstoftwo\else\aftergroup\@secondoftwo\fi
2484 \endgroup}
2485 \DeclareFieldFormat{title}{%
2486 \ifboolexpr{ test {\ifnameundef{labelname}} }
2487 {#1}%
2488 {\mkbibbold{#1}}}
2489 % First word of the title in CAPS for entry-by-title (ABNT 4.2). The titlecase
2490 % format receives the RAW title text (the title format above only sees the
2491 % field-printing commands), so \coppeUCtitle can split off and uppercase the
2492 % first word here.
2493 \DeclareFieldFormat{titlecase}{%
2494 \ifboolexpr{ test {\ifnameundef{labelname}} }

```

```

2495     {\coppeUCtitle{#1}}%
2496     {#1}}
2497 \DeclareFieldFormat[article,incollection,inbook,inproceedings,suppperiodical,review]{title}{
2498 \DeclareFieldFormat[thesis]{title}{\mkbibbold{#1}}
2499 % biblatex's standard style quotes patent (and unpublished) titles; ABNT bolds
2500 % the title of a whole document, so override those back to bold.
2501 \DeclareFieldFormat[patent]{title}{\mkbibbold{#1}}
2502 % Legal documents (NBR 6023 4.2.6): the law's identification/ementa is NOT
2503 % highlighted; the entry leads with the jurisdiction (entity author, CAPS).
2504 \DeclareFieldFormat[legislation,jurisdiction,legal]{title}{#1}
2505 \DeclareFieldFormat[journaltitle]{\mkbibbold{#1}}
2506 \DeclareFieldFormat{booktitle}{\mkbibbold{#1}}
2507 \DeclareFieldFormat{maintitle}{\mkbibbold{#1}}
2508
2509 % --- Pages: ABNT uses "p." (also for ranges); theses use "f." for pagetotal ---
2510 \DeclareFieldFormat{pagetotal}{#1\addnbspace p\adddot}
2511 \DeclareFieldFormat[thesis]{pagetotal}{#1\addnbspace f\adddot}
2512 \DeclareFieldFormat{pages}{p\adddot\addnbspace#1}
2513
2514 % --- Periodical volume / number prefixes ---
2515 \DeclareFieldFormat[article]{volume}{v\adddot\addnbspace#1}
2516 \DeclareFieldFormat[article]{number}{n\adddot\addnbspace#1}
2517
2518 % --- "In:" before host title (incollection/inproceedings/inbook) ---
2519 \renewbibmacro*{in:}{\printtext{\bibstring{in}\addcolon\space}}
2520
2521 % --- Online: "Disponível em: <url>." + "Acesso em: <date>." via lbx strings ---
2522 \DeclareFieldFormat{url}{\bibstring{availablefrom}\addcolon\space\url{#1}}
2523
2524 % --- ABNT article driver: no "In:"; Journal, Local, v., n., p., date, DOI/URL.
2525 % Newspapers (entrysubtype=newspaper, set by the @artigodejornal source map)
2526 % differ: the newspaper name (journaltitle) and city, then either
2527 % "<date>. <caderno>, p. <pages>" (when a caderno/section is given), or
2528 % "p. <pages>, <date>" (page before date, when none),
2529 % per NBR 6023 §4.2.3.6. ---
2530 \DeclareBibliographyDriver{article}{%
2531 \usebibmacro{bibindex}%
2532 \usebibmacro{begentry}%
2533 \usebibmacro{author/translator+others}%
2534 \setunit{\labelnamepunct}\newblock
2535 \usebibmacro{title}%
2536 \newunit\newblock
2537 \printfield{journaltitle}%
2538 \setunit{\addcomma\space}\printlist{location}%
2539 \iffieldequalstr{entrysubtype}{newspaper}
2540 {\iffieldequalstr{caderno}
2541 {\setunit{\addcomma\space}\printfield{pages}%
2542 \setunit{\addcomma\space}\printdate}%
2543 {\setunit{\addcomma\space}\printdate%
2544 \setunit{\adddot\space}\printfield{caderno}%
2545 \setunit{\addcomma\space}\printfield{pages}}}%
2546 {\setunit{\addcomma\space}\printfield{volume}%
2547 \setunit{\addcomma\space}\printfield{number}%
2548 \setunit{\addcomma\space}\printfield{pages}%

```

```

2549 \setunit{\addcomma\space}\printdate}%
2550 \newunit\newblock
2551 \printfield{doi}%
2552 \newunit\newblock
2553 \usebibmacro{url+urldate}%
2554 \usebibmacro{finentry}}
2555
2556 % --- incollection: ABNT puts the host organizer before the booktitle, as
2557 % "AUTOR. Parte. In: ORGANIZADOR (Org.). Booktitle. Local: Editora, ano,
2558 % p. X-Y." The organizer role string is "Org./Orgs." (see the .lbx). ---
2559 \newbibmacro*{coppe:bookeditor}{%
2560 \ifnameundef{editor}
2561 {}
2562 {\printnames{editor}%
2563 \setunit*{\addspace}\printtext{\mkbibparens{\usebibmacro{editorstrg}}}%
2564 \clearname{editor}%
2565 \setunit{\adddot\space}}}
2566 \newbibmacro*{coppe:partdriver}{%
2567 \usebibmacro{bibindex}\usebibmacro{begentry}%
2568 \usebibmacro{author/translator+others}%
2569 \setunit{\labelnamepunct}\newblock
2570 \usebibmacro{title}%
2571 \newunit\newblock
2572 \usebibmacro{in:}%
2573 \usebibmacro{coppe:bookeditor}%
2574 \printfield{booktitle}%
2575 \newunit\newblock
2576 \printfield{edition}%
2577 \newunit\newblock
2578 \printlist{location}%
2579 \setunit*{\addcolon\space}\printlist{publisher}%
2580 \setunit*{\addcomma\space}\printdate%
2581 \newunit\newblock
2582 \printfield{pages}%
2583 \setunit{\addcomma\space}\printfield{chapter}%
2584 \newunit\newblock
2585 \usebibmacro{url+urldate}%
2586 \usebibmacro{finentry}}
2587 \DeclareBibliographyDriver{incollection}{\usebibmacro{coppe:partdriver}}
2588 \DeclareBibliographyDriver{inbook}{\usebibmacro{coppe:partdriver}}
2589
2590 % --- type field: resolve a known bibliography string (babel-aware), else literal ---
2591 \DeclareFieldFormat{type}{\ifbibstring{#1}{\bibstring{#1}}{#1}}
2592
2593 % --- Thesis: "<type> (em <course>)" using the custom course field (ABNT). ---
2594 \DeclareFieldFormat{course}{\mkbibparens{\coppestring{in:}\addspace#1}}
2595 \DeclareBibliographyDriver{thesis}{%
2596 \usebibmacro{bibindex}\usebibmacro{begentry}%
2597 \usebibmacro{author}%
2598 \setunit{\labelnamepunct}\newblock
2599 \usebibmacro{title}%
2600 \newunit\newblock
2601 \printfield{type}\setunit*{\addspace}\printfield{course}%
2602 \newunit\newblock

```



```

2603 \printlist{institution}%
2604 \setunit*{\addcomma\space}\printlist{location}%
2605 \setunit*{\addcomma\space}\printdate%
2606 \newunit\newblock
2607 \printfield{pagetotal}%
2608 \newunit\newblock
2609 \printfield{note}%
2610 \newunit\newblock
2611 \usebibmacro{url+urldate}%
2612 \usebibmacro{finentry}}
2613 % --- Games / audiovisual: render the custom fields (artist, platform,
2614 % director, producer, version) the generic misc driver ignores. ---
2615 \newbibmacro*{coppe:media}{%
2616 \iflistundef{artist}{\setunit{\addperiod\space}%
2617 \printtext{\coppestring{art}\addcolon\space}\printlist{artist}}%
2618 \iffieldundef{platform}{\setunit{\addperiod\space}%
2619 \printtext{\coppestring{platform}\addcolon\space}\printfield{platform}}%
2620 \iflistundef{director}{\setunit{\addperiod\space}%
2621 \printtext{\coppestring{directedby}\addcolon\space}\printlist{director}}%
2622 \iflistundef{producer}{\setunit{\addperiod\space}%
2623 \printtext{\coppestring{producedby}\addcolon\space}\printlist{producer}}}%
2624 \newbibmacro*{coppe:mediadriver}{%
2625 \usebibmacro{bibindex}\usebibmacro{begentry}%
2626 \usebibmacro{author/translator+others}%
2627 \setunit{\labelnamepunct}\newblock
2628 \usebibmacro{title}%
2629 \newunit\newblock
2630 \usebibmacro{coppe:media}%
2631 \newunit\newblock
2632 \printlist{publisher}%
2633 \setunit*{\addcomma\space}\printlist{location}%
2634 \setunit*{\addcomma\space}\printdate%
2635 \newunit\newblock
2636 \printfield{version}%
2637 \newunit\newblock
2638 \printfield{note}%
2639 \newunit\newblock
2640 \usebibmacro{url+urldate}%
2641 \usebibmacro{finentry}}
2642 \DeclareBibliographyDriver{game}{\usebibmacro{coppe:mediadriver}}
2643 \DeclareBibliographyDriver{videogame}{\usebibmacro{coppe:mediadriver}}
2644 \DeclareBibliographyDriver{boardgame}{\usebibmacro{coppe:mediadriver}}
2645 \DeclareBibliographyDriver{movie}{\usebibmacro{coppe:mediadriver}}
2646 \DeclareBibliographyDriver{tvshow}{\usebibmacro{coppe:mediadriver}}
2647 % Cartographic document (map, NBR 6023 4.2.12): like a book (author/title/
2648 % imprenta) plus the mandatory scale ("Escala 1:N").
2649 \DeclareFieldFormat{scale}{\bibstring{coppescale}\adnbspace#1}
2650 \DeclareBibliographyDriver{map}{%
2651 \usebibmacro{bibindex}\usebibmacro{begentry}%
2652 \usebibmacro{author/translator+others}%
2653 \setunit{\labelnamepunct}\newblock
2654 \usebibmacro{title}%
2655 \newunit\newblock
2656 \printlist{location}%

```

```

2657 \setunit*{\addcolon\space}\printlist{publisher}%
2658 \setunit*{\addcomma\space}\printdate%
2659 \newunit\newblock
2660 \printfield{scale}%
2661 \newunit\newblock
2662 \usebibmacro{url+urldate}%
2663 \usebibmacro{finentry}}
2664 % Patent (NBR 6023 4.2.5): inventor. \textbf{title}. Depositante: holder. type
2665 % number. date. The holder (depositante) is printed in normal case, not the
2666 % ALL-CAPS family treatment the bibliography gives the leading author.
2667 \newbibmacro*{coppe:depositor}{%
2668 \ifnameundef{holder}
2669 {}
2670 {\printtext{\bibstring{depositor}\addcolon\space}%
2671 \begingroup
2672 \def\mkbibnamefamily##1{##1}%
2673 \def\mkbibnameprefix##1{##1}%
2674 \printnames{holder}%
2675 \endgroup}}
2676 \DeclareBibliographyDriver{patent}{%
2677 \usebibmacro{bibindex}%
2678 \usebibmacro{begentry}%
2679 \usebibmacro{author/translator+others}%
2680 \setunit{\labelnamepunct}\newblock
2681 \usebibmacro{title}%
2682 \newunit\newblock
2683 \usebibmacro{coppe:depositor}%
2684 \newunit\newblock
2685 \printfield{type}%
2686 \setunit*{\addspace}\printfield{number}%
2687 \newunit\newblock
2688 \printdate%
2689 \newunit\newblock
2690 \usebibmacro{url+urldate}%
2691 \usebibmacro{finentry}}
2692 % Exotic ABNT types that biblatex's standard style leaves without a driver:
2693 % give each a baseline (book-like for published docs: norma/partitura; misc for
2694 % the rest) so they print. Per-type ABNT refinements can follow later.
2695 \DeclareBibliographyDriver{standard}{\usedriver{}{book}}
2696 \DeclareBibliographyDriver{music}{\usedriver{}{book}}
2697 % Legal documents (NBR 6023 4.2.6-4.2.8): jurisdiction/entity (CAPS author),
2698 % the law id + ementa (plain, not bold), then location and date.
2699 \newbibmacro*{coppe:legaldriver}{%
2700 \usebibmacro{bibindex}\usebibmacro{begentry}%
2701 \usebibmacro{author/translator+others}%
2702 \setunit{\labelnamepunct}\newblock
2703 \usebibmacro{title}%
2704 \newunit\newblock
2705 % Jurisprudência: "Relator: <nome>." after the decision title (NBR 6023).
2706 \iffieldundef{relator}{\printtext{Relator\addcolon\space}\printfield{relator}\newunit\ne
2707 \printlist{location}%
2708 \setunit{\addcomma\space}\printdate%
2709 \newunit\newblock
2710 \usebibmacro{url+urldate}%

```

```

2711 \usebibmacro{finentry}}
2712 \DeclareBibliographyDriver{legislation}{\usebibmacro{coppe:legaldriver}}
2713 \DeclareBibliographyDriver{jurisdiction}{\usebibmacro{coppe:legaldriver}}
2714 \DeclareBibliographyDriver{legal}{\usebibmacro{coppe:legaldriver}}
2715 \DeclareBibliographyDriver{audio}{\usedriver{}{misc}}
2716 \DeclareBibliographyDriver{video}{\usedriver{}{misc}}
2717 \DeclareBibliographyDriver{software}{\usedriver{}{misc}}
2718 \DeclareBibliographyDriver{image}{\usedriver{}{misc}}
2719 \DeclareBibliographyDriver{artwork}{\usedriver{}{misc}}
2720 \DeclareBibliographyDriver{performance}{\usedriver{}{misc}}
2721 \DeclareBibliographyDriver{dataset}{\usedriver{}{misc}}
2722
2723 % --- biber source map: Portuguese synonyms (entry types + fields) + thesis family ---
2724 \DeclareSourcemap{%
2725   \maps[datatype=bibtex]{%
2726     % entry-type PT synonyms -> canonical English
2727     \map{\step[tourcesource=livro,typetarget=book]}
2728     \map{\step[tourcesource=artigo,typetarget=article]}
2729     \map{\step[tourcesource=artigodejornal,typetarget=article,final]\step[fieldset=entrysubtyp
2730     \map{\step[tourcesource=partedelivro,typetarget=incollection]}
2731     \map{\step[tourcesource=emlivro,typetarget=inbook]}
2732     \map{\step[tourcesource=verbete,typetarget=inreference]}
2733     \map{\step[tourcesource=anais,typetarget=proceedings]}
2734     \map{\step[tourcesource=trabalhodeevento,typetarget=inproceedings]}
2735     \map{\step[tourcesource=periodico,typetarget=periodical]}
2736     \map{\step[tourcesource=relatorio,typetarget=report]}
2737     \map{\step[tourcesource=relatoriotecnico,typetarget=report]}
2738     \map{\step[tourcesource>manual,typetarget>manual]}
2739     \map{\step[tourcesource=correspondencia,typetarget=letter]}
2740     \map{\step[tourcesource=patente,typetarget=patent]}
2741     \map{\step[tourcesource=diverso,typetarget=misc]}
2742     \map{\step[tourcesource=jogo,typetarget=game]}
2743     \map{\step[tourcesource=jogoeletronico,typetarget=videogame]}
2744     \map{\step[tourcesource=jogodetabuleiro,typetarget=boardgame]}
2745     \map{\step[tourcesource=filme,typetarget=movie]}
2746     \map{\step[tourcesource=serietv,typetarget=tvshow]}
2747     \map{\step[tourcesource=programadettv,typetarget=tvshow]}
2748     % Exotic ABNT types (NBR 6023 4.2.5-4.2.14); targets are biblatex-native
2749     % except map (declared in coppe.dbx).
2750     \map{\step[tourcesource=legislacao,typetarget=legislation]}
2751     \map{\step[tourcesource=atonormativo,typetarget=legislation]}
2752     \map{\step[tourcesource=jurisprudencia,typetarget=jurisdiction]}
2753     \map{\step[tourcesource=documentojuridico,typetarget=legal]}
2754     \map{\step[tourcesource=documentocivil,typetarget=legal]}
2755     \map{\step[tourcesource=norma,typetarget=standard]}
2756     \map{\step[tourcesource=partitura,typetarget=music]}
2757     \map{\step[tourcesource=iconografico,typetarget=image]}
2758     \map{\step[tourcesource=obradearte,typetarget=artwork]}
2759     \map{\step[tourcesource=tridimensional,typetarget=artwork]}
2760     \map{\step[tourcesource=cartografico,typetarget=map]}
2761     \map{\step[tourcesource=mapa,typetarget=map]}
2762     \map{\step[tourcesource=audiovisual,typetarget=video]}
2763     \map{\step[tourcesource=conjuntodedados,typetarget=dataset]}
2764     \map{\step[tourcesource=fasciculo,typetarget=supperperiodical]}

```

```

2765 \map{step[typesource=trabalhoapresentado,typetarget=unpublished]}
2766 % thesis family: retype to thesis and preset the (babel-aware) type word
2767 \map{step[typesource=tese,typetarget=thesis]}
2768 \map{step[typesource=dissertacao,typetarget=thesis]}
2769 \map{step[typesource=masterdissertation,typetarget=thesis,final]\step[fieldset=type,fiel
2770 \map{step[typesource=mscdissertation,typetarget=thesis,final]\step[fieldset=type,fieldv
2771 \map{step[typesource=dissertacaomestrado,typetarget=thesis,final]\step[fieldset=type,fi
2772 \map{step[typesource=dscthesis,typetarget=thesis,final]\step[fieldset=type,fieldvalue=d
2773 \map{step[typesource=tesedoutorado,typetarget=thesis,final]\step[fieldset=type,fieldval
2774 \map{step[typesource=phdthesis,typetarget=thesis,final]\step[fieldset=type,fieldvalue=p
2775 \map{step[typesource=tesephd,typetarget=thesis,final]\step[fieldset=type,fieldvalue=phd
2776 \map{step[typesource=tcc,typetarget=thesis,final]\step[fieldset=type,fieldvalue=tcc]}
2777 % field PT synonyms (unaccented) -> canonical English
2778 \map{step[fieldsource=periodico,fieldtarget=journaltitle]}
2779 \map{step[fieldsource=revista,fieldtarget=journaltitle]}
2780 \map{step[fieldsource=jornal,fieldtarget=journaltitle]}
2781 \map{step[fieldsource=titulo,fieldtarget=title]}
2782 \map{step[fieldsource=subtitulo,fieldtarget=subtitle]}
2783 \map{step[fieldsource=autor,fieldtarget=author]}
2784 \map{step[fieldsource=editora,fieldtarget=publisher]}
2785 \map{step[fieldsource=local,fieldtarget=location]}
2786 \map{step[fieldsource=ano,fieldtarget=year]}
2787 \map{step[fieldsource=data,fieldtarget=date]}
2788 \map{step[fieldsource=edicao,fieldtarget=edition]}
2789 \map{step[fieldsource=paginas,fieldtarget=pages]}
2790 \map{step[fieldsource=numero,fieldtarget=number]}
2791 \map{step[fieldsource=tradutor,fieldtarget=translator]}
2792 \map{step[fieldsource=organizador,fieldtarget=editor]}
2793 \map{step[fieldsource=instituicao,fieldtarget=institution]}
2794 \map{step[fieldsource=tipo,fieldtarget=type]}
2795 \map{step[fieldsource=nota,fieldtarget=note]}
2796 \map{step[fieldsource=dataacesso,fieldtarget=urldate]}
2797 \map{step[fieldsource=curso,fieldtarget=course]}
2798 \map{step[fieldsource=grau,fieldtarget=coppedegree]}
2799 \map{step[fieldsource=artista,fieldtarget=artist]}
2800 \map{step[fieldsource=plataforma,fieldtarget=platform]}
2801 \map{step[fieldsource=diretor,fieldtarget=director]}
2802 \map{step[fieldsource=produtor,fieldtarget=producer]}
2803 \map{step[fieldsource=versao,fieldtarget=version]}
2804 \map{step[fieldsource=escala,fieldtarget=scale]}
2805 }%
2806 }
2807
2808 % --- ABNT list layout: flush left, single spacing, blank line between entries ---
2809 \setlength{\bibhang}{0pt}
2810 \setlength{\bibitemsep}{\baselineskip}
2811
2812 </bbx>
2813 <*cbx>
2814 \ProvidesFile{coppe.cbx}[2026/05/28 v4.0 CoppeTeX ABNT author-date citations]
2815 % Author-date is the CoppeTeX 4.0 default. authoryear-comp gives compressed
2816 % author-year cites with normal-caps names (post-2020 rule) out of the box.
2817 \RequireCitationStyle{authoryear-comp}
2818

```

```

2819 % --- NBR 10520:2023 author separator -----
2820 % Parenthetical citations separate authors with ";" -> (Silva; Souza, 2020).
2821 % Narrative citations (\textcite, natbib \citet) keep comma + "e/and" ->
2822 % Silva, Souza e Oliveira (2020). The reference list keeps its own ";" via
2823 % coppe.bbx ([bib,biblist]); these contextless formats cover the cite side.
2824 \DeclareDelimFormat{multinamedelim}{\addsemicolon\space}
2825 \DeclareDelimFormat{finalnamedelim}{\addsemicolon\space}
2826 \DeclareDelimFormat[textcite]{multinamedelim}{\addcomma\space}
2827 \DeclareDelimFormat[textcite]{finalnamedelim}{\addspace\bibstring{and}\space}
2828
2829 % --- apud: citation of a citation (ABNT NBR 6023 §4.1.2.2 / NBR 10520) -----
2830 % ABNT lists only the work actually CONSULTED. So only the consulted work (#4, a
2831 % .bib key) is cited and enters the reference list; the ORIGINAL work (#2 author,
2832 % #3 year) was not consulted and is given as plain text -- it is never a cite key,
2833 % so it is never added to the bibliography.
2834 % \citetapud[<page>]{<orig author>}{<orig year>}{<consulted key>}
2835 % -> Orig Author (year) apud Consulted (yr[, page]) [narrative]
2836 % \citepapud[<page>]{<orig author>}{<orig year>}{<consulted key>}
2837 % -> (Orig Author, year apud CONSULTED, yr[, page]) [parenthetical]
2838 % #1 = optional postnote (page/loc) on the consulted work; "apud" is latin -> italic.
2839 \DeclareRobustCommand{\citetapud}[4][]{%
2840 #2\space(#3)\space\mkbibemph{apud}\space\textcite[#1]{#4}}
2841 \DeclareRobustCommand{\citepapud}[4][]{%
2842 \mkbibparens{#2\addcomma\space#3\space\mkbibemph{apud}\space
2843 \citeauthor{#4}\addcomma\space\citeyear{#4}%
2844 \ifblank{#1}{ }\addcomma\space\mkpageprefix[pagination]{#1}}}%
2845 \cbx}
2846 \*lbxbr}
2847 \ProvidesFile{brazilian-coppe.lbx}[2026/05/28 v4.0 CoppeTeX brazilian strings]
2848 \InheritBibliographyExtras{brazilian}
2849 \NewBibliographyString{availablefrom,mscdiss,dscthesi,tcc,depositor,coppescale}
2850 \DeclareBibliographyStrings{%
2851 inherit = {brazilian},
2852 availablefrom = {{Dispon'\i vel em}{Dispon'\i vel em}},
2853 urlseen = {{Acesso em}{Acesso em}},
2854 in = {{In}{In}},
2855 editor = {{organizador}{org\adddot}},
2856 editors = {{organizadores}{org\adddot}},
2857 depositor = {{Depositante}{Depositante}},
2858 coppescale = {{Escala}{Escala}},
2859 mscdiss = {{Disserta\c{c}\~{a}o de Mestrado}{Disserta\c{c}\~{a}o de Mestrado}},
2860 dscthesi = {{Tese de Doutorado}{Tese de Doutorado}},
2861 phdthesis = {{Tese de Doutorado}{Tese de Doutorado}},
2862 tcc = {{Trabalho de Conclus\~{a}o de Curso}{Trabalho de Conclus\~{a}o de Curso}}
2863 }
2864 \*lbxbr}
2865 \*lbxen}
2866 \ProvidesFile{english-coppe.lbx}[2026/05/28 v4.0 CoppeTeX english strings]
2867 \InheritBibliographyExtras{english}
2868 \NewBibliographyString{availablefrom,mscdiss,dscthesi,tcc,depositor,coppescale}
2869 \DeclareBibliographyStrings{%
2870 inherit = {english},
2871 availablefrom = {{Available from}{Available from}},
2872 urlseen = {{Access on}{Access on}},

```

```

2873 in = {{In}{In}},
2874 editor = {{editor}{ed\addot}},
2875 editors = {{editors}{ed\addot}},
2876 depositor = {{Depositor}{Depositor}},
2877 coppescale = {{Scale}{Scale}},
2878 msdiss = {{M\addot Sc\addot\ Dissertation}{M\addot Sc\addot\ Diss\addot}},
2879 dscthesi s = {{D\addot Sc\addot\ Thesis}{D\addot Sc\addot\ Thesis}},
2880 phdthesis = {{Ph\addot D\addot\ Thesis}{Ph\addot D\addot\ Thesis}},
2881 tcc = {{Undergraduate Thesis}{Undergraduate Thesis}},
2882 }
2883 </lbxen>
2884 <*lbxes>
2885 \ProvidesFile{spanish-coppe.lbx}[2026/05/28 v4.0 CoppeTeX spanish strings]
2886 \InheritBibliographyExtras{spanish}
2887 \NewBibliographyString{availablefrom,msdiss,dscthesi s,tcc,depositor,coppescale}
2888 \DeclareBibliographyStrings{%
2889 inherit = {spanish},
2890 availablefrom = {{Disponible en}{Disponible en}},
2891 urlseen = {{Consultado el}{Consultado el}},
2892 in = {{En}{En}},
2893 editor = {{editor}{ed\addot}},
2894 editors = {{editores}{ed\addot}},
2895 depositor = {{Solicitante}{Solicitante}},
2896 coppescale = {{Escala}{Escala}},
2897 msdiss = {{Tesis de Maestr'\{i\}a}{Tesis de Maestr'\{i\}a}},
2898 dscthesi s = {{Tesis Doctoral}{Tesis Doctoral}},
2899 phdthesis = {{Tesis Doctoral}{Tesis Doctoral}},
2900 tcc = {{Trabajo de Fin de Carrera}{Trabajo de Fin de Carrera}},
2901 }
2902 </lbxes>
2903 <*lbxfr>
2904 \ProvidesFile{french-coppe.lbx}[2026/05/28 v4.0 CoppeTeX french strings]
2905 \InheritBibliographyExtras{french}
2906 \NewBibliographyString{availablefrom,msdiss,dscthesi s,tcc,depositor,coppescale}
2907 \DeclareBibliographyStrings{%
2908 inherit = {french},
2909 availablefrom = {{Disponible \{a\}}{Disponible \{a\}}},
2910 urlseen = {{Consult\{e\} le}{Consult\{e\} le}},
2911 in = {{Dans}{Dans}},
2912 editor = {{\{\'e\}diteur}{\{\'e\}d\addot}},
2913 editors = {{\{\'e\}diteurs}{\{\'e\}d\addot}},
2914 depositor = {{D\{\'e\}posant}{D\{\'e\}posant}},
2915 coppescale = {{\{\'e\}chelle}{\{\'e\}chelle}},
2916 msdiss = {{M\{\'e\}moire de Master}{M\{\'e\}moire de Master}},
2917 dscthesi s = {{Th\{\'e\}se de Doctorat}{Th\{\'e\}se de Doctorat}},
2918 phdthesis = {{Th\{\'e\}se de Doctorat}{Th\{\'e\}se de Doctorat}},
2919 tcc = {{M\{\'e\}moire de fin d\{\'e\}tudes}{M\{\'e\}moire de fin d\{\'e\}tudes}},
2920 }
2921 </lbxfr>
2922 <*lbxit>
2923 \ProvidesFile{italian-coppe.lbx}[2026/05/28 v4.0 CoppeTeX italian strings]
2924 \InheritBibliographyExtras{italian}
2925 \NewBibliographyString{availablefrom,msdiss,dscthesi s,tcc,depositor,coppescale}
2926 \DeclareBibliographyStrings{%

```

```

2927 inherit      = {italian},
2928 availablefrom = {{Disponibile da}{Disponibile da}},
2929 urlseen       = {{Consultato il}{Consultato il}},
2930 in            = {{In}{In}},
2931 editor        = {{curatore}{cur\adddot}},
2932 editors       = {{curatori}{cur\adddot}},
2933 depositor     = {{Depositante}{Depositante}},
2934 coppescale    = {{Scala}{Scala}},
2935 msdiss        = {{Tesi di Laurea Magistrale}{Tesi di Laurea Magistrale}},
2936 dscthesis     = {{Tesi di Dottorato}{Tesi di Dottorato}},
2937 phdthesis     = {{Tesi di Dottorato}{Tesi di Dottorato}},
2938 tcc           = {{Tesi di Laurea Triennale}{Tesi di Laurea Triennale}},
2939 }
2940 

```

```

2981 \copperdefstring{spanish}{monthname7} {julio}
2982 \copperdefstring{spanish}{monthname8} {agosto}
2983 \copperdefstring{spanish}{monthname9} {septiembre}
2984 \copperdefstring{spanish}{monthname10}{octubre}
2985 \copperdefstring{spanish}{monthname11}{noviembre}
2986 \copperdefstring{spanish}{monthname12}{diciembre}
2987 \copperdefstring{spanish}{algorithm2eopt}{spanish}
2988 \copperdefstring{spanish}{babelname} {spanish}
2989 %% biblatex is loaded by coppe.cls AFTER this .def file is read; defer the
2990 %% language-mapping registration until \begin{document} so biblatex's
2991 %% \DeclareLanguageMapping is available.
2992 \AtBeginDocument{\DeclareLanguageMapping{spanish}{spanish-coppe}}
2993 \

```



```

3035 \copperdefstring{french}{monthname12}{d\'ecembre}
3036 \copperdefstring{french}{algorithm2eopt}{french}
3037 \copperdefstring{french}{babelname} {french}
3038 \AtBeginDocument{\DeclareLanguageMapping{french}{french-coppe}}
3039 \
```

```

3089 \ProvidesFile{coppe-numeric.bbx}[2026/05/28 v4.0 CoppeTeX ABNT numeric bibliography]
3090 % Numeric reference list for the ABNT numeric system (NBR 10520). It reuses the
3091 % full coppe ABNT style (drivers, CAPS surnames, bold titles, "In:", p./f.,
3092 % the PT->EN source map and the language mappings) and only adds the [n] list
3093 % label and the numeric sorting on top.
3094 \RequireBibliographyStyle{coppe}
3095
3096 % ABNT numeric system lists references in the order they are first cited
3097 % (NBR 10520), i.e. unsorted. Override with sorting=nty for an alphabetical
3098 % numbered list. 'labelnumber' generates the [n] field.
3099 \ExecuteBibliographyOptions{labelnumber,sorting=none}
3100
3101 % Bracketed labels, both in the list and for any shorthand entries.
3102 \DeclareFieldFormat{labelnumberwidth}{\mkbibbrackets{#1}}
3103 \DeclareFieldFormat{shorthandwidth}{\mkbibbrackets{#1}}
3104
3105 % Reference list driven by the [n] label (copied from biblatex's numeric.bbx),
3106 % keeping the ABNT blank line between entries via \bibitemsep set in coppe.bbx.
3107 \defbibenvironment{bibliography}
3108 {
  \list
3109   {\printtext[labelnumberwidth]{%
3110     \printfield{labelprefix}%
3111     \printfield{labelnumber}}}
3112   {\setlength{\labelwidth}{\labelnumberwidth}%
3113    \setlength{\leftmargin}{\labelwidth}%
3114    \setlength{\labelsep}{\biblabelsep}%
3115    \addtolength{\leftmargin}{\labelsep}%
3116    \setlength{\itemsep}{\bibitemsep}%
3117    \setlength{\parsep}{\bibparsep}}%
3118   \renewcommand*{\makelabel}[1]{\hss#1}}
3119 {\endlist}
3120 {\item}
3121
3122 </numbbx>
3123 <*numcbx>
3124 \ProvidesFile{coppe-numeric.cbx}[2026/05/28 v4.0 CoppeTeX ABNT numeric citations]
3125 % Numeric citation system (ABNT NBR 10520): in-text references are bracketed
3126 % numbers [1], compressed and sorted within a single cite group. This is the
3127 % opt-in alternative to the author-date default (coppe.cbx); it pairs with
3128 % bibstyle=coppe-numeric. With the biblatex natbib option, \citep -> [1] and
3129 % \citet -> Author [1].
3130 \RequireCitationStyle{numeric-comp}
3131 </numcbx>

```

10.2 The bibliographic databases

The two BibLaTeX/biber databases shipped with the class — `example.bib` (the sample database cited by the example monograph) and `coppe.bib` (the references of this manual) — are generated from this source too, so the whole distribution still has a single editable origin. They live inside an `\iffalse` block below: the documentation driver never typesets them, and only `coppe.ins` extracts them, through the `examplebib` and `coppebib` docstrip targets. Edit the entries here, never the generated `.bib` files.

Acknowledgments

Thanks to all COPPE_{TeX} users who have reported their experience with this class. We also thank to professor Fernando Lizarralde and Heiko Oberdiek for their helpful comments. The authors would like to thank the National Council for Scientific and Technological Development (CNPq) of Brazil.

Index

Numbers written in *italic* refer to the page where the corresponding entry is described; numbers underlined refer to the code line of the definition; numbers in roman refer to the code lines where the entry is used.

Symbols		
<code>\%</code>	<code>\@copperomanfalse</code> 1245	<code>\@wrlos</code> 2066, 2073, 2078
<code>\,</code>	<code>\@copperomantrue</code> . 1246	<code>\{</code> 772
<code>\;</code>	<code>\@date</code> 1981	<code>\sq</code> 122, 377, 554, 567,
<code>\@Roman</code>	<code>\@dedic</code> 1895, 1900	595, 804, 839,
<code>\@advisor</code>	<code>\@degree</code> 1208,	932, 1031, 1552,
1538,	1216, 1225,	1593, 1679,
1540, 1541,	1234, 1574,	1704, 1737,
1542, 1543,	1575, 2294, 2299	1750, 1829,
1544, 1545,	<code>\@degreename</code>	1831, 1847,
1717, 1760,	. . . 1209, 1217,	1877, 1887,
1861, 1888,	1226, 1235,	1888, 1911,
1921, 1959, 2008	1576, 1577,	1949, 1998,
<code>\@afterheading</code> . . .	1831, 1877, 2263	2263, 2268,
1074	<code>\@examiner</code> . . . 1550,	2284, 2285,
<code>\@authname</code> . . . 1556,	1552, 1553, 1781	2291, 2294,
1679, 1704,	<code>\@firstoftwo</code> 2483	2297, 2299,
1750, 1829,	<code>\@footnotetext</code>	2878, 2879, 2880
1845, 1847,	 1143, 1144	<code>\~</code> 236, 266, 291, 325,
1911, 1949, 1998	<code>\@for</code> 1374	344, 364, 500,
<code>\@author</code> 1980	<code>\@ifstar</code> 2129	519, 547, 566,
<code>\@authsurn</code> . . . 1557,	<code>\@labfile</code> 2087,	590, 594, 618,
1679, 1704,	2088, 2100,	694, 705, 851,
1750, 1829,	2105, 2133, 2138	853, 1212, 1221,
1844, 1847,	<code>\@losfile</code> 2061,	1230, 1305,
1911, 1949, 1998	2062, 2074, 2079	1309, 1311,
<code>\@chapapp</code> 1051	<code>\@lst</code> . . 1425, 1452, 1461	1399, 1401,
<code>\@chapter</code> 1047	<code>\@makechapterhead</code> .	1403, 1514,
<code>\@coppeannexfalse</code> 1040	 1071, 1073	1520, 1634,
<code>\@coppeannextrue</code> . 2355	<code>\@nameuse</code> 1266, 1267,	1636, 1652,
<code>\@coppeappendixfalse</code>	1268, 1537,	1658, 1676,
. 1028	2114, 2116, 2125	2265, 2267,
<code>\@coppeappendixtrue</code>	<code>\@roman</code> 2150	2275, 2277,
. 1035	<code>\@secondoftwo</code> 2483	2293, 2859, 2862
<code>\@coppedoublespacefalse</code>	<code>\@tempsymb1</code> . . . 2064,	
. 1247	2066, 2090, 2092	
<code>\@coppedoublespacetrue</code>	<code>\@topnewpage</code> 1071	A
. 1248	<code>\@width</code> 1145	<code>\abbrev</code> . . . 865, 866,
<code>\@coppenumericfalse</code>	<code>\@wrlab</code> . 2092, 2099,	867, 868, 869,
. 1241	2104, 2132, 2137	870, 871, 2089, 2114
<code>\@coppenumerictrue</code> 1242		<code>abstract (env.)</code> . . . 1903

<code>\abstract@top</code>	<code>\AtBeginBibliography</code>	605, 606, 607,
.. 1906, 1993, 2289	... 2196, 2329,	608, 609, 611,
<code>\addbibresource</code> ... 19	2332, 2386, 2453	612, 613, 615, 623
<code>\addcolon</code> 2519,	<code>\Ate</code> 735	<code>\citeauthor</code> 2843
2522, 2579,	<code>\AtEndBibliography</code> 2197	<code>\citep</code> 195, 198,
2617, 2619,	<code>\AtEndOfClass</code> 979	563, 575, 577,
2621, 2623,	<code>\author</code>	578, 579, 580,
2657, 2670, 2706	27, 1555, 1986, 1987	582, 583, 584,
<code>\addcomma</code> 2538, 2541,	<code>\autocite</code> 563, 623	586, 587, 623,
2542, 2543,	<code>\autoref</code> 217,	627, 824, 2310, 3128
2545, 2546,	241, 251, 275,	<code>\citepapud</code>
2547, 2548,	309, 328, 367,	202, 631, 2836, 2841
2549, 2580,	504, 522, 710,	<code>\citet</code> 198, 563,
2583, 2604,	727, 743, 804, 812	603, 605, 606,
2605, 2633,		607, 608, 610,
2634, 2658,		611, 612, 614,
2708, 2826,		615, 623, 824,
2842, 2843, 2844		2310, 2821, 3129
<code>\adddot</code> 2510,		<code>\citetapud</code>
2511, 2512,		202, 631, 2834, 2839
2515, 2516,		<code>\citetitle</code> 108
2544, 2565,		<code>\citeyear</code> 2843
2855, 2856,		<code>\ClassInfo</code> 1379
2874, 2875,		<code>\ClassWarningNoLine</code>
2878, 2879,		 979, 1004
2880, 2893,		<code>\clearname</code>
2894, 2912,		.. 2432, 2441, 2564
2913, 2931, 2932		<code>\color</code> 1418,
<code>\adnbspace</code> ... 2510,		1419, 1420,
2511, 2512,		1421, 1431,
2515, 2516, 2649		1433, 1439,
<code>\addperiod</code> ... 2616,		1440, 1445,
2618, 2620, 2622		1446, 1447,
<code>\addsemicolon</code> . 2414,		1454, 1455, 1456
2415, 2824, 2825		<code>\coppe@algotang</code> ..
<code>\addspace</code> 2430, 2439,		.. 1476, 1477, 1480
2563, 2594,		<code>\coppe@brazilianstr</code>
2601, 2686, 2827		.. 1251, 1273, 1376
<code>\addtocontents</code>		<code>\coppe@cur</code>
..... 1068, 1069		... 1272, 1273,
<code>\addtolength</code> 2203, 3115		1274, 1375,
<code>\addvspace</code>		1376, 1377,
.. 1068, 1069, 1127		1378, 1379, 1381
<code>\advisor</code> 28, 29, 30, 1538		<code>\coppe@englishstr</code> .
<code>\aftergroup</code> 2483		 1194,
<code>\alph</code> 1150		1252, 1274, 1377
<code>\alpha</code> 795, 855, 861		<code>\coppe@lang</code> 1374, 1375
<code>\annex</code>		<code>\coppe@abntdate</code> ...
906, 916, 1027, 2352		.. 2449, 2458, 2459
<code>\appendixname</code> 1031, 2353		<code>\coppe@babelopts</code> ..
<code>\arraybackslash</code> ...		.. 1257, 1258, 1374
... 1713, 1756,		<code>\coppe@bibBegin</code> 2166,
1777, 1857,		2182, 2197, 2332
1917, 1955, 2004		

\coppe@bibEnd .	2162,	\coppefinalfont . . .	2956,	2957,
	2179, 2197, 2336		2958,	2959,
\coppe@date . . .	936, 939	\coppefinalmanual .	2960,	2961,
\coppe@docmonth	944,		2962,	2963,
	1560, 1562, 1565	\coppeforeignstring	2964,	2965,
\coppe@docyear	945,		2966,	2967,
	1561, 1684,	\coppemainstring . .	2968,	2969,
	1737, 1848,		2970,	2971,
	1878, 1913,		2972,	2973,
	1951, 2000, 2285		2974,	2975,
\coppe@foreignlang .		\copperdefstring . .	2976,	2977,
	. . . 1196, 1198,		2978,	2979,
	1200, 1202,		2980,	2981,
	1204, 1206,		2982,	2983,
	1257, 1268, 1943		2984,	2985,
\coppe@hasLof			2986,	2987,
	. . 2030, 2174, 2188		2988,	2999,
\coppe@hypersetup .			3000,	3001,
	 1689, 1798		3002,	3003,
\coppe@mainBegin . .			3004,	3005,
	. . 1616, 2170, 2185		3006,	3007,
\coppe@mainlang	1194,		3008,	3009,
	1195, 1198,		3010,	3011,
	1200, 1202,		3012,	3013,
	1204, 1206,		3014,	3015,
	1257, 1267, 1537		3016,	3017,
\coppe@orig@appendix			3018,	3019,
	 1034, 1035		3020,	3021,
\coppe@orig@footnotetext			3022,	3023,
	 1143, 1144		3024,	3025,
\coppe@orig@printbibliography			3026,	3027,
	 2333, 2335		3028,	3029,
\coppe@selecttitle .			3030,	3031,
	. . . 1537, 1681,		3032,	3033,
	1702, 1748, 1847		3034,	3035,
\coppe@sigla@first .			3036,	3037,
	 2120, 2124		3047,	3048,
\coppe@sigla@full .			3049,	3050,
	. . 2115, 2121, 2128		3051,	3052,
\coppe@sigla@norm .			3053,	3054,
	 2122, 2129		3055,	3056,
\coppe@sigla@reg . .			3057,	3058,
	 2112, 2121		3059,	3060,
\coppe@sigla@star .			3061,	3062,
	 2126, 2129		3063,	3064,
\coppe@sigla@undef .			3065,	3066,
	. . 2117, 2123, 2127		3067,	3068,
\coppe@tocanx	1044, 1054		3069,	3070,
\coppe@tocapp	1041, 1056		3071,	3072,
\coppe@version			3073,	3074,
	. . . 936, 938, 1658		3075,	3076,
\coppefinalengine .			3077,	3078,
	 1630, 1655		3079,	3080,
			3081,	3082,

3083, 3084, 3085	2888, 2907, 2926	\diag 13, 783, 785
\coppestring .. 1042,	\DeclareDatamodelEntryfield	\dots 785, 842
1045, 1108,	... 2376, 2377,	\draw 543
1109, 1125,	2379, 2380, 2381	
1262, 1266,	\DeclareDatamodelEntrytypes	E
1408, 1409,	 2365	\endfirsthead 374
2321, 2325,	\DeclareDatamodelFields	\endfoot 382
2326, 2353,	... 2367, 2368,	\endhead 379
2594, 2617,	2369, 2370,	\endlastfoot 387
2619, 2621, 2623	2371, 2372,	\endwrite 2146
\CoppeTeX 146,	2373, 2374, 2375	\Enquanto 727
916, 962, 1171, 1657	\DeclareDelimFormat	\enquote ... 1384, 1387
\coppetexfinalpage .	 2414,	environments:
... 922, 937, 1630	2415, 2824,	abstract 1903
\coppeUC@w	2825, 2826, 2827	brazilianabstract
.. 2480, 2481, 2482	\DeclareFieldFormat	 1989
\coppeUCifart	 2485,	foreignabstract 1940
.. 2471, 2477, 2478	2493, 2497,	longquote 2198
\coppeUCkeep .. 2467,	2498, 2501,	\es@scroman 2151
2473, 2475, 2476	2504, 2505,	\examiner 35,
\coppeUCstop	2506, 2507,	36, 37, 38, 39, 1550
... 2469, 2470,	2510, 2511,	\ExecuteBibliographyOptions
2472, 2473,	2512, 2515,	 2396, 3099
2474, 2475, 2476	2516, 2522,	
\coppeUCtitle . 2464,	2591, 2594,	F
2469, 2491, 2495	2649, 3102, 3103	\fancyhead
\coppeUCtitleA	\DeclareFixedFont .	.. 1159, 1607, 1619
..... 2469, 2470	 1395, 1396	\fancyhf 1159,
\coppeUCtitleB	\DeclareLanguageMapping	1607, 1611, 1619
..... 2472, 2474	 2406,	\fancypagestyle 1159,
\cpfonte 210, 1123, 1129	2407, 2991,	1607, 1611, 1619
\cpsource	2992, 3038, 3086	\filcenter 1079
210, 1123, 1126,	\DeclareMathOperator	\floatname . 1111, 1136
1129, 1130, 1131	.. 12, 13, 782, 783	\floatstyle . 1102, 1135
\cpsourcenam 1125, 1128	\DeclareNameAlias .	\fonte 210, 236, 266,
	 2410,	291, 325, 344,
D	2411, 2412, 2413	364, 500, 519,
\Dados 732	\DeclareRobustCommand	547, 590, 618,
\date 41,	 2839, 2841	694, 705, 1121, 1131
940, 942, 1559, 1988	\DeclareSourcemap 2724	\foo 839
\DeclareBibliographyDriver	\dedication ... 50, 1894	\footnote
... 2530, 2587,	\defbibenvironment 3107	. 155, 161, 808, 824
2588, 2595,	\defbibheading ... 2324	\footnotemark 808
2642, 2643,	\DefineVerbatimEnvironment	\footnoterule 1145
2644, 2645,	 1465, 1471	\footnotetext 808
2646, 2650,	\department ... 40, 1488	\foreign@ 1185
2676, 2695,	\det 757	\foreign@advisorstring
2696, 2712,	\detokenize	.. 1547, 1956, 2249
2713, 2714,	... 2147, 2164,	\foreign@degname ..
2715, 2716,	2165, 2168,	... 1211, 1219,
2717, 2718,	2169, 2172,	1228, 1237,
2719, 2720, 2721	2173, 2176,	1580, 1581, 2299
\DeclareBibliographyStrings	2177, 2181,	\foreign@deptname .
... 2850, 2869,	2184, 2187, 2190	 1491,

1494, 1497, 1500, 1503, 1506, 1509, 1512, 1515, 1518, 1521, 1524, 1527, 1976	\foreign@deptstring 1976, 2257	\hookrightarrow 1423, 1435, 1442, 1450, 1459	\itshape ... 1089, 2206
\foreign@doctype 1213, 1222, 1231, 1239, 1584, 1585, 2297	\foreign@monthname 1565, 1951	I \i 1658, 2852, 2897 \if@coppeannex 1040, 1053 \if@coppeappendix . .. 1028, 1031, 1052	J \javaexternal 1463 \javainline .. 727, 1462 \javastyle ... 1453, 1460, 1462, 1463
\foreign@title 1534, 1947, 1982, 1983	foreignabstract (env.) 1940	\if@coppedoublespace 1247, 2305 \if@coppenumeric 1241, 2311 \if@copperoman 1245, 1606	K \keyword 43, 44, 45, 1568
\foreign@toptext 1944, 2296	\foreign@title 26, 1280, 1533	\if@english . 1190, 1256 \if@mainmatter ... 1049 \if@openright 1624 \if@twoside . 1165, 1642 \ifbibstring 2591 \ifblank 2472, 2473, 2475, 2844	L \l@figure 1115 \l@quadro .. 1114, 1115 \labelnamepunct 2534, 2569, 2598, 2627, 2653, 2680, 2702 \labelnumberwidth 3112 \lambda 785 \language name 1266 \Large 1030, 1079 \large 1083 \LaTeXe 965 \ldots 743 \left 772 \leftarrow 734, 735, 736 \leftskip 2203 \linewidth 509, 514, 1669, 1696
\freeconfig 1573	\frontcover@maintext 1709, 2262	\ifboolexpr ... 2428, 2437, 2486, 2494 \iffieldequalstr . 2539 \iffieldundef . 2450, 2451, 2452, 2540, 2618, 2706 \ifin@ 2483 \iflistundef 2616, 2620, 2622 \ifLuaTeX 1632 \ifnameundef .. 2428, 2437, 2486, 2494, 2560, 2668 \ifodd 1166, 1643 \ifPDFTeX 977, 993, 1631 \ifstrempty . 2064, 2090 \ifuseeditor 2428, 2437 \ifXeTeX 1633 \in@ 2482 \index 820 \InheritBibliographyExtras ... 2848, 2867, 2886, 2905, 2924 \InputIfFileExists 1275, 1378 \int 795 \itemsep 3116	\listof 212, 1133 \listofalgorithms . 97 \listoffigures 93, 2029 \listofprogramas 96, 710, 1390, 1410, 1411 \listofquadros 95, 212, 1116 \listoftables . 94, 2043 \listquadroname 1109, 1117, 1118, 1119 \listsymbolname 2058, 2245 \local@advisorstring ... 1546, 1714, 1757, 1858, 1918, 2005, 2248 \local@approvedname 2251, 2253 \local@cityname 1683, 1736, 1847, 2258, 2284
\frontmatter 49, 1154, 1590, 1595, 1742	\frontpage@bottomtext 1797, 2282		
\frontpage@maintext 1753, 2271			
G \ge 773 \glossaryname 2058, 2084, 2214, 2215, 2216, 2217, 2246			
H \hat 785 \hbar 8, 743, 1007 \headheight 1157 \headrulewidth 1160, 1608, 1611, 1620 \hfill .. 512, 1670, 1697 \hologo ... 161, 532, 559, 561, 743, 789, 829, 961, 963, 964, 965, 966, 967, 968, 969, 970, 971, 1631, 1632, 1633			

	2658, 2680,	\stripzeros . 2450, 2452	\ttm 1396, 1417, 1428,
	2686, 2702, 2708	\subsection ... 305,	1437, 1444, 1453
\sigla	876, <u>2109</u>	625, 629, 633,	
\singlespacing	1128,	641, 652, 661, 1085	U
	1144, 1673,	\subsubsection ... 1088	\uppercase 2152
	2205, 2385, 2454	\symbl 174, 175, 856,	\url 112, 816, 1659, 2522
\smallskipamount .	1127	857, 858, 859,	\usebibmacro . . 2431,
\source 210, 212, 248,		860, 861, 862, 2063	2440, 2531,
	554, 719, 1091,		2532, 2533,
	1121, 1130, 1133	T	2535, 2553,
\step 2727, 2728, 2729,		\tablename 377	2554, 2563,
	2730, 2731,	\tableofcontents . . 100	2567, 2568,
	2732, 2733,	\text 773, 774	2570, 2572,
	2734, 2735,	\textcite	2573, 2585,
	2736, 2737,	563, 623, 2821, 2840	2586, 2587,
	2738, 2739,	\textcolor 1423, 1435,	2588, 2596,
	2740, 2741,	1442, 1450, 1459	2597, 2599,
	2742, 2743,	\textendash	2611, 2612,
	2744, 2745,	. . 1031, 1043, 1046	2625, 2626,
	2746, 2747,	\textsc 1652	2628, 2630,
	2750, 2751,	\thechapter	2640, 2641,
	2752, 2753,	. . . 1031, 1051,	2642, 2643,
	2754, 2755,	1054, 1056, 1059	2644, 2645,
	2756, 2757,	\theFancyVerbLine 1472	2646, 2651,
	2758, 2759,	\thefield	2652, 2654,
	2760, 2761,	. . 2450, 2451, 2452	2662, 2663,
	2762, 2763,	\thesection 1083	2677, 2678,
	2764, 2765,	\thesubsection . . . 1086	2679, 2681,
	2767, 2768,	\thesubsubsection 1089	2683, 2690,
	2769, 2770,	\thetable 377	2691, 2700,
	2771, 2772,	\title . . . 25, 1280, <u>1529</u>	2701, 2703,
	2773, 2774,	\titleformat	2710, 2711,
	2775, 2776,	. . . 1029, 1078,	2712, 2713, 2714
	2778, 2779,	1082, 1085, 1088	\usecoppelanguage .
	2780, 2781,	\titlein . . . 1278, 1282	 1187, 1271
	2782, 2783,	\titlespacing 1033, 1080	\usedriver 2695, 2696,
	2784, 2785,	\today 941, 1656	2715, 2716,
	2786, 2787,	\topmargin 1020	2717, 2718,
	2788, 2789,	\toprule 282,	2719, 2720, 2721
	2790, 2791,	317, 336, 357, 602	
	2792, 2793,	\ttb 1395, 1419, 1420,	V
	2794, 2795,	1431, 1433,	\varnothing 8, 743, 1007
	2796, 2797,	1439, 1446,	
	2798, 2799,	1447, 1455, 1456	X
	2800, 2801,	\ttfamily 1397, 1418,	\XeLaTeX 970
	2802, 2803, 2804	1429, 1438,	\xmlstyle . . 1428, 1436
\stretch 1678, 1680, 1682		1445, 1454, 1472	\xspace 839

Change History

v0.0

General: Creation Date. 1

v0.1

General: Documentation:

	bibliography fixed, title translation.	1		
	Sourceforge submission.	1		
v0.2	General: Unification of the code for the list of symbols and abbreviations.	1	v3.4	General: Some examples for figures, tables, longtables, etc. . .
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v0.3	General: Added 'draft' option. . . .	1	v3.4.1	General: Fix english option to consider brazilian a second language, describe it in document
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	\maketitle: Added number of examiners test.	47	v3.5	General: New command annex . .
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v0.4	General: Beta documentation. . . .	1	v3.5.1	General: New command annex now supports brazilian or english . .
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v0.5	abstract: Changed from macro to environment.	51	v4.0	General: Catalog-label writers (coppe@mainBegin, coppe@bibBegin, coppe@bibEnd, coppe@hasLof) made language-agnostic: detokenize wrap on label names protects ':' and '.' from babel shorthand activation, and the raw etex primitives romannumeral/number replace LaTeX's roman/arabic to bypass spanish-babel's small-caps @roman redefinition that would otherwise crash pdftex with "Unbalanced write command" on every es-main document.
	\backmatter: Added mainmatter pages counter.	45		1
	foreignabstract: Changed from macro to environment.	52		Class-wide string dispatcher (copperdefstring, copperstring, coppemainstring, copperforeignstring) replaces scattered iflanguage and if english branches; pt/en outputs remain byte-identical. . .
v1.0	General: First COPPETEX release. . .	1		1
v2.0	General: COPPETEX release 2.0. . .	1		Multilingual model: three fixed slots (main, foreign, optional third); five built-in language options (brazilian, english, spanish, french, italian); plug-in mechanism for any other Babel language via coppe-lang-lang.def plus lang-coppe.lbx.
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v2.1	General: COPPETEX release 2.1: Matching the new rules.	1		New user API: titlein, usecoppelanguage, brazilianabstract environment. . .
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	\department: Added new course on Nanotechnology.	42		
	\examiner: Examiners expansion without degree.	43		
v3.1	\department: Included a sort key in symb	42		
v3.2	General: Fixed version problem between cls and dtx.	1		
v3.3	General: Extend abbrev to work like symb and accept a sorting key . .	1		